# Vector Bundles on Curves

Taught by Jiabin Du

# Contents

# Introduction

The course aims at introducing vector bundles on curve and Koszul cohomology. You may benefit from the following references

- *Ample Vector Bundles*, Hartshorne
- *Koszul Cohomology and Algebraic Geometry*, Aprodu-Nagel
- *Lectures on Vector Bundles*, J.Le Potier
- *The Geometry of Moduli Space of Sheaves*, Huybrechts-Lehn.

# 1 Vector Bundles

## 1.1 Sheaf theory

Recall the definition of sheaves.

**Definition 1.1.** *Let $X$ be a topological space and $\mathcal{C}$ be a category. Assume $op(X)$ is the category of open subsets of $X$ with morphisms are inclusions. A presheaf on $X$ valued on $\mathcal{C}$ (or say of $\mathcal{C}$) is a contravariant functor $\mathcal{F} : op(X) \longrightarrow \mathcal{C}$.*

**Remark 1.1.** *Though we can consider this general definition of presheaf, we would only consider presheaves at least valued on Ab, where Ab is the category of abelian groups. In addition, we need these presheaves map $\varnothing$ to $\{0\}$.*

**Definition 1.2.** *Let $X$ be a topological space, $\mathcal{F}$ presheaf of abelian groups on $X$. For $V \subseteq V$, we call the correspondence homomorphism $\mathcal{F}(U) \longrightarrow \mathcal{F}(V)$ a restriction map, denoted by $\cdot\rho_{UV}$. Elements in $\mathcal{F}(U)$ is called a section of $\mathcal{F}$ on $U$. In particular, if $U = X$, we call elements of $\mathcal{F}(X)$ global sections, otherwise local sections.*

**Remark 1.2.** *For convenience, for $s \in \mathcal{F}(U)$, we would write $s\big|_V$ to denote image of $s$ instead of $\rho_{UV}(s)$. Also, sometimes $\mathcal{F}(U)$ can be rewritten as $\Gamma(U, \mathcal{F})$.*

**Definition 1.3.** *Let $X$ be a topological space, $\mathcal{F}$ presheaf of abelian groups on $X$. $\mathcal{F}$ is a sheaf on $X$ if it satisfies that*
*(1)For each open subset $U$ with open covering $\{U_i\}_{i\in I}$ and $s \in \mathcal{F}(U)$, if $s\big|_{U_i} = 0$ for all $i$, then $s = 0$.*
*(2)For each open subset $U$ with open covering $\{U_i\}_{i\in I}$, if there exists $s_i \in \mathcal{F}(U_i)$ for all $i$, satisfying $s_i\big|_{U_i\cap U_j} = s_j\big|_{U_i\cap U_j}$ for all $i, j$, then there exists $s \in \mathcal{F}(U)$ such that $s\big|_{U_i} = s_i$ for all $i$.*

**Remark 1.3.** *The two conditions in fact give a exact sequence for each open subset $U$ with open covering $\{U_i\}_{i\in I}$.*

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \prod_{i\in I} \mathcal{F}(U_i) \longrightarrow \prod_{i,j\in I} \mathcal{F}(U_i \cap U_j) \tag{1}$$

**Example 1.1.** *(1)Let $X$ be a topological space. Then $U \longmapsto \{$continuous functions $U \longrightarrow \mathbb{R}\}$ give a sheaf on $X$.*
*(2)Let $X = \mathbb{R}^n$. Then $U \longmapsto \{C^\infty$-functions $U \longrightarrow \mathbb{R}\}$ gives a sheaf on $X$.*
*(3)Let $X$ be a topological space and $A$ be an abelian group. View $A$ as a topological space with discrete topology. Then $U \longmapsto \{$locally constant functions $U \longrightarrow A\}$ gives a sheaf on $X$, which is called the constant sheaf of $A$ on $X$.*

**Definition 1.4.** *Let $X$ be a topological space, $\mathcal{F}$ (pre)sheaf of abelian groups on $X$. For $x \in X$, define the stalk of $\mathcal{F}$ at $x$ to be $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$. More concretely, an element of $\mathcal{F}_x$ is represented by a pair $(U, s) = s_x$ where $U$ is an open neighbourhood of $x$ and $s \in \mathcal{F}(U)$, called a germ. Two germs $(U, s)$ and $(V, t)$ represent the same element if there exists open neighbourhood $W$ of $x$ such that $W \subseteq U \cap V$ and $s\big|_W = t\big|_W$.*

**Proposition 1.1.** *Let $X$ be a topological space, $\mathcal{F}$ sheaf of abelian groups on $X$. Suppose $U$ is an open subset of $X$ and $s, t \in \mathcal{F}(U)$. Then $s = t$ if and only if $s_x = t_x$ for all $x \in U$.*

*Proof.* "$\Rightarrow$ ": Obviously.

"$\Leftarrow$ ": Since for all $x \in U$, we have $s_x = t_x$, there exists open neighbourhood $W_x$ of $x$ such that $W_x \subseteq U$ and $s\big|_{W_x} = t\big|_{W_x}$. Note that $\{W_x\}_{x \in U}$ form an open covering of $U$. By property of sheaf, we get $s = t$. □

**Definition 1.5.** *Let $X$ be a topological space, $\mathcal{F}, \mathcal{G}$ (pre)sheaves of abelian groups on $X$. A morphism $\varphi$ from $\mathcal{F}$ to $\mathcal{G}$ is a family of group homomorphisms $\varphi(U) : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$ such that for $V \subseteq U$, there is a commutative diagram*

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, "\varphi(U)"] \arrow[d, "\rho_{UV}"] & \mathcal{G}(U) \arrow[d, "\rho_{UV}"] \\ \mathcal{F}(V) \arrow[r, "\varphi(V)"] & \mathcal{G}(V) \end{tikzcd} \tag{2}$$

**Remark 1.4.** *By property of direct limit, morphism $\varphi$ gives homomorphism $\varphi_x : \mathcal{F}_x \longrightarrow \mathcal{G}_x$ for all $x \in X$.*

**Definition 1.6.** *Let $X$ be a topological space, $\varphi : \mathcal{F} \longrightarrow \mathcal{G}$ morphism of (pre)sheaves of abelian groups on $X$. We say $\varphi$ is isomorphic if there exists morphism $\psi : \mathcal{G} \longrightarrow \mathcal{F}$ such that for each open subsets $U$, $\psi(U)\varphi(U) = \mathrm{id}_{\mathcal{F}(U)}$ and $\varphi(U)\psi(U) = \mathrm{id}_{\mathcal{G}(U)}$.*

**Proposition 1.2.** *Let $X$ be a topological space, $\varphi : \mathcal{F} \longrightarrow \mathcal{G}$ morphism of sheaves of abelian groups on $X$. Then $\varphi$ is isomorphic if and only if $\varphi_x$ is isomorphic for all $x \in X$.*

*Proof.* "$\Rightarrow$": For all $(U, t) \in \mathcal{F}_x$, since $\varphi(V)$ is isomorphic, there exists $s \in \mathcal{F}(U)$ such that $\varphi(V)(s) = t$. Thus $\varphi_x((U, s)) = (U, t)$, get $\varphi_x$ is surjective. For all $(U, s) \in \ker(\varphi_x)$, $(U, \varphi(U)(s)) = 0$ in $\mathcal{G}_x$. Then there exists open neighbourhood $V$ of $x$ such that $V \subseteq U$ and $\varphi(U)(s)\big|_V = 0$. By commutative diagram, $\varphi(U)(s)\big|_V = \varphi(V)(s\big|_V)$. Since $\varphi(V)$ is isomorphic, get $s\big|_V = 0$ and so $(U, s) = 0$ in $\mathcal{F}_x$. Get $\varphi_x$ is isomorphic.

"$\Leftarrow$": For each open subset $U$, want to show that $\varphi(U)$ is isomorphic. For $s \in \ker \varphi(U)$, $\varphi(U)(s) = 0$. Then for all $x \in U$, $(U, \varphi(U)(s)) = 0$ in $\mathcal{G}_x$. As $(U, \varphi(U)(s)) = \varphi_x((U, s))$ and

$\varphi_x$ is isomorphic, get $(U, s) = 0$ in $\mathcal{F}_x$. Thus there exists open neighbourhood $W_x$ of $x$ such that $W_x \subseteq U$ and $s\big|_{W_x} = 0$. By property of sheaf, get $s = 0$ and so $\varphi(U)$ is injective.

For all $t \in \mathcal{G}(U)$, $(U, t)$ is an element of $\mathcal{G}_x$ for all $x \in U$. Then there exists $(V_x, s(x)) \in \mathcal{F}_x$ such that $\varphi_x((V_x, s(x))) = (U, t)$ and $\varphi(V_x)(s(x)) = t\big|_{V_x}$. Note that $\varphi(V_x \cap V_y)(s(x)\big|_{V_x \cap V_y}) = (t\big|_{V_x})\big|_{V_x \cap V_y} = (t\big|_{V_y})\big|_{V_x \cap V_y} = \varphi(V_x \cap V_y)(s(y)\big|_{V_x \cap V_y})$. Since $\varphi(V_x \cap V_y)$ is injective, get $s(x)\big|_{V_x \cap V_y} = s(y)\big|_{V_x \cap V_y}$. By property of sheaf, there exists $s \in \mathcal{F}(U)$ such that $s\big|_{V_x} = s(x)$. So $\varphi(U)(s)\big|_{V_x} = \varphi(V_x)(s(x)) = t\big|_{V_x}$. By property of sheaf, get $\varphi(U)(s) = t$ and so $\varphi(U)$ is isomorphic. $\square$

**Definition 1.7.** *Let $X$ be a topological space, $\varphi : \mathcal{F} \longrightarrow \mathcal{G}$ morphism of presheaves of abelian groups on $X$. We have several notions*

- *Define the kernel of $\varphi$ to be* $\ker(\varphi) : op(X) \longrightarrow Ab \quad U \longmapsto \ker(\varphi(U))$.
- *Define the image of $\varphi$ to be* $im(\varphi) : op(X) \longrightarrow Ab \quad U \longmapsto im(\varphi(U))$.
- *Define the cokernel of $\varphi$ to be* $coker(\varphi) : op(X) \longrightarrow Ab \quad U \longmapsto coker(\varphi(U))$.

**Remark 1.5.** *The definition of cokernel in fact induce the definition of quotient presheaf.*

**Definition 1.8** (***Sheafification***)**.** *Let $X$ be a topological space, $\mathcal{F}$ presheaf of abelian groups on $X$. Define a sheaf $\mathcal{F}^+$ of abelian groups on $X$ by $U \longmapsto \mathcal{F}^+(U)$, where $\mathcal{F}^+(U) = \{(s_x)_{x\in U} \big|$ for all $x \in U$, $\exists$ open $V_x \ni x$ and $t(x) \in \mathcal{F}(V_x)$ such that $t(x)_y = s_y$, $\forall y \in V_x\}$ is a subset of $\prod_{x\in U} \mathcal{F}_x$.*

**Proposition 1.3.** *Let $X$ be a topological space, $\mathcal{F}$ presheaf of abelian groups on $X$. Show that there is a natural morphism of presheaves $\varphi : \mathcal{F} \longrightarrow \mathcal{F}^+$ satisfies that*

- *$\varphi_x$ is identity homomorphism.*
- *If $\mathcal{F}$ is sheaf, then $\varphi$ is isomorphism.*
- *For any morphism of presheaves $\psi : \mathcal{F} \longrightarrow \mathcal{G}$ where $\mathcal{G}$ is a sheaf, there exists unique morphism $\phi : \mathcal{F}^+ \longrightarrow \mathcal{G}$ such that the following diagram commutes.*

$$\begin{tikzcd} \mathcal{F} \arrow[rr, "\psi"] \arrow[dr, "\varphi"'] & & \mathcal{G} \\ & \mathcal{F}^+ \arrow[ur, "\phi"'] & \end{tikzcd} \tag{3}$$

**Reason 1.1.** *The construction of $\varphi$ is just $\varphi(U) : s \longmapsto (s_x)_{x\in U}$.*

**Definition 1.9.** *Let $X$ be a topological space, $\varphi : \mathcal{F} \longrightarrow \mathcal{G}$ morphism of sheaves of abelian groups on $X$.*
*(1)Define the kernel of $\varphi$ to be* $\ker(\varphi) : op(X) \longrightarrow Ab \quad U \longmapsto \ker(\varphi(U))$.
*(2)Define the image of $\varphi$ to be sheafification of $op(X) \longrightarrow Ab \quad U \longmapsto im(\varphi(U))$, denoted by $im(\varphi)$.*
*(3)Define the cokernel of $\varphi$ to be sheafification of $op(X) \longrightarrow Ab \quad U \longmapsto coker(\varphi(U))$, denoted by $coker(\varphi)$.*

**Remark 1.6.** *The definition of cokernel in fact induce the definition of quotient sheaf.*

**Definition 1.10.** *Let $X$ be a topological space, $\varphi : \mathcal{F} \longrightarrow \mathcal{G}$ morphism of sheaves of abelian groups on X.*
*(1)We say that $\varphi$ is injective if* $\ker(\varphi) = 0$.
*(2)We say that $\varphi$ is surjective if $im(\varphi) = \mathcal{G}$*

**Proposition 1.4.** *Let $X$ be a topological space, $\varphi : \mathcal{F} \longrightarrow \mathcal{G}$ morphism of sheaves of abelian groups on X. Then*
*(1)$\varphi$ is injective if and only if $\varphi(U)$ is injective for all open subset $U$.*
*(2)$\varphi$ is injective if and only if $\varphi_x$ is injective for all $x \in X$.*
*(3)$\varphi$ is surjective if and only if $\varphi_x$ is surjective for all $x \in X$.*
*(4)$\varphi$ is isomorphic if and only if $\varphi$ is injective and surjective.*

**Remark 1.7.** *The proof is similar to proof of Proposition 1.2. In addition, Hartshorne exercise II 1.2 gives a more explicit sketch.*

**Definition 1.11** (***Direct Image and Inverse Image***)**.** *Let $f : X \longrightarrow Y$ be continuous map of topological spaces, $\mathcal{F}$ sheaf of abelian groups on X, $\mathcal{G}$ sheaf of abelian groups on Y.*
*(1)Define dierct image to be the sheaf $f_*\mathcal{F} : V \longmapsto f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$.*
*(2)Define inverse image to be the sheafification of $U \longmapsto \varinjlim_{V \supseteq f(U)} \mathcal{G}(V)$, denoted by $f^{-1}\mathcal{G}$.*

**Remark 1.8.** *In general, we don't have $(f_*\mathcal{F})_{f(x)} \neq \mathcal{F}_x$, while there is a canonical homomorphism $(f_*\mathcal{F})_{f(x)} \longrightarrow \mathcal{F}_x$. On the other hand, we always have $(f^{-1}\mathcal{G})_x = \mathcal{G}_{f(X)}$.*

**Example 1.2.** *Let $i : Z \hookrightarrow X$ be an inclusion of topological spaces, $\mathcal{F}$ sheaf of abelian groups on X. We often write $i^{-1}\mathcal{F}$ as $\mathcal{F}\big|_Z$.*

## 1.2 Sheaf modules

**Definition 1.12.** *Let $X$ be a scheme or $(X, \mathcal{O}_X)$ be a ringed space. A sheaf of $\mathcal{O}_X$-modules is a sheaf of abelian groups $\mathcal{F}$ on $X$ such that for all $U \subseteq X$ open, $\mathcal{F}(U)$ is a module over $\mathcal{O}_X(U)$ which is compatible with restriction maps.*

**Definition 1.13.** *Let $X$ be a scheme or $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{F}, \mathcal{G}$ two $\mathcal{O}_X$-modules. A morphism $\varphi$ from $\mathcal{F}$ to $\mathcal{G}$ is a morphism of sheaves compatible with the module structure i.e. for all $U \subseteq X$ open, $\varphi(U) : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$ is a homomorphism of $\mathcal{O}_X(U)$-modules.*

**Remark 1.9.** *Kernel, image and cokernel of a morphism of $\mathcal{O}_X$-modules are still $\mathcal{O}_X$-modules. Can also define submodule, quotient module and direct sum.*

**Definition 1.14.** *Let $X$ be a scheme or $(X, \mathcal{O}_X)$ be a ringed space. An sheaf ideal on $X$ is an $\mathcal{O}_X$-submodule of $\mathcal{O}_X$.*

**Definition 1.15** (***Tensor Products***)**.** *Let $X$ be a scheme or $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{F}, \mathcal{G}$ two $\mathcal{O}_X$-modules. Define tensor product $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ to be the sheafification of $U \longmapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$.*

**Remark 1.10.** *The stalk of* $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ *at* $x \in X$ *is just* $\mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{G}_x$.

**Definition 1.16** (***Free*** $\mathcal{O}_X$***-modules***)**.** *Let* $X$ *be a scheme or* $(X, \mathcal{O}_X)$ *be a ringed space,* $\mathcal{F}$ *an* $\mathcal{O}_X$*-module. We say that* $\mathcal{F}$ *is free if* $\mathcal{F}$ *is isomorphic to a direct sum of* $\mathcal{O}_X$*. In addition, we say that* $\mathcal{F}$ *is free of rank* $r$ *if* $\mathcal{F} \cong \mathcal{O}_X^r$.

**Definition 1.17** (***Locally Free*** $\mathcal{O}_X$***-modules***)**.** *Let* $X$ *be a scheme or* $(X, \mathcal{O}_X)$ *be a ringed space,* $\mathcal{F}$ *an* $\mathcal{O}_X$*-module. We say that* $\mathcal{F}$ *is locally free if* $X$ *can be covered by open subsets* $\{U_i\}_{i\in I}$ *such that* $\mathcal{F}\big|_{U_i}$ *is free as* $\mathcal{O}_{U_i}$*-module. In addition, we say that* $\mathcal{F}$ *is locally free of rank* $r$ *if* $X$ *can be covered by open subsets* $\{U_i\}_{i\in I}$ *such that* $\mathcal{F}\big|_{U_i} \cong \mathcal{O}_{U_i}^r$ *as* $\mathcal{O}_{U_i}$*-module.*

**Definition 1.18** (***Invertible*** $\mathcal{O}_X$***-modules***)**.** *Let* $X$ *be a scheme or* $(X, \mathcal{O}_X)$ *be a ringed space,* $\mathcal{F}$ *an* $\mathcal{O}_X$*-module. We say that* $\mathcal{F}$ *is invertible if it is locally free of rank 1.*

**Remark 1.11.** *If take* $\mathcal{G} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X)$*, then* $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} \cong \mathcal{O}_X$.

**Lemma 1.1.** *Let* $f : X \longrightarrow Y$ *be a morphism of schemes or ringed spaces,* $\mathcal{F}$ *an* $\mathcal{O}_X$*-module,* $\mathcal{G}$ *an* $\mathcal{O}_Y$*-module. Then the direct image* $f_*\mathcal{F}$ *of* $\mathcal{F}$ *is an* $\mathcal{O}_Y$*-module induced by* $f^\sharp$*. On the other hand, the inverse image* $f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$ *of* $\mathcal{G}$ *is an* $\mathcal{O}_X$*-module where* $f^{-1}\mathcal{O}_Y \longrightarrow \mathcal{O}_X$ *is induced by* $f^\sharp$.

## 1.3 Quasi-coherent and coherent sheaves

**Definition 1.19.** *Let* $A$ *be a ring,* $X = \operatorname{Spec} A$*,* $M \in \mathbf{Mod}_A$*. Define an* $\mathcal{O}_X$*-module* $\widetilde{M}$ *on* $X$ *by the same process as* $\mathcal{O}_X$ *replacing* $A$ *by* $M$.

**Remark 1.12.** *The definition gives a fully faithful functor from* $\mathbf{Mod}_A$ *to category of* $\mathcal{O}_{Spec(A)}$*-modules* $M \longmapsto \widetilde{M}$.

**Proposition 1.5.** *Let* $A$ *be a ring,* $X = \operatorname{Spec} A$*,* $M \in \mathbf{Mod}_A$*. Then*
*(1)*$\widetilde{M}(X) = M$,
*(2)*$\widetilde{M}(D(f)) = M_f = M \otimes_A A_f$
*(3)*$\widetilde{M}_{\mathfrak{p}} = M_{\mathfrak{p}} = M \otimes_A A_{\mathfrak{p}}$.

**Proposition 1.6.** *Let* $A$ *be a ring,* $X = \operatorname{Spec} A$*. Then*
*(1)Let* $M_i$ *be A-modules. Then* $\widetilde{\oplus_i M_i} \cong \oplus_i \widetilde{M_i}$.
*(2)Let M,N be A-modules. Then* $\widetilde{M \otimes_A N} \cong \widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N}$.
*(3)Let* $L, M, N$ *be A-modules. Then* $L \longrightarrow M \longrightarrow N$ *is exact if and only if* $\widetilde{L} \longrightarrow \widetilde{M} \longrightarrow \widetilde{N}$ *is exact.*
*(4)Let* $f : \operatorname{Spec}(B) \longrightarrow \operatorname{Spec} A$ *be a morphism of affine schemes corresponding to ring homomorphism* $\varphi : A \longrightarrow B$*,* $M \in Mod_A$*,* $N \in Mod_B$*. Then* $f_*\widetilde{N} \cong \widetilde{N}$*, where* $N$ *is viewed as A-module, and* $f^*\widetilde{M} \cong \widetilde{M \otimes_A B}$.

**Definition 1.20** (***Quasi-coherent Sheaves***)**.** *Let* $X$ *be a scheme,* $\mathcal{F}$ *a sheaf of abelian groups on* $X$*. We say that* $\mathcal{F}$ *is quasi-coherent if for all affine open subset* $U = \operatorname{Spec} A \subseteq X$*,* $\mathcal{F}\big|_U \cong \widetilde{M}$ *for some A-module* $M$.

**Definition 1.21** (***Coherent Sheaves***). *Let $X$ be a noetherian scheme, $\mathcal{F}$ a sheaf of abelian groups on $X$. We say that $\mathcal{F}$ is coherent if for all affine open subset $U = \operatorname{Spec} A \subseteq X$, $\mathcal{F}\big|_U \cong \widetilde{M}$ for some finite $A$-module $M$.*

**Theorem 1.1.** *Let $X$ be a scheme, $\mathcal{F}$ an $\mathcal{O}_X$-module. Suppose $X$ admits an affine open subset covering $\{U_i = \operatorname{Spec}(A_i)\}_i$ such that for all $i$, $\mathcal{F}\big|_{U_i} \cong \widetilde{M_i}$ for some $A_i$-module $M_i$. Then $\mathcal{F}$ is quasi-coherent. If moreover $X$ is noetherian and $M_i$ are finite over $A_i$, then $\mathcal{F}$ is coherent.*

*Proof.* Let $U = \operatorname{Spec} A \subseteq X$ be an affine open subset. Cover each $U \cap U_i$ with $\{U_{ik}\}$ such that $U_{ik} \subseteq U_i$ of the form $D(g_{ik})$. Then $\mathcal{F}\big|_{U_{ik}} \cong \widetilde{M_{i g_{ik}}}$. Can assume $X = U = \operatorname{Spec} A$ and finitely many affine open subset $U_i$ such that $\mathcal{F}\big|_{U_i} \cong \widetilde{M_i} = \widetilde{\mathcal{F}(U_i)}$.

To prove $\mathcal{F} \cong \widetilde{\mathcal{F}(X)}$, it suffices to show that for all $f \in A$, $\mathcal{F}(X)_f = \mathcal{F}(X) \otimes_A A_f \longrightarrow \mathcal{F}(D(f))$ is isomorphic. Set $V_i = U_i \cap D(f) = D(f\big|_{U_i})$. As tensor product commutes with finite direct sum, there is commutative diagram with exact rows,

$$\begin{array}{ccccc} 0 \longrightarrow & \mathcal{F}(X)_f & \longrightarrow & \prod_i \mathcal{F}(U_i)_f & \longrightarrow & \prod_{i,j} \mathcal{F}(U_i \cap U_j)_f \\ & \downarrow & & \downarrow & & \downarrow \\ 0 \longrightarrow & \mathcal{F}(D(f)) & \longrightarrow & \prod_i \mathcal{F}(V_i) & \longrightarrow & \prod_{i,j} \mathcal{F}(V_i \cap V_j) \end{array} \tag{4}$$

Similar to proof of Lemma 2.6, get $\mathcal{F}$ is quasi-coherent.

Further if $A$ is noetherian, $\mathcal{F} = \widetilde{M}$. Can cover $X$ by finitely many $\{D(f_i)\}$ such that $\mathcal{F}(D(f_i)) = M_{f_i}$ is finite over $A_{f_i}$ Assume $M_{f_i} = \sum_j \frac{m_{ij}}{1} A_{f_i}$. Take $N = \sum_{i,j} m_{i,j} A$. Then $N$ is a finitely generated A-submodule of $M$ and for all $i$, $N_{f_i} = M_{f_i}$. Thus there is a natural morphism $\widetilde{N} \longrightarrow \widetilde{M}$ and $\widetilde{N}\big|_{D(f_i)} \xrightarrow{\sim} \widetilde{M}\big|_{D(f_i)}$ is isomorphic for all $i$. As $d(f)_i$ cover $X$, get $\widetilde{N} \longrightarrow \widetilde{M}$ is isomorphic so that $N = M$ and M is finitely generated. Thus $\mathcal{F}$ is coherent. □

**Proposition 1.7.** *Let $X$ be a scheme. Then*
*(1)Kernel, image and cokernel of morphism of quasi-coherent sheaves are still quasi-coherent.*
*(2)Direct sum of quasi-coherent sheaves is quasi-coherent.*
*(3)Tensor product of two quasi-coherent sheaves is quasi-coherent.*

**Remark 1.13.** *If $X$ is moreover noetherian, then (1) and (3) still hold for coherent and (2) is also ok if the direct sum is finite.*

**Proposition 1.8.** *Let $X$ be an affine scheme. Then for all exact sequence of quasi-coherent sheaves*

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0 \tag{5}$$

*the sequence of global sections*

$$0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{G}(X) \longrightarrow \mathcal{H}(X) \longrightarrow 0 \tag{6}$$

*is exact.*

**Proposition 1.9.** *Let $X$ be a scheme, $\mathcal{F}$ an $\mathcal{O}_X$-module. Then $\mathcal{F}$ is quasi-coherent if and only if for all $x \in X$, there exists open neighbourhood $U \ni x$ such that there is an exact sequence*

$$\mathcal{O}_U^{\oplus J} \longrightarrow \mathcal{O}_U^{\oplus I} \longrightarrow \mathcal{F}\big|_U \longrightarrow 0 \tag{7}$$

*If moreover $X$ is noetherian, then $\mathcal{F}$ is coherent if and only if for all $x \in X$, there exists open neighbourhood $U \ni x$ such that there is an exact sequence with $I, J$ finite*

$$\mathcal{O}_U^{\oplus J} \longrightarrow \mathcal{O}_U^{\oplus I} \longrightarrow \mathcal{F}\big|_U \longrightarrow 0 \tag{8}$$

*Proof.* "$\Rightarrow$": For $\mathcal{F}$ quasi-coherent and all $x \in X$, there is affine open neighbourhood $U = \operatorname{Spec} A \ni x$ such that $\mathcal{F}\big|_U \cong \widetilde{M}$ for some $A$-module $M$. Note that there is an exact sequence for some index sets $I, J$

$$A^{\oplus J} \longrightarrow A^{\oplus I} \longrightarrow M \longrightarrow 0 \tag{9}$$

By Proposition 1.6 (3), get exact sequence

$$\mathcal{O}_U^{\oplus J} \longrightarrow \mathcal{O}_U^{\oplus I} \longrightarrow \mathcal{F}\big|_U \longrightarrow 0 \tag{10}$$

"$\Leftarrow$": For all $x \in X$, there exists open neighbourhood $U \ni x$ such that there is an exact sequence

$$\mathcal{O}_U^{\oplus J} \longrightarrow \mathcal{O}_U^{\oplus I} \longrightarrow \mathcal{F}\big|_U \longrightarrow 0 \tag{11}$$

Can assume that $U$ is affine. By Proposition 1.9, get exact sequence

$$\mathcal{O}_U(U)^{\oplus J} \longrightarrow \mathcal{O}_U(U)^{\oplus I} \longrightarrow \mathcal{F}\big|_U(U) \longrightarrow 0 \tag{12}$$

By Proposition 1.6, get exact sequence

$$\mathcal{O}_U^{\oplus J} \longrightarrow \mathcal{O}_U^{\oplus I} \longrightarrow \widetilde{\mathcal{F}\big|_U(U)} \longrightarrow 0 \tag{13}$$

Thus $\mathcal{F}\big|_U \cong \widetilde{\mathcal{F}\big|_U(U)}$. By Theorem 1.1, $\mathcal{F}$ is quasi-coherent. For coherent, the same argument would make sense. □

**Example 1.3.** *(1)Let $X$ be a scheme. Then locally free $\mathcal{O}_X$-modules are quasi-coherent.*
*(2)Let $X$ be a noetherian scheme. Then locally free $\mathcal{O}_X$-modules of rank $< \infty$ are coherent.*
*(3)Let $i : Y \longrightarrow X$ be a closed immersion with $X, Y$ noetherian. Then $i_*\mathcal{O}_Y$ is coherent on $X$.*
*(4)Let $X$ be an integral noetherian scheme with function field $K$. Then constant sheaf $K_X$ is quasi-coherent but not coherent in general.*
*(5)Let $X$ be a scheme, $U \subseteq X$ open subset with inclusion $j : U \longrightarrow X$, $\mathcal{F}$ sheaf on $U$. Define $j_!\mathcal{F}$ to be the sheafification of $V \longmapsto \begin{cases} \mathcal{F}(V) & \text{if } V \subseteq U \\ 0 & \text{otherwise} \end{cases}$. Then stalk $(j_!\mathcal{F})_x = \begin{cases} \mathcal{F}_x & \text{if } x \in U \\ 0 & \text{otherwise} \end{cases}$. Now let $X = \operatorname{Spec} A$ be an affine scheme with $A$ integral domain, $U \subsetneq X$ nonempty open subset with inclusion $j : U \longrightarrow X$. Then $j_!\mathcal{O}_U$ is an $\mathcal{O}_X$-module but not quasi-coherent.*

Now assume that $S$ is a graded ring. Let $M = \oplus_{d\in\mathbb{Z}} M_d$ be a graded $C$-module. Define $\widetilde{M}$ on $X = \mathrm{Proj}(S)$ by the same process on $\mathcal{O}$ replacing $S$ by $M$. In particular, $\widetilde{M}\big|_{D_+(f)} = \widetilde{M_{(f)}}$ and $\widetilde{M}_{\mathfrak{p}} = M_{(\mathfrak{p})}$ for $\mathfrak{p} \in X$. Then $\widetilde{M}$ is quasi-coherent on $X$ and coherent if $X$ is noetherian and $M$ is finitely generated. In addition, we should note that $\widetilde{M}$ doesn't determine $M$. For instance, if $M = \oplus_{d\geq 0} M_d$ and $N = \oplus_{d\geq d_0} M_d$ for some $d_0 > 0$, then $\widetilde{M} = \widetilde{N}$.

**Definition 1.22** (***Twisting***)**.** *Let $S$ be a graded ring and $X = \mathrm{Proj}(S)$, $\mathcal{F}$ an $\mathcal{O}_X$-module. Set $\mathcal{O}_X(n) = \widetilde{S(n)}$ where $S(n)$ is the graded $C$-module given by $S(n)_d = S_(n + d)$. Also set $\mathcal{F}(n) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$.*

**Proposition 1.10.** *Let $S$ be a graded ring and $X = \mathrm{Proj}(S)$. Assume that $S$ is generated by $S_1$ as $S_0$-algebra. Then $\mathcal{O}_X$ is an invertible sheaf. In addition, For all graded $S$-module $M$, we have $\widetilde{M(n)} = \widetilde{M}(n)$. In particular, $\mathcal{O}_X(n) \otimes_{\mathcal{O}_X} \mathcal{O}_X(m) \cong \mathcal{O}_X(m + n)$.*

*Proof.* Since $S$ is generated by $S_1$ as $S_0$-algebra, $\{D_+(f)\}_{f\in S_1}$ cover $X$. Suffice to show for all $f \in S_1$, $S(n)_{(f)}$ is free of rank 1 over $S_{(f)}$. Note that there is a canonical isomorphism $S(n)_{(f)} \longrightarrow S_{(f)} \quad s \longmapsto f^{-n}s$, done! □

**Lemma 1.2** (***qcqs Lemma***)**.** *Let $X$ be a scheme, $\mathcal{F}$ quasi-coherent sheaf on $X$, $\mathcal{L}$ invertible sheaf on $X$, $s \in \mathcal{L}(X)$. Assume that $X$ is either noetherian or quasi-compact and separated over* $\mathrm{Spec}(\mathbb{Z})$*. Then*
*(1)Let $f \in \mathcal{F}(X)$. If $f\big|_{X_s} = 0$, then there exists $n > 0$ such that $f \otimes s^n = 0$ in $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)(X)$, where $X_s = \{x \in X \big| s_x\mathcal{O}_{X,x} = \mathcal{L}_x\}$ is open and $\mathcal{L}^n$ denotes $\mathcal{L}^{\otimes n}$.*
*(2)Let $g \in \mathcal{F}(X_s)$. Then there exists $n_0 > 0$ such that for all $n \geq n_0$, $g \otimes (s^n\big|_{X_s})$ extends to a section in $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)(X)$.*

*Proof.* Cover $X$ by finitely many affine open subsets $\{U_i\}$ such that $\mathcal{L}\big|_{U_i}$ is free and generated by $e_i \in \mathcal{L}(U_i)$. Then $s_i = s\big|_{U_i} = h_ie_i$ for some $h_i \in \mathcal{O}_X(U_i)$. Get $X_s \cap U_i = D(h_i) \subseteq U_i$.

(1): Note that $\mathcal{F}\big|_{U_i} \cong \widetilde{\mathcal{F}(U_i)}$. Since $f\big|_{D(h_i)}$, there exists $n > 0$ such that for all i, $f_i = f\big|_{U_i}$ satisfies that $h_i^n f_i = 0$ in $\mathcal{F}(U_i)$. Thus $(f \otimes s)\big|_{U_i} = f_i \otimes s_i^n = h_i^n f_i \otimes e_i^n = 0$. As $U_i$ cover $X$, get $f \otimes s^n = 0$.

(2): Set $g_i = g\big|_{D(h_i)}$, Then there exists $m > 0$ such that for all $i$, there exists $g_i' \in \mathcal{F}(U_i)$ such that $h_i^m g_i = g_i'$. Set $t_i = g_i' \otimes e_i^m \in (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^m)(U_i)$. Note that $t_i$ and $t_j$ coincide on $X_s \cap U_i \cap U_j$. Thus $(t_i\big|_{U_i\cap U_j} - t_j\big|_{U_i\cap U_j})\big|_{X_s\cap U_i\cap U_j}$. As $U_i \cap U_j$ is either noetherian or affine, by (1), there exists $p > 0$ such that $(t_i\big|_{U_i\cap U_j} - t_j\big|_{U_i\cap U_j}) \otimes s\big|^p_{U_i\cap U_j} = 0$ in $((\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^m)\big|_{U_i\cap U_j} \otimes_{\mathcal{O}_{U_i\cap U_j}} (\mathcal{L}\big|_{U_i\cap U_j})^p)(U_i \cap U_j)$. While $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^m)\big|_{U_i\cap U_j} \otimes_{\mathcal{O}_{U_i\cap U_j}} (\mathcal{L}\big|_{U_i\cap U_j})^p \cong (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{m+p})\big|_{U_i\cap U_j}$, get $t_i \otimes s_i^p$ and $t_j \otimes s_j^p$ coincide on $U_i \cap U_j$. Thus we can glue them to obtain $t \in (\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{m+p})(X)$ such that $t\big|_{X_s} = g \otimes (s\big|^{m+s}_{X_s})$. Take $n_0 = m + p$, done! □

**Definition 1.23.** *Let $X$ be a scheme, $\mathcal{F}$ an $\mathcal{O}_X$-module. We say that $\mathcal{F}$ is generated by global sections (or say globally generated) if there exists a family of global sections $\{s_i\}$ such that for all $x \in X$, $\{(s_i)_x\}$ generate $\mathcal{F}_x$.*

**Remark 1.14.** *Note that $\mathcal{F}$ is generated by global sections if and only if $\mathcal{F}$ is a quotient of a free $\mathcal{O}_X$-module.*

**Example 1.4.** *(1)Let $X = \mathbb{P}^n_k$ be projective $k$-scheme. Then $\mathcal{O}_X(-1)$ is not generated by global sections since there is no global sections at all.*
*(2)Let $X = \operatorname{Spec} A$ be an affine scheme. Then all quasi-coherent sheaves on $X$ are generated by global sections.*

**Definition 1.24.** *Let $A$ be a ring and $X$ be a projective scheme over $A$ with commutative diagram*

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[dr] & \mathbb{P}^d_A \arrow[d] \\ & \operatorname{Spec} A \end{tikzcd} \tag{14}$$

*Set $\mathcal{O}_X(n) = i^*\mathcal{O}_{\mathbb{P}^d_A}(n)$. For $\mathcal{F}$ an $\mathcal{O}_X$-module, also set $\mathcal{F}(m) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$.*

**Lemma 1.3** (***Projection Formula for Affine Case***)**.** *Let $f : X \longrightarrow Y$ be an affine morphism, $\mathcal{F}$ a quasi-coherent sheaf on $X$, $\mathcal{G}$ a quasi-coherent sheaf on $Y$. Then $f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}) \cong f_*\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$.*

*Proof.* Can assume that $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec} A$, $\mathcal{F} = \widetilde{N}$ and $\widetilde{M}$. Then $f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}) \cong \widetilde{P}$, where $P = N \otimes_B (M \otimes_A B)$ as $A$-module. On the other hand, $f_*\mathcal{F} \otimes \mathcal{G} \cong \widetilde{Q}$, where $Q = N \otimes_A M$ as $A$-module. As $P \cong Q$, done! $\square$

**Theorem 1.2** (***Serre's Theorem***)**.** *Let $A$ be a noetherian ring and $X$ be a projective scheme over $A$ with commutative diagram,*

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[dr] & \mathbb{P}^d_A \arrow[d] \\ & \operatorname{Spec} A \end{tikzcd} \tag{15}$$

*Assume that $\mathcal{F}$ is a coherent sheaf on $X$. Then there exists $n_0 > 0$ such that for all $n \geq n_0$, $\mathcal{F}(n)$ is generated by finitely many global sections.*

*Proof.* Since $i$ is a closed immersion, $i$ is finite so that $i_*\mathcal{F}$ is coherent. By Lemma 1.3, $i_*(\mathcal{F}(n)) \cong (i_*\mathcal{F})(n)$. Thus global sections of $i_*(\mathcal{F}(n))$ are equal to global sections of $(i_*\mathcal{F})(n)$ and stalks of $i_*(\mathcal{F}(n))$ are equal to stalks of $(i_*\mathcal{F})(n)$. It suffices to prove for $X = \mathbb{P}^d_A = \operatorname{Proj}(A[x_0, \cdots, x_d])$.

Cover $X$ by $\{U_i = D_+(x_i)\}$. Each $\mathcal{F}\big|_{U_i}$ is generated by finitely many sections $g_{ij} \in \mathcal{F}(U_i)$. By Lemma 1.2, there exists $n_0 > 0$ such that $g_{ij} \otimes x_i^n$ extends to a global section for all $i, j$. Thus $\mathcal{F}(n)$ is generated by these extensions. $\square$

**Corollary 1.1.** *Let $A$ be a noetherian ring and $X$ be a projective scheme over $A$ with commutative diagram,*

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[dr] & \mathbb{P}^d_A \arrow[d] \\ & \operatorname{Spec} A \end{tikzcd} \tag{16}$$

*Assume that $\mathcal{F}$ is a coherent sheaf on $X$. Then there exists $m \in \mathbb{Z}$ and $r < \infty$ such that $\mathcal{F}$ is a quotient of $\mathcal{O}_X(m)^{\oplus r}$.*

**Reason 1.2.** *Since $\mathcal{F}(n)$ is generated bv global sections for large enough $n$, $\mathcal{F}(n)$ is quotient of $\mathcal{O}_X^{\oplus I}$ with finite index set. Tensor by $\mathcal{O}_X(-n)$, done!*

## 1.4 Vector bundles

Let $X$ be an algebraic variety which we mean a separated scheme of finite type over $\mathbb{C}$. And all morphisms mentioned are morphisms of varieties.

**Definition 1.25.** *A complex linear fibration over $X$ is a surjection $p : E \twoheadrightarrow X$ from variety $E$ such that for all $x \in X$, the fiber $E_x$ is an affine space. In convention, we call $E$ the total space and $X$ the base variety.*

**Example 1.5.** *We call $X \times \mathbb{A}^r_{\mathbb{C}} \to X$ the trivial fibration of rank $r$.*

**Lemma 1.4.** *Given a linear fibration $p : E \to X$, for any open subset $U \subseteq X$, the restriction $p^{-1}(U) \to U$ is also a linear fibration. Denote $E\big|_U = p^{-1}(U)$.*

**Definition 1.26.** *Given two linear fibrations $E \xrightarrow{p} X$ and $F \xrightarrow{q} X$, a morphism $f : E \to F$ is said to be a morphism of linear fibrations if the following diagram commutes*

$$\begin{tikzcd} E \arrow[rr, "f"] \arrow[dr, "p"'] & & F \arrow[dl, "q"] \\ & X & \end{tikzcd} \tag{17}$$

*and for all $x \in X$, $E_x \xrightarrow{f_x} F_x$ is a linear transformation.*

**Definition 1.27.** *An algebraic vector bundle of rank $r$ on $X$ is a linear fibration $E \xrightarrow{p} X$ which is locally trivial i.e. for all $x \in X$, there is an open neighbourhood $U$ of $x$ and isomorphism of linear fibrations $\varphi_U$ such that the following diagram commutes*

$$\begin{tikzcd} E\big|_U \arrow[rr, "\varphi_U"] \arrow[dr, "p"'] & & U \times \mathbb{A}^r_{\mathbb{C}} \arrow[dl] \\ & U & \end{tikzcd} \tag{18}$$

*where $\varphi_U$ are called the local charts or local trivialization.*

*And for two such $U_i$ and $U_j$ with local charts $\varphi_i$ and $\varphi_j$, there is an invertible matrix $g_{ij}$ with entries regular functions on $U_{ij}$, called transition function, satisfying that $g_{ii} = I_r$, $g_{ij} \cdot g_{jk} = g_{ik}$ and for all affine open subset $W \subseteq U_{ij}$, $\varphi_i \circ \varphi_j^{-1}\big|_W$ is linear transformation defined by $g_{ij}\big|_W$. Equivalently, $g_{ij}$ would correspond to a morphism $\widetilde{g}_{ij} : U_{ij} \to \mathrm{GL}_r(\mathcal{O}_X(U_{ij}))$.*

**Lemma 1.5.** *Given data $\{g_{ij}\}$ satisfies that $g_{ii} = I_r$ and $g_{ij} \cdot g_{jk} = g_{ik}$, we can recover a vector bundle.*

*Proof.* Construct $E$ by gluing up $U_i \times \mathbb{A}^r_{\mathbb{C}}$ along $U_{ij} \times \mathbb{A}^r_{\mathbb{C}} \xrightarrow{\varphi_{ij}} U_{ij} \times \mathbb{A}^r_{\mathbb{C}}$ defined by $g_{ij}$. □

**Definition 1.28.** *Let $E, F$ be two vector bundles on $X$ of rank $r$ and $s$ respectively. We say a morphism $f : E \to F$ of linear fibrations is a morphism of vector bundles if there is a $r \times s$*

*matrix $g_U$ with entries regular functions on $U$ such that for all affine open subset $W \subseteq U$, restriction of the following morphism to $W$ is linear transformation defined by $g_U\big|_W$*

$$U \times \mathbb{A}^r_{\mathbb{C}} \xrightarrow{\varphi_U^{-1}} E\big|_U \xrightarrow{f|_U} F\big|_U \xrightarrow{\psi_U} U \times \mathbb{A}^s_{\mathbb{C}} \tag{19}$$

*where $\varphi_U$ and $\psi_U$ are local charts. Equivalently, $g_U$ would correspond to a morphism $\widetilde{g}_U : U \to \mathrm{GL}_r(\mathcal{O}_X(U))$.*

**Remark 1.15.** *With Lemma 1.5, we can easily define $E \oplus F$, $E \otimes F$, $\mathcal{H}om(E, F)$, $E^\vee$, $\wedge^k E$ and $S^k E$.*

**Proposition 1.11.** *Let $f : E \to F$ be a morphism of vector bundles. Assume that for $x_0 \in X$, $f_{x_0} : E_{x_0} \to F_{x_0}$ is invertible. Then there exists an open neighbourhood $U_0$ of $x_0$ such that $f\big|_U : E\big|_U \to F\big|_U$ is an isomorphism.*

*Proof.* Assume $x_0 \in U$ with local charts $\varphi_U : E\big|_U \to U \times \mathbb{A}^r_{\mathbb{C}}$ and $\psi_U : F\big|_U \to \mathbb{A}^s_{\mathbb{C}}$. By definition, there is a matrix $g_U$ and $f_{x_0}$ is defined by $g_U(x_0)$. This immediately comes from the fact that there exists an open neighbourhood $U_0 \subseteq U$ of $x_0$ such that $g_U\big|_{U_0}$ is invertible. □

**Definition 1.29.** *Let $E$ be a vector bundle of rank $r$ on $X$. A subbundle of rank $m$ of $E$ is a subvariety $F \subseteq E$ such that fiber $F_x$ is an $m$-dimensional affine linear subspace of $E_x$ and the induced fibration $p\big|_F : F \to X$ is locally trivial.*

**Example 1.6.** *Let $X = \mathbb{P}^n_{\mathbb{C}}$ be $n$-dimensional projective space with homogeneous coordinate $x_0, \cdots, x_n$. Consider the subvariety $\Sigma := V(x_i y_j - x_j y_i)$ of $X \times \mathbb{A}^{n+1}_{\mathbb{C}}$, where $y_0, \cdots, y_n$ are coordinate of $\mathbb{A}^{n+1}_{\mathbb{C}}$. Then the fiber is a $1$-dimensional affine space and hence $\Sigma$ is a subbundle of rank $1$ of the trivial bundle, denoted by $\mathcal{O}(-1)$.*

**Proposition 1.12.** *Let $F \subseteq E$ be a subbundle of rank $m$. Then there exists unique vector bundle structure on $E/F$ satisfying the universal property that any map of vector bundles $E \to G$ vanishing over $F$ would factor through $E/F$ uniquely.*

*Proof.* Firstly cover $X$ with affine open subsets trivializing $F$ and $E$, we may assume that $X = \operatorname{Spec} A$, $E = \mathbb{A}^n_A$ and $F = \mathbb{A}^m_A$. Since $F \subseteq E$ is a subbundle, by definition, for all $x \in X$, $E_x = \mathbb{A}^n_{k(x)}$ and $F_x = \mathbb{A}^m_{k(x)}$ with coordinates $x_1, \cdots, x_n$ and $y_1, \cdots, y_m$ respectively, satisfying that $y_i$ can be linearly represented by $x_i$.

It is clear that these relations still hold over a neighbourhood $U = D(f)$ of $x$. Then we extend $y_i$ to a basis $y_1, \cdots, y_m, z_1, \cdots, z_{n-m}$ and hence $\operatorname{Spec} A_f[x_1, \cdots, x_n]/(y_1, \cdots, y_m) \cong \operatorname{Spec} A_f[z_1, \cdots, z_{n-m}]$ is a linear subspace. For different $D(f)$ and $D(g)$, since $z_1, \cdots, z_{n-m}$ and $z'_1, \cdots, z'_{n-m}$ span same linear subspace, there is an invertible transformation matrix $M \in \mathrm{Mat}_{n-m}(A_{fg})$ i.e. the transition function. Thus we defines a vector bundle structure on $E/F$.

Now given a map of vector bundles $E \to G$ vanishing over $F$. Still assume that $X = \operatorname{Spec} A$ trivializing all those bundles. Then $E \to G$ corresponds to a linear transformation $\varphi : A[w_1, \cdots, w_r] \to A[x_1, \cdots, x_n]$, which is defined by an $n \times r$ matrix with entries in $A$.

By assumption, we know $y_i$ are not in the image of $\varphi$ and hence $\operatorname{im}\varphi$ lies in the subspace spanned by $z_i$. Thus $\varphi$ factors a canonical map $A[w_1, \cdots, w_r] \to A[z_1, \cdots, z_{n-m}]$. Finally, by universal property, we know the vector bundle structure on $E/F$ is also unique. $\square$

**Proposition 1.13** (***Rigidity Property***)**.** *Let $f : E \to F$ be a map of vector bundles. Suppose that rank of fiber morphisms $f_x$ is constant as $x$ varies over $X$. Then $\ker f$ and $\operatorname{im} f$ are subbundles of $E$ and $F$ respectively.*

*Proof.* Here we only show $\ker f$ is a subbundle of $E$ and similar argument makes sense for $\operatorname{im} f$. We may assume that $X = \operatorname{Spec} A$ trivializing both $E$ and $F$. Now $f$ corresponds to a linear transformation $\varphi : A[y_1, \cdots, y_m] \to A[x_1, \cdots, x_n]$ defined by an $n \times m$ matrix $M$ with entries in $A$.

Since the rank of fiber morphism is constant, we know both the dimensions of $\operatorname{im}\varphi_x$ are constant, denoted by $n - r$. Then we may choose $y_1, \cdots, y_r$ spanning the complementary subspace of $\operatorname{im}\varphi_x$, which can be linearly represented by $x_i$. And these relation still hold over an neighbourhood $U = D(f)$ of $x$. For different $D(f)$ and $D(g)$, since both $y_1, \cdots, y_r$ and $y'_1, \cdots, y'_r$ span complementary subspace of $\operatorname{im}\varphi_{fg}$, there is an invertible transformation matrix $M \in \operatorname{Mat}_r(A_{fg})$ i.e. the transition function. Thus we give a subbundle structure on $\ker f$. $\square$

**Remark 1.16.** *Though proof for $\operatorname{im} f$ follows by similar argument, there is still a little tricky thing in that case. When we talk about $\ker f_x$, it is clear the points living outside $\operatorname{im}\varphi_x$. But when we talk about $\operatorname{im} f_x$, what are they? As $X$ is locally noetherian, by a proposition in Some Useful Results, we know they are just the points living in coimage of $\varphi_x$.*

*In addition, since this note is translated by me from language of varieties to language of schemes, there must be some ambiguity about the terminology. For example, here by $\ker f$, we possibly mean the subset of points whoes image under $f$ are same to image of the origin.*

**Corollary 1.2.** *Let $f : E \to F$ be a map of vector bundles. When $f$ is surjective, $\ker f$ is a subbundle of $E$. And when $f$ is injective, $\operatorname{im} f$ is a subbundle of $F$.*

**Reason 1.3.** *When $f$ is surjective, as remark above, we know $\ker\varphi = 0$ so that rank of $f_x$ is constant. When $f$ is injective, as proof above, we know $\operatorname{im}\varphi = k[x_1, \cdots, x_n]$ so that rank of $f_x$ is constant.*

**Definition 1.30.** *Let $p : E \to X$ be a vector bundle on $X$ of rank $r$. A (regular) section of $E$ over open subset $U \subseteq X$ is a morphism $s : U \to E$ such that $(p \circ s)\big|_U = \mathrm{id}_U$. Denote $\Gamma(U, E) = \{$regular sections of $E$ over $U\}$.*

**Lemma 1.6.** *Let $p : E \to X$ be a vector bundle on $X$ of rank $r$. Then $\Gamma(U, E)$ naturally admits an $\mathcal{O}(U)$-module structure*

$$\mathcal{O}(U) \times \Gamma(U, E) \longrightarrow \Gamma(U, E) \tag{20}$$

*Moreover, $\mathcal{O}_X(E)(U) = \Gamma(U, E)$ gives a locally free sheaf $\mathcal{O}_X(E)$ of rank $r$.*

*Proof.* For $\alpha \in \mathcal{O}(U)$ and section $s : U \to E$, locally $s$ is defined by a homomorphism $\varphi : A[x_1, \cdots, x_r] \to A$. Define $\alpha \cdot s$ by gluing up $\alpha \cdot \varphi : x_i \mapsto \alpha \cdot \varphi(x_i)$. In particular, locally $\Gamma(U, E) \cong A^{\oplus r}$ hence $\mathcal{O}_X(E)$ is a locally free sheaf of rank $r$. $\square$

**Proposition 1.14.** *Let $\mathcal{C}$ be the category of locally free sheaves of finite rank on $X$. There is a functor $\theta : \underline{\mathbf{Vect}}_X \to \mathcal{C}$ sending $E$ to $\mathcal{O}_X(E)$ and $E \to F$ to $\{\Gamma(U, E) \to \Gamma(U, F)\}$, which is an equivalence of category.*

*Proof.* First to prove $\theta$ fully faithful i.e. $\mathrm{Hom}_{\underline{\mathbf{Vect}}_X}(E, F) \to \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X(E), \mathcal{O}_X(F))$ is bijection for all $E, F$. We may assume that $X = \operatorname{Spec} A$ and $E, F$ are both trivial. Then $f : \mathbb{A}^n_A \to \mathbb{A}^m_A$ corresponds to a matrix $M \in \mathrm{Mat}_{n\times m}(A)$. Also, $(s : U \to E) \to (f \circ s : U \to F)$ also corresponds to such a matrix. Hence the bijection is clear.

Next to prover $\theta$ dense (or say essentially surjective) i.e. given $\mathcal{E} \in \mathcal{C}$, there exists $E \in \underline{\mathbf{Vect}}_X$ with isomorphism $\mathcal{O}_X(E) \xrightarrow{\sim} \mathcal{E}$. Assume $U_i = \operatorname{Spec} A_i$ is an affine cover of $X$ trivializing $\mathcal{E}$. Denote the trivialization $\varphi_i : \mathcal{E}\big|_{U_i} \to \mathbb{A}^n_{A_i}$. Consider the transition functions $g_{ij} = \varphi_i \circ \varphi_j^{-1}\big|_{U_{ij}}$. Then thoes transition functions give a vector bundle $E$.

Define $\psi_i : \mathcal{O}_X(E)(U_i) = \Gamma(U_i, E) \to \mathcal{E}\big|_{U_i}$ sending $s : U_i \to \mathbb{A}^n_{U_i}$ to $\varphi_i^{-1}(v_s)$, where $v_s$ is the vector defining $s$. Then

$$\psi_i\big|_{U_{ij}}(s\big|_{U_{ij}}) = \varphi_i^{-1}(v_s) = \varphi_i^{-1}(g_{ij}v'_s) = \varphi_i^{-1} \circ \varphi_i \circ \varphi_j^{-1}(v'_s) = \varphi_j^{-1}(v'_s) = \psi_j\big|_{U_{ij}}(s\big|_{U_{ij}}) \quad (21)$$

Hence $\psi_i$ glue up and get our desired isomorphism. $\square$

## 1.5 Bundles associated to a representation

**Lemma 1.7.** *Let $f : Y \to X$ be a morphism, $p : E \to X$ vector bundle of rank $r$ on $X$. Consider the fibered product $F = Y \times_X E$. Then the first projection $q : F \to Y$ gives a bundle on $Y$ of rank $r$ and $\mathcal{O}_Y(F) = f^*\mathcal{O}_X(E)$.*

*Proof.* Locally we have trivializations $\varphi_i : E\big|_{U_i} \to U_i \times \mathbb{A}^r_{\mathbb{C}}$ and transition functions $g_{ij} : U_{ij} \times \mathbb{A}^r_{\mathbb{C}} \to U_{ij} \times \mathbb{A}^r_{\mathbb{C}}$ satisfying cocycle condition. Then after base change we get $\widetilde{\varphi}_i : F\big|_{V_i} \to V_i \times \mathbb{A}^r_{\mathbb{C}}$ and $\widetilde{g}_{ij} : U_{ij} \times \mathbb{A}^r_{\mathbb{C}} \to U_{ij} \times \mathbb{A}^r_{\mathbb{C}}$ still satisfying cocycle condition. Hence $q : F \to Y$ defines a bundle on $Y$ of rank $r$.

Now for all open subset $V \subseteq Y$, define $\sigma(V) : f^*\mathcal{O}_X(E)(V) \to \mathcal{O}_Y(F)(V)$ sending $(U, s) \otimes 1$ to $\widetilde{s}\big|_V$, where $\widetilde{s} : f^{-1}(U) \to F$ is the pull back of $s : U \to E$. For all $y \in Y$, want to show that $\sigma_y : \mathcal{O}_X(E)_{f(y)} \otimes_{\mathcal{O}_{X,f(y)}} \mathcal{O}_{Y,y} \to \mathcal{O}_Y(F)_y$ is isomorphism.

For injectivity, if $\sum_{i=1}^r (U, s_i) \otimes b_i \in \ker \sigma_y$, then locally we get $\sum_{i=1}^r b_i \cdot f^\sharp \circ s_i^\sharp(x_j) = 0$ for all $1 \leq j \leq r$, so that also $\sum_{i=1}^r (s_i^\sharp(x_1), \cdots, s_i^\sharp(x_r)) \otimes b_i = 1 \otimes (\sum_{i=1}^r b_i \cdot f^\sharp \circ s_i^\sharp(x_1), \cdots, \sum_{i=1}^r b_i \cdot f^\sharp \circ s_i^\sharp(x_r)) = 0$. Then $(U, s) = \oplus_{i=1}^r a_i \otimes 1 = 0$. For surjectivity, given $(b_1, \cdots, b_r) \in B^{\oplus r}$, clearly it is the image of $\sum_{i=1}^r (U, s_i) \otimes b_i$ where $s_i$ is defined by $e_i \in A^{\oplus r}$. $\square$

Let $p : E \to X$ be a vector bundle of rank $r$. To avoid ambiguity, we denote the general linear group by $G_r$ and the group variety by $\mathrm{GL}_r(\mathbb{C})$, which is the open subset $D(\det)$ of $\mathbb{A}^{r^2}_{\mathbb{C}}$.

Consider the variety $R = X \times \mathrm{GL}_r(\mathbb{C})$. We have a group action of $G_r$ on $R$ defined locally as following, for $M = (a_{ij})$

$$\mathrm{id} \times (\circ M^{-1}) : A \times \mathrm{GL}_r(\mathbb{C}) \to A \times \mathrm{GL}_r(\mathbb{C}) \tag{22}$$

where $\circ M^{-1}$ corresponds to the homomorphism defined by $(x_{ij}) \mapsto (\sum_k x_{ik} a_{kj})$. Then $R/G_r = X$ and the orbits can be identified with the fibers of $R \to X$. In addition, each matirx $M$ gives a morphism $\mathbb{A}^r_{\mathbb{C}} \to E\big|_U$ satisfying that composition with trivialization $\varphi_U$ is a linear transformation defined by $M$. Then it is natural to consider transition function $g_{ij}$ which gives a $\mathrm{GL}_r(\mathbb{C})$-bundle on $X$, denoted by $\mathrm{Fr}(E)$, the frame bundle of $E$. And there is an group action of $G_r$ on $R \times \mathbb{A}^r_{\mathbb{C}}$ defined locally as following, for $M = (a_{ij})$

$$\mathrm{id} \times (\circ M^{-1}) \times M : A \times \mathrm{GL}_r(\mathbb{C}) \times \mathbb{A}^r_{\mathbb{C}} \to A \times \mathrm{GL}_r(\mathbb{C}) \times \mathbb{A}^r_{\mathbb{C}} \tag{23}$$

where $M$ corresponds to the linear transformation defined by $M$. Consider $R \times \mathrm{GL}_r(\mathbb{C})/G_r$. It is clear that $A[x_{ij}]_{(\det)}[y_k]^{G_r} = A[\sum_{k=1}^r x_{ik} y_k]$ and hence the quotient scheme is isomorphic to $\mathbb{A}^r_{\mathbb{C}}$. With transition functions $g_{ij}$, gluing up we get original $E$.

Now given a representation $\psi : \mathrm{GL}_r(\mathbb{C}) \to \mathrm{GL}_m(\mathbb{C})$, which is in fact a morphism of group schemes, consider the group action of $G_r$ on $R \times \mathbb{A}^m_{\mathbb{C}}$ defined locally as following, for $M = (a_{ij})$

$$\mathrm{id} \times (\circ M^{-1}) \times \psi(M) : A \times \mathrm{GL}_r(\mathbb{C}) \times \mathbb{A}^m_{\mathbb{C}} \to A \times \mathrm{GL}_r(\mathbb{C}) \times \mathbb{A}^m_{\mathbb{C}} \tag{24}$$

Since the representabtion corresponds to a homomorphism $A[y_{ij}] \to A[x_{ij}]$, similarly the invariant subring is generated by $\psi(x_{ij})(y_k)$, where the $(i,j)$-entry of $\psi(x_{ij})$ is the image of $y_{ij}$ in $A[x_{ij}]$. Then locally the quotient scheme is isomorphic to $\mathbb{A}^m_A$ with natural transition functions $\psi(g_{ij})$. These information give a vector bundle $E_\psi$ of rank $m$ on $X$, called the bundle associated to representation $\psi$.

**Remark 1.17.** *The idea of frame bundle also comes from moduli thery. Consider functor $\mathcal{F} : \mathbf{Sch}_X \to \mathbf{Set}$ defined as following*

$$(f : T \to X) \longmapsto \{f^*\mathcal{O}(E) \xrightarrow{\sim} \mathcal{O}_T^r\} \tag{25}$$

*And in fact this functor is representable by frame bundle above.*

# 2 Sheaf Cohomology

## 2.1 Cohomology group

There are lots of abelian categories in Algebraic Geometry.

**Example 2.1.** *(1)$\mathbf{Ab}$ is the category of abelian groups.*
*(2)$\mathbf{Mod}_A$ is the category of $A$-modules.*
*(3)$\mathbf{Ab}(X)$ is the category of sheaves of abelian groups on a topological space $X$.*
*(4)$\mathbf{Mod}(\mathcal{O}_X)$ is the category of $\mathcal{O}_X$-modules on a ringed space $(X, \mathcal{O}_X)$.*
*(5)$\mathbf{Qcoh}(X)$ is the category of quasi-coherent sheaves on a scheme $X$.*
*(6)$\mathbf{Coh}(X)$ is the category of coherent sheaves on a noetherian scheme $X$.*

There are some special functors in Algebraic Geometry.

**Example 2.2.** *(1)*$\mathrm{Hom}_{\mathcal{C}}(A, \cdot) : \mathcal{C} \longrightarrow \mathbf{Ab}$ *is left exact covariant functor.*
*(2)*$\mathrm{Hom}_{\mathcal{C}}(\cdot, A) : \mathcal{C} \longrightarrow \mathbf{Ab}$ *is left exact contravariant functor.*
*(3)*$\Gamma(X, \cdot) : \mathbf{Ab}(X) \longrightarrow \mathbf{Ab}$ *is left exact covariant functor.*
*(4)*$M \otimes_A \cdot : \mathbf{Mod}_A \longrightarrow \mathbf{Mod}_A$ *is right exact covariant functor. In particular,* $M \otimes_A \cdot$ *is exact if and only if* $M$ *is flat* $A$*-module.*

**Definition 2.1.** *Let* $\mathcal{C}$ *be a category with enough injective objects,* $\mathcal{F}$ *a left exact covariant functor,* $J \in \mathcal{C}$*. We say that* $J$ *is acyclic for the functor* $\mathcal{F}$ *(or say* $\mathcal{F}$*-acyclic) if* $R^i\mathcal{F}(J) = 0$ *for all* $i > 0$*.*

**Proposition 2.1.** *Let* $\mathcal{C}$ *be a category with enough injective objects,* $\mathcal{F}$ *a left exact covariant functor,* $A \in \mathcal{C}$*. Assume that there is an* $\mathcal{F}$*-acyclic resolution* $J^*$ *of* $A$ *i.e.* $0 \longrightarrow A \longrightarrow J_0 \longrightarrow J_1 \longrightarrow \cdots$ *exact with each* $J_i$ $\mathcal{F}$*-acyclic. Then* $R^i\mathcal{F}(A) \cong H^i(\mathcal{F}(J^*))$*.*

**Proposition 2.2.** *Let* $(X, \mathcal{O}_X)$ *be a ringed space. Then* $\mathbf{Mod}(\mathcal{O}_X)$ *has enough injective objects. In particular,* $\mathbf{Ab}(X)$ *has enough injective objects if viewing* $X$ *as* $(X, \mathbb{Z}_X)$ *ringed space, where* $\mathbb{Z}_X$ *is the constant sheaf.*

*Proof.* Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. For all $x \in X$, can embed $\mathcal{F}_x$ in an injective $\mathcal{O}_{X,x}$-module $I_x$ since $\mathbf{Mod}_A$ has enough injective objects for all ring $A$. View $I_x$ as a sheaf on $\{x\}$. Set $\mathcal{I} = \prod_{x\in X} j_*I_X$ where $j : \{x\} \longrightarrow X$ is inclusion. For all $\mathcal{G}$ $\mathcal{O}_X$-module, we have $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, \mathcal{I}) = \prod_{x\in X} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, j_*I_x) = \prod_{x\in X} \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{G}_x, I_x)$. With $\mathcal{F}_x \hookrightarrow I_x$, get a monomorphism $\mathcal{F} \longrightarrow \mathcal{I}$ with $\mathcal{I}$ injective. $\square$

**Definition 2.2** (***Sheaf Cohomology***)**.** *Let* $X$ *be a topological space,* $\Gamma(X, \cdot) : \mathbf{Ab}(X) \longrightarrow \mathbf{Ab}$*. Define the cohomology functor* $H^i(X, \cdot) = R^i\Gamma(X, \cdot)$*. For* $\mathcal{F} \in \mathbf{Ab}(X)$*, we call* $H^i(X, \mathcal{F})$ *the ith cohomology group of* $\mathcal{F}$*.*

**Definition 2.3.** *Let* $X$ *be a topological space,* $\mathcal{F} \in \mathbf{Ab}(X)$*. We say that* $\mathcal{F}$ *is flasque if for all* $V \subseteq U$ *inclusion of open subsets, the restriction map* $\mathcal{F}(U) \longrightarrow \mathcal{F}(V)$ *is surjective.*

**Proposition 2.3.** *(1)Direct products of flasque sheaves are flasque.*
*(2)Direct image of flasque sheaves are flasque.*
*(3)quotient of flasque sheaves by flasque sheaves are flasque.*
*(4)If* $0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0$ *is exact with* $\mathcal{F}$ *flasque, then* $0 \longrightarrow \mathcal{F}(U) \longrightarrow \mathcal{G}(U) \longrightarrow \mathcal{H}(U) \longrightarrow 0$ *is exact for all open subset* $U \subseteq X$*.*

**Proposition 2.4.** *Let* $(X, \mathcal{O}_X)$ *be a ringed space. Then every injective* $\mathcal{O}_X$*-module is flasque.*

*Proof.* Consider $U \subseteq X$ is an inclusion of open subsets. Set $\mathcal{G}_U = j_!\mathcal{O}_U$. Let $\mathcal{F}$ be an injective $\mathcal{O}_X$-module, $V \subseteq U$ open subset. Then the natural morphism $\mathcal{G}_V \longrightarrow \mathcal{G}_U$ is monic so that $0 \longrightarrow \mathcal{G}_V \longrightarrow \mathcal{G}_U$ is exact. Since $\mathcal{F}$ is injective, get $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}_U, \mathcal{F}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}_V, \mathcal{F}) \longrightarrow 0$ exact. Note that $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}_U, \mathcal{F})\big|_U = \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{O}_U, \mathcal{F}\big|_U) = \mathcal{F}(U)$. Thus $\mathcal{F}(U) \longrightarrow \mathcal{F}(V) \longrightarrow 0$ is exact so that $\mathcal{F}$ is flasque. $\square$

**Corollary 2.1.** *Let $X$ be a topological space. Then every injective object in $\mathbf{Ab}(X)$ is flasque by viewing $X$ as ringed space $(X, \mathbb{Z}_X)$ where $\mathbb{Z}_X$ is the constant sheaf.*

**Proposition 2.5.** *Let $X$ be a topological space. Then every flasque sheaf on $X$ is acyclic for the functor $\Gamma(X, \cdot)$.*

*Proof.* Let $\mathcal{F}$ be a flasque sheaf on $X$. Consider exact sequence $0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I} \longrightarrow \mathcal{G} \longrightarrow 0$ where $\mathcal{I}$ is an injective object in $\mathbf{Ab}(X)$. By Corollary 2.1, $\mathcal{I}$ is flasque. Thus $\mathcal{G}$ is flasque since it is quotient of flasque sheaf by flasque sheaf. Take the long exact sequence $0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{I}(X) \longrightarrow \mathcal{G}(X) \longrightarrow H^1(X, \mathcal{F}) \longrightarrow H^1(X, \mathcal{I}) \longrightarrow \cdots$. While $\mathcal{F}$ is flasque, we have $0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{I}(X) \longrightarrow \mathcal{G}(X) \longrightarrow 0$ is exact. Also, as $\mathcal{I}$ is injective, $H^i(X, \mathcal{I}) = 0$ for all $i > 0$. Thus $H^i(X, \mathcal{F}) = 0$ for all $i > 0$ so that $\mathcal{F}$ is acyclic for the functor $\Gamma(X, \cdot)$. $\square$

**Corollary 2.2.** *Let $(X, \mathcal{O}_X)$ be a ringed space. Then the right derived functor of $\Gamma(X, \cdot) : \mathbf{Mod}(\mathcal{O}_X) \longrightarrow \mathbf{Ab}$ coincide with $H^i(X, \cdot)$.*

**Reason 2.1.** *Take injective resolution in $\mathbf{Mod}(\mathcal{O}_X)$. By Proposition 2.4 and Proposition 2.5, it is a $\Gamma(X, \cdot)$-acyclic resolution. Thus by Proposition 2.1, the two right derived functors coincide.*

**Corollary 2.3.** *Let $X$ be a topological space, $Y \subseteq X$ closed subset, $i : Y \longrightarrow X$ inclusion. Then for all $\mathcal{F} \in Ab(Y)$, we have $H^i(Y, \mathcal{F}) \cong H^i(X, j_*\mathcal{F})$.*

**Reason 2.2.** *Take injective resolution $J^*$ of $\mathcal{F}$ in $\mathbf{Ab}(Y)$. Then $f_*J^*$ is a flasque resolution of $f_*\mathcal{F}$.*

**Remark 2.1** (***Gabber Theorem***)**.** *Let $X$ be a scheme. Then $\mathbf{Qcoh}(X)$ has enough injective objects.*

**Remark 2.2.** *Let $X$ be a scheme. Suppose that $X$ is either noetherian or quasi-compact and separated. Then the right derived functor of $\Gamma(X, \cdot) : \mathbf{Qcoh}(X) \longrightarrow \mathbf{Ab}$ coincide with $H^i(X), \cdot$. In fact, this property is equivalent to each injective object is $\Gamma(X, \cdot)$-acyclic.*

**Theorem 2.1** (***Grothendieck Vanishing***)**.** *Let $X$ be a noetherian topological space of dimension $n < \infty$. Then for all $\mathcal{F} \in \mathbf{Ab}(X)$, we have that $H^i(X, \mathcal{F}) = 0$ for all $i > n$.*

## 2.2 Cohomology of noetherian affine schemes

**Theorem 2.2** (***Artin-Rees***)**.** *Let $A$ be a noetherian ring, $M \subseteq N$ finitely generated $A$-modules, $\mathfrak{a} \subseteq A$ ideal. Then for all $n > 0$, there exists $m > n$ such that $\mathfrak{a}M \supseteq M \cap a^m N$.*

**Remark 2.3.** *This theorem says that the $\mathfrak{a}$-adic topology of $M$ is compatible with the induced topology of $\mathfrak{a}$-adic topology of $N$ on $M$.*

**Lemma 2.1.** *Let $A$ be a noetherian ring, $I$ an injective $A$-module, $\mathfrak{a} \subseteq A$ ideal. Assume $J \subseteq I$ is the $A$-submodule consist of $x \in I$ such that $\mathfrak{a}^n x = 0$ for some $n > 0$. Then $J$ is injective.*

*Proof.* It suffices to show that for all $\mathfrak{b} \subseteq A$ ideal and $\varphi : \mathfrak{b} \longrightarrow J$ homomorphism of $A$-modules, there exists $\psi : A \longrightarrow J$ extending $\varphi$ satisfies the following diagram commutes

$$\begin{tikzcd} \mathfrak{b} \arrow[r, hook] \arrow[d, "\varphi"] & A \arrow[dl, "\psi"] \\ J & \end{tikzcd} \tag{26}$$

Since $A$ is noetherian, $B$ is finitely generated. Thus there exists $n > 0$ such that $\mathfrak{a}^n\varphi(\mathfrak{b}) = 0$. By Artin-Rees, there exists $m > n$ such that $\mathfrak{a}^n\mathfrak{b} \supseteq \mathfrak{b} \cap \mathfrak{a}^m$. Thus $\varphi(\mathfrak{b} \cap \mathfrak{a}^m) = 0$ so that $\varphi$ factors through $\mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^m)$. Consider $\varphi_I : \mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^m) \longrightarrow J \hookrightarrow I$. As $I$ is injective, $\varphi_I$ extends to $\psi_I : A/\mathfrak{a}^m \longrightarrow I$. Note that $im(\psi_I) \subseteq J$ since $\mathfrak{a}^m im(\psi I) = 0$, get $\psi$. □

**Lemma 2.2.** *Let $A$ be a noetherian ring, $I$ an injective $A$-module. Then for all $f \in A$, the localization $\theta : I \longrightarrow I_f$ is surjective.*

*Proof.* For all $i > 0$, set $\mathfrak{b}_i = \mathrm{ann}_A(f^i)$. Get an increasing sequence of ideals $\mathfrak{b}_1 \subseteq \mathfrak{b}_2 \subseteq \cdots$. As $A$ is noetherian, $\{\mathfrak{b}_i\}$ stabilize from $\mathfrak{b}_r$ for some $r < \infty$. For $x \in I_f$, we can write $x = \frac{\theta(y)}{f^n}$ where $y \in I$ and $n > 0$. There is a canonical homomorphism $\varphi : (f^{n+r}) \longrightarrow I \quad f^{n+r} \longmapsto f^r y$. Easy to check that $\varphi$ is well defined. Since $I$ is injective, $\varphi$ extends to $\psi : A \longrightarrow I$. Set $z = \psi(1)$, then $f^{n+r}z = f^r y$. Thus $\theta(z) = \frac{z}{1} = \frac{f^{n+r}z}{f^{n+r}} = \frac{f^r y}{f^{n+r}} = \frac{\theta(y)}{f^n} = x$ so that $\theta$ is surjective. □

**Proposition 2.6.** *Let $A$ be a noetherian ring, $I$ an injective $A$-module. Then $\widetilde{I}$ is flasque on $X = \operatorname{Spec} A$.*

*Proof.* Set $Y = Supp(\widetilde{I}) = \{x \in X \big| \widetilde{I}_x \neq 0\} \subseteq X$ closed. If $Y = \varnothing$, then $\widetilde{I}$ is flasque. If $Y \neq \varnothing$, since $A$ is noetherian, we can assume for all injective $A$-module $J$ with support $\mathrm{Supp}(\widetilde{J}) \subsetneq Y$, $\widetilde{J}$ is flasque. Want to show $\widetilde{I}(X) \longrightarrow \widetilde{I}(U)$ is surjective for all open subset $U \subseteq X$.

If $Y \cap U = \varnothing$, then $\widetilde{I}(U) = 0$, done! If $Y \cap U \neq \varnothing$, take standard open subset $D(f) \subseteq U$ and $D(f) \cap Y \neq \varnothing$. Set $Z = X \setminus D(f)$. Let $s \in \widetilde{I}(U)$, then $s\big|_{D(f)} \subseteq \widetilde{I}(D(f)) = I_f$. By Lemma 2.2, we can lift $s\big|_{D(f)}$ to $t \in \widetilde{I}(X) = I$. Thus $(s - t\big|_U)\big|_{D(f)} = 0$ so that $(s - t\big|_U) \in \Gamma_{Z \cap U}(U, \widetilde{I}\big|_U)$ i.e. support of it is in $Z$. Remains to show $\Gamma_Z(X, \widetilde{I}) \longrightarrow \Gamma_{Z \cap U}(U, \widetilde{I}\big|_U)$.

Note that $J = \Gamma_Z(X, \widetilde{I})$ is exactly the submodule consist of $x \in I$ such that $f^n x = 0$ for some $n > 0$. By Lemma 2.1, $J$ is injective $A$-module with $Supp(\widetilde{J}) \subsetneq Y$. By induction hypothesis, $\widetilde{J}$ is flasque. Thus $\widetilde{J}(X) \longrightarrow \widetilde{J}(U)$ is surjective. While $\widetilde{J}(X) = \Gamma_Z(X, \widetilde{I})$ and $\widetilde{J}(U) = \Gamma_{Z \cap U}(U, \widetilde{I}\big|_U)$, done! □

**Theorem 2.3** (***Serre Vanishing***)**.** *Let $A$ be a noetherian ring and $X = \operatorname{Spec} A$. Then for all $\mathcal{F}$ quasi-coherent sheaf in $X$, we have that $H^i(X, \mathcal{F}) = 0$ for all $i > 0$.*

*Proof.* For $\mathcal{F}$ quasi-coherent, set $M = \mathcal{F}(X)$. Take injective resolution $I^*$ of $M$ in $Mod_A$. Thus $0 \longrightarrow \mathcal{F} \longrightarrow \widetilde{I^0} \longrightarrow \cdots$ is exact, which is a $\Gamma(X, \cdot)$-acyclic resolution by Proposition 2.6. As $\widetilde{I^i}$ is $\Gamma(X, \cdot)$-acyclic, can apply $\Gamma(X, \cdot)$ to calculate $H^i(X, \mathcal{F})$. But $\Gamma(X, \cdot)$ gives back to $0 \longrightarrow M \longrightarrow I^0 \longrightarrow \cdots$ exact, get $H^i(X, \mathcal{F}) = 0$ for all $i > 0$. □

**Corollary 2.4.** *Let $X$ be a noetherian affine scheme. Then for all exact sequence $0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0$ of $\mathcal{O}_X$-module with $\mathcal{F}$ quasi-coherent, we have that $0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{G}(X) \longrightarrow \mathcal{H}(X) \longrightarrow 0$ is exact.*

**Corollary 2.5.** *Let $X$ be a noetherian scheme. Suppose that $0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0$ is an exact sequence of $\mathcal{O}_X$-module. If $\mathcal{F}$ and $\mathcal{H}$ are both quasi-coherent (resp. coherent), then $\mathcal{G}$ is quasi-coherent (resp. coherent).*

*Proof.* Can assume that $X = \operatorname{Spec} A$ is affine. We have a commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & \widetilde{\mathcal{F}(X)} \arrow[r] \arrow[d, "\sim"] & \widetilde{\mathcal{G}(X)} \arrow[r] \arrow[d] & \widetilde{\mathcal{H}(X)} \arrow[r] \arrow[d, "\sim"] & 0 \\ 0 \arrow[r] & \mathcal{F} \arrow[r] & \mathcal{G} \arrow[r] & \mathcal{H} \arrow[r] & 0 \end{tikzcd} \tag{27}$$

Thus by 5 Lemma, $\mathcal{G}$ is quasi-coherent.

If moreover $\mathcal{F}$ and $\mathcal{H}$ are coherent, then $\mathcal{F}(X)$ and $\mathcal{H}(X)$ are finitely generated. Then $\mathcal{G}(X)$ is also finitely generated so that $\mathcal{G}$ is coherent. □

**Lemma 2.3.** *Let $X$ be a noetherian scheme. Assume that $H^1(X, \mathcal{I}) = 0$ for all $\mathcal{I}$ coherent ideal sheaf. Then we can cover $X$ by affine open subsets of the form $X_f$ for $f \in A = \Gamma(X, \mathcal{O}_X)$.*

*Proof.* Suffice to show that every closed point $x \in X$ admits an affine open neighbourhood $X_f$. For $x \in X$ closed point, choose affine open neighbourhood $U$ of $x$ and set $Y = X \setminus U$. Get $0 \longrightarrow \mathcal{I}_{Y\cup\{x\}} \longrightarrow \mathcal{I}_Y \longrightarrow i_*\mathcal{O}_{\{x\}} \longrightarrow 0$ is exact with reduced scheme structures on each set, where $i : \{x\} \longrightarrow X$ is inclusion. By assumption, $\Gamma(X, \mathcal{I}_Y) \longrightarrow \Gamma(X, i_*\mathcal{O}_{\{x\}})$ is surjective. Take $f \in \Gamma(X, \mathcal{I}_Y)$ such that $f$ is mapped to 1. Then $x \in X_f$. On the other hand, $X_f \subseteq U$, get $X_f$ is affine. □

**Lemma 2.4.** *Let $X$ be a noetherian scheme. Assume that $H^1(X, \mathcal{I}) = 0$ for all $\mathcal{I}$ coherent ideal sheaf. Then for all $e > 0$ and $\mathcal{F} \subseteq \mathcal{O}_X^{\oplus r}$ coherent, we have that $H^1(X, \mathcal{F}) = 0$.*

*Proof.* Define filtration $\mathcal{F} \supseteq \mathcal{F} \cap \mathcal{O}_X^{\oplus(r-1)} \supseteq \cdots \supseteq \mathcal{F} \cap \mathcal{O}_X \supseteq 0$ such that the associated quotients are coherent ideal sheaves. By long exact sequence and induction, get $H^1(X, \mathcal{F} \cap \mathcal{O}_X^i) = 0$ for all $i > 0$ so that $H^1(X, \mathcal{F}) = 0$. □

**Theorem 2.4** (***Serre, Cohomology Criterion of Affineness***). *Let $X$ be a noetherian scheme. Then the following conditions are equivalent:*
*(1)$X$ is affine*
*(2)$H^i(X, \mathcal{F}) = 0$ for all $i > 0$ and $\mathcal{F}$ quasi-coherent.*
*(3)$H^1(X, \mathcal{I}) = 0$ for all $\mathcal{I}$ coherent ideal sheaf.*

*Proof.* See in the Hartshorne Chapter III p.215-216. □

**Remark 2.4.** *In fact, noetherian condition is not necessary.*

## 2.3 Čech cohomology

**Definition 2.4.** *Let $X$ be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ open covering of $X$ with well-ordered index set $I$, $\mathcal{F}$ sheaf of abelian groups on $X$. For $i_0, \cdots, i_p \in I$, set $U_{i_0,\cdots,i_p} = U_{i_0} \cap \cdots \cap U_{i_p}$. Define complex $C^*(\mathcal{I}, \mathcal{F})$ to be*

$$C^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0,\cdots,i_p} \mathcal{F}(U_{i_0,\cdots,i_p}) \tag{28}$$

*with coboundary map*

$$\begin{aligned} d^p : C^p(\mathcal{U}, \mathcal{F}) &\longrightarrow C^{p+1}(\mathcal{U}, \mathcal{F}) \\ \alpha = (\alpha_{i_0,\cdots,i_p})_{i_0,\cdots,i_p} &\longmapsto (\sum_{k=0}^{p+1} (-1)^k \alpha_{i_0,\cdots,\widehat{i_k},\cdots,i_{p+1}}|_{U_{i_0,\cdots,i_{p+1}}})_{i_0,\cdots,i_{p+1}} \end{aligned} \tag{29}$$

**Remark 2.5.** *If we don't assume that $I$ is well-ordered, we need to set $U_{i_0,\cdots,i_p} = 0$ if there are repeated index and $\alpha_{i_0,\cdots,i_p} = (-1)^{sgn(\sigma)} \alpha_{i_{\sigma(0)},\cdots,i_{\sigma(p)}}$.*

**Definition 2.5.** *Let $X$ be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ open covering of $X$ with well-ordered index set $I$, $\mathcal{F}$ sheaf of abelian groups on $X$. Define $\check{H}^p(\mathcal{U}, \mathcal{F}) = h^p(C^*(\mathcal{U}, \mathcal{F}))$, called the pth Čech cohomology of $\mathcal{F}$ with respect to $\mathcal{U}$.*

**Example 2.3.** *(1)For any $\mathcal{F}$ and $\mathcal{U}$, $\check{H}^0(\mathcal{U}, \mathcal{F}) = \Gamma(X, \mathcal{F})$.*
*(2)For $\mathcal{U} = \{X\}$, $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$.*

**Lemma 2.5.** *Let $X$ be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ open covering of $X$ with well-ordered index set $I$, $\mathcal{F}$ sheaf of abelian groups on $X$. Suppose that $\mathcal{U}$ contains $U_i = X$ for some $i$. Then $0 \longrightarrow \mathcal{F}(X) \longrightarrow C^*(\mathcal{U}, \mathcal{F})$ is exact.*

*Proof.* Obviously, exactness at $\mathcal{F}(X)$ and $C^0(\mathcal{U}, \mathcal{F})$ is ok. For $p \geq 1$, without loss of generality, may assume that $i = 0$ which is the smallest element. Define homotopy $k^p : C^p(\mathcal{U}, \mathcal{F}) \longrightarrow C^{p-1}(\mathcal{U}, \mathcal{F}) \quad \alpha \longmapsto (\alpha_{0,i_0,\cdots,i_{p-1}})_{i_0,\cdots,i_{p-1}}$. Check that $d^{p-1} \circ k^p + k^{p+1} \circ d^p = \mathrm{id}_{C^*(\mathcal{U},\mathcal{F})}$. Thus identity is homotopic to zero map, which inducing $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$.. ☐

## 2.4 Comparison Theorem

**Definition 2.6.** *Let $X$ be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ open covering of $X$ with well-ordered index set $I$, $\mathcal{F}$ sheaf of abelian groups on $X$. Define a complex of sheaves on $X$, denoted by $\mathcal{C}^*(\mathcal{U}, \mathcal{F})$, as following*

$$\mathcal{C}^p(\mathcal{U}, \mathcal{F}) = \prod_{i_0,\cdots,i_p} f_*(\mathcal{F}|_{U_{i_0,\cdots,i_p}}) \tag{30}$$

*where $f$ is the inclusion, with coboundary map*

$$d^p : \mathcal{C}^p(\mathcal{U}, \mathcal{F}) \longrightarrow \mathcal{C}^{p+1}(\mathcal{U}, \mathcal{F}) \tag{31}$$

*defined accordingly.*

**Remark 2.6.** *Obviously, $\Gamma(X, \mathcal{C}^p(\mathcal{U}, \mathcal{F})) = C^p(\mathcal{U}, \mathcal{F})$.*

**Proposition 2.7.** *Let $X$ be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ open covering of $X$ with well-ordered index set $I$, $\mathcal{F}$ sheaf of abelian groups on $X$. Then $0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{C}^*(\mathcal{U}, \mathcal{F})$ is exact.*

**Reason 2.3.** *Since exactness of complex of sheaves is equivalent to exactness at all stalks. Set $x \in X$, we can represent elements in stalks by sections on $V$, which is small enough to be contained in $U_j$ for some $j$. Apply Lemma 2.5, get exactness of complex of sections on $V$.*

**Proposition 2.8.** *Let $X$ be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ open covering of $X$ with well-ordered index set $I$, $\mathcal{F}$ flasque sheaf of abelian groups on $X$. Then $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$.*

*Proof.* Since flasque is stable under restriction to open subset, direct limit and product, $\mathcal{C}^p(\mathcal{U}, \mathcal{F})$ is flasque for all $p$. Thus $\mathcal{C}^*(\mathcal{U}, \mathcal{F})$ is a flasque resolution of $\mathcal{F}$. By Proposition 2.5 and Proposition 2.1, get

$$\check{H}^p(\mathcal{U}, \mathcal{F}) = h^p(\Gamma(X, \mathcal{C}^*(\mathcal{U}, \mathcal{F}))) = H^p(X, \mathcal{F}) = 0 \tag{32}$$

hence we are done! □

**Lemma 2.6.** *Let $X$ be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ open covering of $X$ with well-ordered index set $I$, $\mathcal{F}$ sheaf of abelian groups on $X$. Then for all $p \geq 0$, there is a natural homomorphism $\check{H}^p(\mathcal{U}, \mathcal{F}) \longrightarrow H^p(X, \mathcal{F})$.*

*Proof.* Let $0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I}^*$ be an injective resolution of $\mathcal{F}$ in $Ab(X)$. By Comparison Theorem in Homological Algebra, there exists a morphism of complexes which is unique up to homotopy, inducing the natural homomorphisms we want. □

**Theorem 2.5** (***Comparison Theorem***)**.** *Let $X$ be a topological space, $\mathcal{U} = \{U_i\}_{i\in I}$ open covering of $X$ with well-ordered index set $I$, $\mathcal{F}$ sheaf of abelian groups on $X$. Suppose that for all $p$ and all $i_0, \cdots, i_p$, we have that $H^q(U_{i_0,\cdots,i_p}, \mathcal{F}|_{U_{i_0,\cdots,i_p}}) = 0$ for all $q > 0$. Then the natural homomorphism $\check{H}^r(\mathcal{U}, \mathcal{F}) \longrightarrow H^r(X, \mathcal{F})$ is isomorphic for all $r \geq 0$.*

*Proof.* For $r = 0$, the natural homomorphism is identity map of global sections of $\mathcal{F}$. For $r > 0$, embed $\mathcal{F}$ into an injective sheaf $\mathcal{I}$. There is an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I} \longrightarrow \mathcal{G} \longrightarrow 0 \tag{33}$$

On the one hand, by definition of Čech sheaf resolution, we have an exact sequence of complexes

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F} \arrow[r] \arrow[d] & \mathcal{I} \arrow[r] \arrow[d] & \mathcal{G} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{C}^\bullet(\mathcal{U}, \mathcal{F}) \arrow[r] & \mathcal{C}^\bullet(\mathcal{U}, \mathcal{I}) \arrow[r] & \mathcal{C}^\bullet(\mathcal{U}, \mathcal{G}) \arrow[r] & 0
\end{tikzcd} \tag{34}$$

Taking products, get

$$0 \longrightarrow C^*(\mathcal{U}, \mathcal{F}) \longrightarrow C^*(\mathcal{U}, \mathcal{I}) \longrightarrow C^*(\mathcal{U}, \mathcal{G}) \longrightarrow 0 \tag{35}$$

On the other hand, by Horseshoe Lemma, we may take injective resolution $\mathcal{I}^\bullet_{\mathcal{F}}$, $\mathcal{I}^\bullet_{\mathcal{I}}$ and $\mathcal{I}^\bullet_{\mathcal{G}}$ of $\mathcal{F}$, $\mathcal{I}$ and $\mathcal{G}$ respectively such that

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{F} \arrow[r] \arrow[d] & \mathcal{I} \arrow[r] \arrow[d] & \mathcal{G} \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & \mathcal{I}^\bullet_{\mathcal{F}} \arrow[r] & \mathcal{I}^\bullet_{\mathcal{I}} \arrow[r] & \mathcal{I}^\bullet_{\mathcal{G}} \arrow[r] & 0 \end{tikzcd} \tag{36}$$

is an exact sequence of complexes and $\mathcal{I}^i_{\mathcal{I}} = \mathcal{I}^i_{\mathcal{F}} \oplus \mathcal{I}^i_{\mathcal{G}}$ for all $i \geq 0$. Then by Comparison Theorem for injective resolution, we get morphism of chain complexes $\alpha : \mathcal{C}^\bullet(\mathcal{U}, \mathcal{F}) \to \mathcal{I}^\bullet_{\mathcal{F}}$, $\beta : \mathcal{C}^\bullet(\mathcal{U}, \mathcal{I}) \to \mathcal{I}^\bullet_{\mathcal{I}}$ and $\gamma : \mathcal{C}^\bullet(\mathcal{U}, \mathcal{G}) \to \mathcal{I}^\bullet_{\mathcal{G}}$, which are unique up to homotopy.

Consider the following diagram

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{C}^i(\mathcal{U}, \mathcal{F}) \arrow[r, hook, "f"] \arrow[d, "\alpha"] & \mathcal{C}^i(\mathcal{U}, \mathcal{I}) \arrow[r, "g"] \arrow[d, "\beta"] & \mathcal{C}^i(\mathcal{U}, \mathcal{G}) \arrow[r] \arrow[d, "\gamma"] & 0 \\ 0 \arrow[r] & \mathcal{I}^i_{\mathcal{F}} \arrow[r, hook, "u"] & \mathcal{I}^i_{\mathcal{I}} \arrow[r, "v"] & \mathcal{I}^i_{\mathcal{G}} \arrow[r] & 0 \end{tikzcd} \tag{37}$$

Want to modify $\alpha$, $\beta$ and $\gamma$. Since the bottom row is split, we have inclusion $\sigma : \mathcal{I}^\bullet_{\mathcal{G}} \to \mathcal{I}^\bullet_{\mathcal{I}}$ and projection $\pi : \mathcal{I}^\bullet_{\mathcal{I}} \to \mathcal{I}^\bullet_{\mathcal{F}}$. Take $\beta' = \beta + \pi \circ (\gamma \circ g - v \circ \beta)$ and $\alpha' = \sigma \circ (\beta' \circ f - u \circ \alpha)$. Then by our choice of $\alpha$, $\beta$ and $\gamma$, it is clear that the new diagram is commutative.

By assumption, for all $i_0, \cdots, i_p$,

$$0 \longrightarrow \mathcal{F}(U_{i_0,\cdots,i_p}) \longrightarrow \mathcal{I}(U_{i_0,\cdots,i_p}) \longrightarrow \mathcal{G}(U_{i_0,\cdots,i_p}) \longrightarrow 0 \tag{38}$$

is exact. Taking global sections and by Homological Algebra, there is commutative diagram between two long exact sequences

$$\begin{tikzcd} \cdots \arrow[r] & \check{H}^r(\mathcal{U}, \mathcal{F}) \arrow[r] \arrow[d] & \check{H}^r(\mathcal{U}, \mathcal{I}) \arrow[r] \arrow[d] & \check{H}^r(\mathcal{U}, \mathcal{G}) \arrow[r] \arrow[d] & \check{H}^{r+1}(\mathcal{U}, \mathcal{F}) \arrow[r] \arrow[d] & \cdots \\ \cdots \arrow[r] & H^r(X, \mathcal{F}) \arrow[r] & H^r(X, \mathcal{I}) \arrow[r] & H^r(X, \mathcal{G}) \arrow[r] & H^{r+1}(X, \mathcal{F}) \arrow[r] & \cdots \end{tikzcd} \tag{39}$$

Since $\mathcal{I}$ is injective, get $\mathcal{I}$ flasque and $\check{H}^r(\mathcal{U}, \mathcal{I}) = 0$ for all $r > 0$. Thus $\check{H}^{r+1}(\mathcal{U}, \mathcal{F}) = \check{H}^r(\mathcal{U}, \mathcal{G})$ and $H^{r+1}(X, \mathcal{F}) = H^r(X, \mathcal{G})$ for all $r \geq 1$. In particular, we have commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & \check{H}^0(\mathcal{U}, \mathcal{F}) \arrow[r] \arrow[d, equal] & \check{H}^0(\mathcal{U}, \mathcal{I}) \arrow[r] \arrow[d, equal] & \check{H}^0(\mathcal{U}, \mathcal{G}) \arrow[r] \arrow[d, equal] & \check{H}^1(\mathcal{U}, \mathcal{F}) \arrow[r] \arrow[d] & 0 \\ \cdots \arrow[r] & H^0(X, \mathcal{F}) \arrow[r] & H^0(X, \mathcal{I}) \arrow[r] & H^0(X, \mathcal{G}) \arrow[r] & H^1(X, \mathcal{F}) \arrow[r] & 0 \end{tikzcd} \tag{40}$$

Hence by 5 Lemma, we get the natural map $\check{H}^1(\mathcal{U}, \mathcal{F}) \to H^1(X, \mathcal{F})$ is isomorphic. Finally, by induction, done! $\square$

**Remark 2.7.** *Here our modifying assumption is in fact same as the proof of injectivity of a split exact sequence consists of injective objects.*

**Corollary 2.6.** *Let $X$ be a (noetherian) separated scheme, $\mathcal{U}$ affine open covering of $X$, $\mathcal{F}$ quasi-coherent sheaf on $X$. Then the natural homomorphism $\check{H}^p(\mathcal{U}, \mathcal{F}) \longrightarrow H^p(X, \mathcal{F})$ is isomorphic for all $p$.*

**Remark 2.8.** *This corollary immediately comes from Serre Vanishing Theorem.*

**Corollary 2.7.** *Let $X$ be a (noetherian) separated scheme covered by $r+1$ affine open subsets. Then for all $\mathcal{F}$ quasi-coherent on $X$, we have that $H^i(X,\mathcal{F})=0$ for all $i>r$. In particular, when $r=0$, this specializes to Serre Vanishing Theorem.*

### 2.5 Cohomology of projective space

**Theorem 2.6.** *Let $A$ be a noetherian ring, $S=A[x_0,\cdots,x_r]$, $X=\mathrm{Proj}(S)=\mathbb{P}^r_A$, $r\geq 1$. Then we have that*
*(1)$H^i(X,\mathcal{O}_X(n))=0$ for all $0<i<r$ and all $n\in\mathbb{Z}$*
*(2)$H^r(X,\mathcal{O}_X(-r-1))\cong A$ is free $A$-module of rank $1$*
*(3)the natural pairing $H^0(X,\mathcal{O}_X(n))\times H^r(X,\mathcal{O}_X(-n-r-1))\longrightarrow H^r(X,\mathcal{O}_X(-r-1))$ is a perfect pairing of finite free $A$-modules for all $n\in\mathbb{Z}$.*

**Remark 2.9.** *In fact, we have already known that $H^0(X,\mathcal{O}_X(n))=\begin{cases}S_n & n\geq 0\\ 0 & n<0\end{cases}$ and $H^i(X,\mathcal{O}_X(n))=0$ for all $i>r$ and all $n\in\mathbb{Z}$. Thus with this theorem, we have figured out all the cohomological groups of $\mathbb{P}^r_A$. Especially, all of them are finite free $A$-modules.*

*Proof.* Consider $\mathcal{F}=\oplus_{n\in\mathbb{Z}}\mathcal{O}_X(n)$ quasi-coherent. Since cohomology commutes with direct sum, $H^i(X,\mathcal{F})=\oplus_{n\in\mathbb{Z}}H^i(X,\mathcal{F})$. In particular, $H^0(X,\mathcal{F})=\Gamma_*(\mathcal{O}_\mathcal{X})\cong S$. Note that $X$ is noetherian, separated and covered by $r+1$ affine open subsets. By Corollary 2.7, get $\check{H}^p(\mathcal{U},\mathcal{F})\cong H^p(X,\mathcal{F})$ for all $p$.

For all $i_0,\cdots,i_p$, note that $U_{i_0,\cdots,i_p}=D_+(x_{i_0},\cdots,x_{i_p})$. Get $\mathcal{F}(U_{i_0,\cdots,i_p})\cong S_{x_{i_0},\cdots,x_{i_p}}$. Thus $C^*(\mathcal{U},\mathcal{F})$ is given by

$$0\longrightarrow\prod_{i_0}S_{x_{i_0}}\longrightarrow\cdots\longrightarrow S_{x_0,\cdots,x_r}. \tag{41}$$

Since $\check{H}^r(\mathcal{U},\mathcal{F})$ is the cokernel of $d^{r-1}:\prod_k S_{x_0,\cdots,\widehat{x_k},\cdots,x_r}\longrightarrow S_{x_0,\cdots,x_r}$. Note that $S_{x_0,\cdots,x_r}$ is free $A$-module spanned by $\{x_0^{l_0}..x_r^{l_r}\}$, where $l_0,\cdots,l_r\in\mathbb{Z}$. In addition, $im(d^{r-1})$ is free $A$-module spanned by $\{x_0^{l_0}\cdots x_r^{l_r}\}$, where $l_0,\cdots,l_r\in\mathbb{Z}$ and at least one $l_i\geq 0$. Get $H^r(\mathcal{U},\mathcal{F})$ is free $A$-module spanned by $\{x_0^{l_0}\cdots x_r^{l_r}\}$, where $l_i<0$. While the only "monomial" of degree $-r-1$ is $x_0^{-1}\cdots x_r^{-1}$, get $H^r(X,\mathcal{O}_X(-r-1))\cong A$.

For (3), we also find $H^r(X,\mathcal{O}_X(-n-r-1))=0$ if $n<0$. On the other hand, $H^0(X,\mathcal{O}_X(n))=0$ if $n<0$. Thus, (3) is trivially true for $n<0$. For $n\geq 0$, $H^0(X,\mathcal{O}_X(n))$ has basis $\{x_0^{m_0}\cdots x_r^{m_r}\}$, where $m_i\geq 0$ and $\sum_{i=0}^r m_i=n$. The natural pairing is generated by $(x_0^{m_0}\cdots x_r^{m_r},x_0^{l_0}\cdots x_r^{l_r})\longmapsto x_0^{m_0+l_0}\cdots x_r^{m_r+l_r}$, here that $x_i^{m_i+l_i}=0$ if $m_i+l_i\geq 0$. Then it is a perfect pairing since it is easy to see that dual basis of $x_0^{m_0}\cdots x_r^{m_r}$ is $x_0^{-m_0-1}\cdots x_r^{-m_r-1}$.

For (1), take induction on $r$. When $r=1$, there is nothing to prove. Suppose that $r\geq 2$. Consider localization of $C^*(\mathcal{U},\mathcal{F})$ with respect to $x_r$. By Theorem 2.1, the localization gives the Čech complex of $\mathcal{F}\big|_{U_r}$ with respect to $\mathcal{U}\big|_{U_r}$. As localization is exact and $U_r$ is affine, $H^i(X,\mathcal{F})_{x_r}=H^i(U_r,\mathcal{F}\big|_{U_r})=0$. Suffice to show that for all $0<i<r$, $x_r$ acts injectively on

$H^i(X, \mathcal{F})$. View $x_r$ as a homomorphism of $A$-modules like below

$$x_r : H^i(X, \mathcal{F}(-1)) \longrightarrow H^i(X, \mathcal{F}) \tag{42}$$

here $\mathcal{F}(-1)$ is same sheaf on $\mathcal{F}$ with grading shifted. Consider exact sequence of graded $C$-modules

$$0 \longrightarrow S(-1) \xrightarrow{x_r} S \longrightarrow S/(x_r) \longrightarrow 0 \tag{43}$$

Get exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_X(-1) \xrightarrow{x_r} \mathcal{O}_X \longrightarrow j_*\mathcal{O}_H \longrightarrow 0 \tag{44}$$

where $H$ is the hyperplane given by $x_r = 0$ and $j : H \hookrightarrow X$ is closed immersion. Take direct sum, get exact sequence

$$0 \longrightarrow \mathcal{F}(-1) \xrightarrow{x_r} \mathcal{F} \longrightarrow \mathcal{F}_H \longrightarrow 0 \tag{45}$$

where $\mathcal{F}_H = \oplus_{n\in\mathbb{Z}} j_*(\mathcal{O}_H(n))$. Get long exact sequence

$$\cdots \longrightarrow H^{i-1}(X, \mathcal{F}_H) \longrightarrow H^i(X, \mathcal{F}(-1)) \xrightarrow{x_r} H^i(X, \mathcal{F}) \longrightarrow H^i(X, \mathcal{F})_H \longrightarrow \cdots \tag{46}$$

Now $H \cong \mathbf{P}_A^{r-1}$ and since $j$ is closed immersion, $H^i(X, \mathcal{F}_H) = H^i(H, \oplus_{n\in\mathbb{Z}} O_H(n))$. By induction hypothesis, $H^i(X, \mathcal{F}_H) = 0$ for all $0 < i < r - 1$. Thus $x_r$ is injective on $H^i(X, \mathcal{F}(-1))$ for all $1 < i < r$. When $i = 1$, find that $H^0(X, \mathcal{F}) = S \longrightarrow H^0(X, \mathcal{F}_H) = S/(x_r)$ is surjective. Get $x_r$ is injective on $H^i(X, \mathcal{F}(-1))$ for all $0 < i < r$, done! □

**Theorem 2.7** (***Serre***)**.** *Let $A$ be a noetherian ring, $X$ projective scheme over $A$, $\mathcal{F}$ coherent sheaf on $X$. Assume $X \to A$ factor through closed immersion $j : X \hookrightarrow \mathbb{P}_A^r$. Then*
*(1)for all $i \geq 0$, $H^i(X, \mathcal{F})$ is a finite $A$-module.*
*(2)there exists integer $n_0$ relative to $\mathcal{F}$ such that $H^i(X, \mathcal{F}(n)) = 0$ for all $i > 0$ and $n \geq n_0$.*

*Proof.* Since $j$ is finite, $j_*(\mathcal{F})$ is coherent on $\mathbb{P}_A^r$ and by Projection Formula, $j_*(\mathcal{F}(n)) = (j_*(\mathcal{F}))(n)$. What's more, since $j$ is closed immersion, $H^i(X, \mathcal{F}) = H^i(\mathbb{P}_A^r, j_*(\mathcal{F}))$. Thus we can reduce this problem to $X = \mathbb{P}_A^r$. For $\mathcal{F} = \mathcal{O}_X(m)$, this is just what Theorem 2.6 said. Also, for general $\mathcal{F}$, as $\mathbb{P}_A^r$ can be covered by $r + 1$ affine open subsets, $H^i(X, \mathcal{F}(n)) = 0$ for all $i > r$ and $n \in \mathbb{Z}$.

For (1), by induction, suppose theorem is correct for $i+1$. By Corollary 1.1, there exists $m \in \mathbb{Z}$ and $r < \infty$ such that $\mathcal{F}$ is quotient of $\mathcal{O}_X(m)^r$, inducing the following exact sequence of coherent sheaves

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{O}_X(m)^r \longrightarrow \mathcal{F} \longrightarrow 0 \tag{47}$$

Consider the long exact sequence

$$\cdots \longrightarrow H^i(X, \mathcal{O}_X(m)^r) \longrightarrow H^i(X, \mathcal{F}) \longrightarrow H^{i+1}(X, \mathcal{G}) \longrightarrow \cdots \tag{48}$$

By hypothesis, $H^{i+1}(X, \mathcal{G})$ is finite $A$-module. As $A$ is noetherian, $H^i(X, \mathcal{F})$ is also finite.

For (2), by above discuss, we only need to check for $0 < i \leq r$. Induct on $i$, assume for $i + 1$ and all coherent sheaf $\mathcal{F}$, there exists $n_{i+1}(\mathcal{F})$ such that $H^{i+1}(X, \mathcal{F}(n)) = 0$ for all

$n \geq n_{i+1}(\mathcal{F})$. Take same short exact sequence. Tensor the short exact sequence by $\mathcal{O}_X(n)$. Get exact sequence

$$0 \longrightarrow \mathcal{G}(n) \longrightarrow \mathcal{O}_X(m+n)^r \longrightarrow \mathcal{F}(n) \longrightarrow 0 \tag{49}$$

Consider the long exact sequence

$$\cdots \longrightarrow H^i(X, \mathcal{O}_X(m+n)^r) \longrightarrow H^i(X, \mathcal{F}(n)) \longrightarrow H^{i+1}(X, \mathcal{G}(n)) \longrightarrow \cdots \tag{50}$$

By hypothesis, there exists $n_{i+1}(\mathcal{G})$ such that $H^{i+1}(X, \mathcal{G}(n)) = 0$ for all $n \geq n_{i+1}(\mathcal{G})$. By Theorem 2.6, $H^i(X, \mathcal{O}_X(m+n)^r) = 0$ for large enough $n$. Thus there exists $n_i(\mathcal{F})$ such that $H^i(X, \mathcal{F}(n)) = 0$ for all $n \geq n_i(\mathcal{F})$. As there are finitely many $i$, we can take a uniform upper $n_0(\mathcal{F})$ bound for $n_i(\mathcal{F})$. $\square$

**Corollary 2.8.** *Let $f : X \longrightarrow Y$ be a projective morphism with $X, Y$ noetherian, $\mathcal{F}$ coherent sheaf on $X$. Then $f_*\mathcal{F}$ is coherent on $Y$.*

**Reason 2.4.** *Can assume $Y = \operatorname{Spec} A$. Then $f_*\mathcal{F} \cong \widetilde{H^0(X, \mathcal{F})}$ which is a finite $A$-module by Theorem 2.7.*

**Corollary 2.9.** *Let $X$ be a geometrically integral scheme projective over field $k$. Then $\Gamma(X, \mathcal{O}_X) = k$.*

**Reason 2.5.** *By Theorem 2.7, $L = \Gamma(X, \mathcal{O}_X)$ is finite over $k$. While $X$ is integral, get $L$ is integral domain. Thus by Noetherian Normalization Lemma, $L/k$ is finite field extension. Note that geometrically integral if and only if $k$ is algebraically closed in function field $k(X)$. Get $L = k$.*

# 3 Riemann-Roch Formula

## 3.1 Serre's duality

**Definition 3.1.** *Let $X$ be a variety, $\mathcal{F} \in \mathbf{Ab}(X)$. Define the Euler characteristic of $\mathcal{F}$ to be $\chi(\mathcal{F}) := \sum_{i \geq 0} (-1)^i \cdot h^i(X, \mathcal{F})$, which is in fact a finite sum by Grothendieck Vanishing Theorem.*

**Lemma 3.1.** *Let $X$ be a variety. Then for all short exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ in $\mathbf{Ab}(X)$, we have that $\chi(\mathcal{F}) = \chi(\mathcal{F}') + \chi(\mathcal{F}'')$.*

*Proof.* Applying global section functor, we get a long exact sequence

$$0 \to H^0(X, \mathcal{F}') \to H^0(X, \mathcal{F}) \to H^0(X, \mathcal{F}'') \to H^1(X, \mathcal{F}') \to \cdots \tag{51}$$

By Grothendieck Vanishing Theorem, this sequence would end for some $n$ large enough. Then we can split it into several short exact sequences. Summing the equations of dimensions given by these short exact sequence, we are done! $\square$

**Theorem 3.1** (***Serre's Duality***)**.** *Let $X$ be a smooth projective variety of dimension $n$, $\omega_X$ canonical sheaf. Then for all locally free sheaf $\mathcal{F}$ on $X$, we have canonical isomorphism $H^i(X, \mathcal{F}^\vee \otimes \omega_X) \xrightarrow{\sim} H^{n-i}(X, \mathcal{F})^\vee$.*

**Remark 3.1.** *In class, Professor Du gave a proof as following. First take injective resolutions $\mathcal{I}^\bullet$ of $\mathcal{F}$, then tensor with $\omega_X$. Since $\omega_X$ is flat, we $0 \to \mathcal{F} \otimes \omega_X \to \mathcal{I}^\bullet \otimes \omega_X$ is still exact. However, we do not know if $\mathcal{I} \otimes \omega_X$ is still injective.*

**Corollary 3.1.** *Let $X$ be a smooth projective variety of dimension $n$, $\omega_X$ canonical sheaf. Then for all locally free sheaf $\mathcal{F}$ on $X$, $\chi(\mathcal{F}) = (-1)^n \cdot \chi(\mathcal{F}^\vee \otimes \omega_X)$.*

*Proof.* Immediately comes from Grothendieck Vanishing and Serre's duality. □

## 3.2 Divisors

**Proposition 3.1.** *Let $X$ be an integral variety, $\mathcal{F} \in \mathbf{Coh}(X)$. Then there exists open dense subset $U \subseteq X$ such that $\mathcal{F}\big|_U$ is locally free. Moreover, we can well define $\operatorname{rank}(\mathcal{F}) := \operatorname{rank}(\mathcal{F}\big|_U)$.*

*Proof.* Consider the subset $\{x \in X \big| \dim_{k(x)} \mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x \leq d\}$, which by Nakayama's Lemma is an open subset. Then we may take $d$ minimal so that $U := \{x \in X \big| \dim_{k(x)} \mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x = d\}$ is an open subset. Since $X$ is integral, $U$ is irreducible.

Now we want to show that $\mathcal{F}\big|_U$ is locally free. Take affine subset $W = \operatorname{Spec} A$ such that $\mathcal{F}\big|_W = \widetilde{M}$ for some finitely generated $A$-module $M$. Then for all $\mathfrak{p} \in \operatorname{Spec} A$, $M_\mathfrak{p}$ is generated by exact $d$ elements.

Narrowing $W$, we may assume that there exists $m_1, \cdots, m_d$ in $M$ such that $M_\mathfrak{p}$ can be generated by them for all $\mathfrak{p} \in \operatorname{Spec} A$. Then if there is some relation $\sum_{i=1}^d a_i m_i = 0$, then $a_i \in \mathfrak{R}(A) = (0)$ for all $i$. Otherwise, there exists $\mathfrak{q} \in \operatorname{Spec} A$ such that $a_i \notin \mathfrak{q}$ for some $i$. And then this relation becomes a relation in $M_\mathfrak{q} \otimes A_\mathfrak{q}/\mathfrak{q}A_\mathfrak{q}$, contradicting to $\mathfrak{q} \in U$. Hence $\mathcal{F}\big|_W$ is free and $\mathcal{F}\big|_U$ is locally free. □

**Lemma 3.2.** *Let $X$ be an integral variety. Then for all short exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ in $\mathbf{Coh}(X)$, we have that $\operatorname{rank}(\mathcal{F}) = \operatorname{rank}(\mathcal{F}') + \operatorname{rank}(\mathcal{F}'')$.*

*Proof.* We may take open dense subset $U \subseteq X$ such that $\mathcal{F}\big|_U$, $\mathcal{F}'\big|_U$ and $\mathcal{F}''\big|_U$ are all locally free. Then it is clear that $\operatorname{rank}(\mathcal{F}\big|_U) = \operatorname{rank}(\mathcal{F}'\big|_U) + \operatorname{rank}(\mathcal{F}''\big|_U)$. □

**Definition 3.2.** *Let $X$ be an integral variety, $\mathcal{F} \in \mathbf{Coh}(X)$. Define $\deg(\mathcal{F}) = \chi(\mathcal{F}) - \operatorname{rank} \cdot \chi(\mathcal{O}_X)$.*

**Lemma 3.3.** *Let $X$ be an integral variety. Then for all short exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ in $\mathbf{Coh}(X)$, we have that $\deg(\mathcal{F}) = \deg(\mathcal{F}') + \deg(\mathcal{F}'')$.*

**Reason 3.1.** *Immediately comes from Lemma 3.1 and Lemma 3.2*

Now let $C$ be a smooth connected projective curve. Define the (arithmatic) genus of $C$ to be $g(C) := 1 - \chi(C) = h^1(C, \mathcal{O}_C)$.

**Proposition 3.2.** *Let $C \subseteq \mathbb{P}^2_\mathbb{C}$ be a smooth curve of degree $n$. Then $C$ is connected and $g(C) = \frac{(n-1)(n-2)}{2}$.*

*Proof.* Assume $C = V(f)$ with $\deg f = n$. Consider exact sequence $0 \to \mathcal{O}_{\mathbb{P}^2_{\mathbb{C}}}(-n) \xrightarrow{\cdot f} \mathcal{O}_{\mathbb{P}^2_{\mathbb{C}}} \to \mathcal{O}_C \to 0$. Then $\chi(\mathcal{O}_C) = \chi(\mathcal{O}_{\mathbb{P}^2_{\mathbb{C}}}) - \chi(\mathcal{O}_{\mathbb{P}^2_{\mathbb{C}}}(-n))$. Since $\chi(\mathcal{O}_{\mathbb{P}_{\mathbb{C}^2}}) = h^0(\mathcal{O}_{\mathbb{P}_{\mathbb{C}^2}}) = 1$ and by Serre's duality, $\chi(\mathcal{O}_{\mathbb{P}_{\mathbb{C}^2}}(-n)) = -h^1(\mathcal{O}_{\mathbb{P}_{\mathbb{C}^2}}(-n)) = -h^1(\mathcal{O}_{\mathbb{P}_{\mathbb{C}^2}}(-n))^\vee = -h^0(\mathcal{O}_{\mathbb{P}_{\mathbb{C}^2}}(n-3)) = -\frac{(n-1)(n-2)}{2}$, we get that $g(C) = \frac{(n-1)(n-2)}{2}$. □

**Remark 3.2.** *Connectedness comes from the fact that every two different curves in $\mathbb{P}^2_{\mathbb{C}}$ would intersect, in which case smoothness implies irreducibility. In general, we would just assume curve to be integral.*

Let $D = \sum_i a_i x_i$ be an effective Weil divisor on $C$. Define the degree of $D$ to be $\deg D = \sum_i a_i$.

**Proposition 3.3.** *Every effective Weil divisor $D$ would correspond to an invertible sheaf $\mathcal{O}_C(D)$. Moreover, $\mathcal{O}_C(D_1) \cong \mathcal{O}_C(D_2)$ if and only if $D_1 \sim D_2$.*

*Proof.* Consider $C_i = \operatorname{Spec} \mathcal{O}_{C,x_i}/\mathfrak{m}_{x_i}^{a_i}$. Then there are closed immersions $\varphi_i : C_i \hookrightarrow \operatorname{Spec} C$. Taking disjoint product $Y := \sqcup C_i$, we get a closed subscheme of $C$. Consider the ideal sheaf $\mathcal{O}_C(-D) := \mathcal{I}_{Y/C}$. And our desired $\mathcal{O}_C(D)$ can be defined as dual of $\mathcal{O}_C(-D)$.

Now we are supposed to show that $\mathcal{O}_C(-D)$ is locally free of rank 1. Affine locally, we have $U = \operatorname{Spec} A$ and $C_i = \operatorname{Spec} A/\mathfrak{p}_i^{a_i}$. Then $Y = \sqcup C_i = \operatorname{Spec} \bigoplus A/\mathfrak{p}_i^{a_i} \cong \operatorname{Spec} A/\prod \mathfrak{p}_i^{a_i}$ and $\mathcal{I}_{Y/C}\big|_U = \prod \mathfrak{p}_i^{a_i}$.

Note that $C$ is smooth, for all $i$, $A_{\mathfrak{p}_i}$ is a DVR. Take uniformizers $\varpi_i \in A_{\mathfrak{p}_i}$. Then there exists open subset $W = D(s) \subseteq U$ such that $\varpi_i \in A_s$. Thus $\mathfrak{p}_i A_s$ is a principle ideal generated by $\varpi_i$ and so is $\prod \mathfrak{p}_i^{a_i}$. Conclude that $\mathcal{O}_C(-D)$ is locally free of rank 1. □

**Remark 3.3.** *For arbitrary Weil divisor $D$, we can write $D = D_1 - D_2$ with $D_1$ and $D_2$ effective. Then define $\mathcal{O}_C(D) = \mathcal{O}_C(D_1) \otimes \mathcal{O}_C(-D_2)$. In addition, the second part comes from careful argument about transition functions.*

**Lemma 3.4.** $\deg D = \deg(\mathcal{O}_C(D))$ *for all Weil divisor $D$.*

*Proof.* First prove for effective $D$. Take $\ell : Y \to C$ as above, then there is an exact sequence

$$0 \longrightarrow \mathcal{O}_C(-D) \longrightarrow \mathcal{O}_C \longrightarrow \ell_* \mathcal{O}_Y \longrightarrow 0 \tag{52}$$

Tensoring by $\mathcal{O}_C(D)$, we get exact sequence

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{O}_C(D) \longrightarrow \ell_* \mathcal{O}_Y(D) \longrightarrow 0 \tag{53}$$

Then by Lemma 3.1, $\deg(\mathcal{O}_C(D)) = \chi(\mathcal{O}_C(D)) - \chi(\mathcal{O}_C) = \chi(\ell_* \mathcal{O}_Y(D))$. Is suffices to show $\chi(\ell_* \mathcal{O}_Y \otimes \mathcal{O}_C(D)) = \deg D$. Note that we can take affine open covering $\mathcal{U}$ of $C$ trivializing $\mathcal{O}_C(D)$ such that for all $i$, there is a unique $U_i = \operatorname{Spec} A_i$ in $\mathcal{U}$ containing $x_i$.

Use such a covering to compute cohomology. Since $\ell_* \mathcal{O}_Y = \oplus \ell_* \mathcal{O}_{C_i}$, $\chi(\ell_* \mathcal{O}_Y(D)) = \sum \chi(\ell_* \mathcal{O}_{C_i}(D))$. It is clear that $\Gamma(C, \ell_* \mathcal{O}_{C_i}(D)) \cong \Gamma(U_i, \ell_* \mathcal{O}_{C_i}(D))$. Thus $\chi(\ell_* \mathcal{O}_{C_i}(D)) = a_i - 0 = a_i$ and hence $\chi \ell_* \mathcal{O}_Y(D) = \sum a_i = \deg D$.

For arbitrary $D$, write $D = D_1 - D_2$ with $D_1$ and $D_2$ effective. Then $\deg D = \deg D_1 - \deg D_2 = \deg(\mathcal{O}_C(D_1)) - \deg(\mathcal{O}_C(D_2)) = \chi(\mathcal{O}_C(D_1)) - \chi(\mathcal{O}_C(D_2))$. Consider exact sequence

$$0 \longrightarrow \mathcal{O}_C(D_1 - D_2) \longrightarrow \mathcal{O}_C(D_1) \longrightarrow \mathcal{O}_{D_2}(D_1) \longrightarrow 0 \tag{54}$$

By similar argument, we know $\chi(\mathcal{O}_C(D)) = \chi(\mathcal{O}_C(D_1)) - \deg D_2 = \chi(\mathcal{O}_C(D_1)) - \chi(\mathcal{O}_C(D_2)) + \chi(\mathcal{O}_C)$. Thus $\deg D = \deg(\mathcal{O}_C(D))$. □

**Corollary 3.2.** *The map* $\deg : \mathrm{Pic}(C) \to \mathbb{Z}$ *is a group homomorphism.*

**Reason 3.2.** *Immediately comes from the fact that* $\deg : WeilDiv \to \mathbb{Z}$ *is a group homomorphism and Lemma 3.4.*

**Lemma 3.5.** *Let $\mathcal{L}$ be an invertible sheaf on $C$. Then for all nonzero global section $s$ of $\mathcal{L}$, we have that* $\mathcal{O}_C(\mathrm{div}(s)) \cong \mathcal{L}$.

*Proof.* Assume $g_{ij}$ are transition functions of $\mathcal{L}$ and $s\big|_{U_i} = f_i e_i$ where $e_i$ has image 1 under trivialization. Then we have that $f_j e_j = f_j g_{ij} e_i$ so that $f_i = g_{ij} f_j$ on $U_i \cap U_j$. By definition, $\mathrm{div}(s) = (U_i, f_i)$ and $\mathcal{O}_C(\mathrm{div}(s))$ is defined by transition functions $\frac{f_i}{f_j} = g_{ij}$. Hence $\mathcal{O}_C(\mathrm{div}(s)) \cong \mathcal{L}$. □

**Corollary 3.3.** *Let $\mathcal{L}$ be an invertible sheaf on $C$. If $H^0(C, \mathcal{L}) \neq 0$, then* $\deg \mathcal{L} \geq 0$.

**Reason 3.3.** *By Lemma 3.5, pick a nonzero global section $s$ of $\mathcal{L}$, then we know* $\mathcal{L} \cong \mathcal{O}_C(\mathrm{div}(s))$. *Note that $s$ has no pole,* $\mathrm{div}(s)$ *is an effective Weil divisor so that* $\deg \mathcal{L} = \deg \mathcal{O}_C(\mathrm{div}(s)) = \deg(\mathrm{div}(s)) \geq 0$.

**Definition 3.3.** *Let $C$ be a smooth connected projective curve. Define a relation $\geq$ in* $\mathrm{Pic}(C)$ *by $\mathcal{L} \geq \mathcal{L}'$ if $H^0(C, \mathcal{L} \otimes \mathcal{L}'^\vee) \neq 0$. Or equivalently, $\mathcal{L} \geq \mathcal{L}'$ if there is a nonzero morphism $\mathcal{L}' \hookrightarrow \mathcal{L}$.*

**Corollary 3.4.** *If $\mathcal{L} \geq \mathcal{L}'$, then* $\deg \mathcal{L} \geq \deg \mathcal{L}'$.

*Proof.* If $\mathcal{L} \geq \mathcal{L}'$, then $H^0(C, \mathcal{L} \otimes \mathcal{L}'^\vee) \neq 0$. Hence by Corollary 3.3, $\deg(\mathcal{L} \otimes \mathcal{L}'^\vee) \geq 0$ so that $\deg(\mathcal{L}) \geq \deg(\mathcal{L}')$. □

## 3.3 Grothendieck group

**Definition 3.4.** *Let $X$ be a variety. Define the Grothendieck group $K(X)$ (resp. $K_0(X)$) to be the quotient group of the free group spanned by the coherent sheaves (resp. locally free sheaves of finite rank) by the subgroup spanned by elements of the form $\mathcal{F} - \mathcal{F}' - \mathcal{F}''$ for short exact sequence* $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$.

**Remark 3.4.** *There is a natural homomorphism $i : K_0(X) \to K(X)$ sending $\overline{\mathcal{E}}$ to $\overline{\mathcal{E}}$.*

**Lemma 3.6.** *Let $X$ be a regular projective curve, $\mathcal{F} \in$* $\mathbf{Coh}(X)$. *Then there is an exact sequence $0 \to \mathcal{E}_1 \to \mathcal{E}_2 \to \mathcal{F} \to X$ with $\mathcal{E}_1$ and $\mathcal{E}_2$ locally free of finite rank.*

*Proof.* Since $X$ is projective, there is a very ample line bundle $\mathcal{O}_X(1)$. Then by Corollary 1.1, there is a surjection $\mathcal{O}_X(m)^{\oplus r} \twoheadrightarrow \mathcal{F}$. Set $\mathcal{E}_1 = \mathcal{O}_X(m)^{\oplus r}$, consider kernel $\mathcal{E}_2$. Want to show that $\mathcal{E}_2$ is locally free. In fact, since for all closed point $x \in X$, $\mathcal{O}_{X,x}$ is a DVR and hence PID. As $\mathcal{E}_{2,x}$ is a submodule of $\mathcal{E}_{1,x}$ which is free, we get $\mathcal{E}_{2,x}$ also free. Hence $\mathcal{E}_2$ is locally free of finite rank. □

**Proposition 3.4.** *Let $X$ be a regular projective curve. Then $i : K_0(X) \to K(X)$ is a group isomorphism.*

*Proof.* First to show that $i$ is surjective. For all $\mathcal{F} \in \mathbf{Coh}(X)$, by Lemma 3.5, there is a short exact sequence

$$0 \longrightarrow \mathcal{E}_2 \xrightarrow{\varphi_2} \mathcal{E}_1 \xrightarrow{\varphi_1} \mathcal{F} \longrightarrow 0 \tag{55}$$

with $\mathcal{E}_1$ and $\mathcal{E}_2$ locally free of finite rank. Then we get $\overline{\mathcal{F}} = \overline{\mathcal{E}_1 - \mathcal{E}_2}$ so that $i$ is surjective.

Conversely, it is natural to define $j : K(X) \to K_0(X)$ sending $\overline{\mathcal{F}}$ to $\overline{\mathcal{E}_1 - \mathcal{E}_2}$. Given another exact sequence

$$0 \longrightarrow \mathcal{E}_2' \xrightarrow{\psi_2} \mathcal{E}_1' \xrightarrow{\psi_1} \mathcal{F} \longrightarrow 0 \tag{56}$$

By universal property of resolution, there is a chain map

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{E}_2 \arrow[r, "\varphi_2"] & \mathcal{E}_1 \arrow[r, "\varphi_1"] & \mathcal{F} \arrow[r] & 0 \\
0 \arrow[r] & \mathcal{E}_2' \arrow[r, "\psi_2"] \arrow[u, "\alpha_2"] & \mathcal{E}_1' \arrow[r, "\psi_1"] \arrow[u, "\alpha_1"] & \mathcal{F} \arrow[r] \arrow[u, equal] & 0
\end{tikzcd} \tag{57}$$

Then we can construct a commutative diagram with exact rows and exact columns except the medium column

$$\begin{tikzcd}
 & 0 & 0 & 0 & \\
0 \arrow[r] & \mathcal{E}_2 \arrow[r, "\varphi_2"] \arrow[u] & \mathcal{E}_1 \arrow[r, "\varphi_1"] \arrow[u] & \mathcal{F} \arrow[r] \arrow[u] & 0 \\
0 \arrow[r] & \mathcal{E}_2 \arrow[r] \arrow[u, equal] & \mathcal{E}_2 \oplus \mathcal{E}_1' \arrow[r] \arrow[u, "\varphi_2+\alpha_1"] & \mathcal{E}_1' \arrow[r] \arrow[u, "\psi_1"] & 0 \\
 & 0 \arrow[r] \arrow[u] & \mathcal{E}_2' \arrow[r, equal] \arrow[u, "-\alpha_2\oplus\psi_2"] & \mathcal{E}_2' \arrow[r] \arrow[u, "\psi_2"] & 0 \\
 & & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd} \tag{58}$$

By diagram chasing, the medium column also exact. Hence $\overline{\mathcal{E}_2 + \mathcal{E}_1' - \mathcal{E}_1 - \mathcal{E}_2'} = 0$ in $K_0(X)$ so that $j$ is well defined. Left to show that $j$ is a group homomorphism. For exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$, by Horseshoe Lemma, we can construct a locally free resolution of $\mathcal{F}$ as direct sum of thoes resolutions of $\mathcal{F}'$ and $\mathcal{F}''$. Thus $j$ is indeed a group homomorphism. Note that $i \circ j = \mathrm{id}_{K(X)}$ and $j \circ i = \mathrm{id}_{K_0(X)}$, we conclude that $i$ is isomorphic. □

**Corollary 3.5.** *Let $X$ be a regular projective curve. Then $K(X)$ admits a ring structure with multiplication induced by tensor product of locally free sheaves, callled the Grothendieck ring of $X$.*

**Corollary 3.6.** *Let $X$ be a regular projective curve. Then $K(X) \xrightarrow{j} K_0(X) \xrightarrow{\det} \mathrm{Pic}(X)$ sending $\overline{\mathcal{L}_1 + \mathcal{L}_2}$ to $\det \mathcal{L}_1 \otimes \det \mathcal{L}_2$ is a group homomorphism.*

**Theorem 3.2.** *Let $X$ be a regular projective curve, $\mathcal{F} \in \mathbf{Coh}(X)$. Then $\deg \mathcal{F} = \deg(\det \mathcal{F})$.*

*Proof.* Note that $\deg : K(X) \to \mathbb{Z}$ is a group homomorphism, suffices to check for generators of $K(X)$ i.e. locally free sheaves of finite rank. For each vector bundle $\mathcal{E}$ on $X$, we have a filtration of $\mathcal{E}$ by line bundle

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E} \tag{59}$$

such that $\mathcal{E}_{i+1}/\mathcal{E}_i = \mathcal{L}_i$ are line bundles. Then $\det \mathcal{E}_{i+1} = \det \mathcal{E}_i \otimes \mathcal{L}_i$ for all $i$. Combining together, we get $\det \mathcal{E} = \bigotimes_{i=0}^{r-1} \mathcal{L}_i$. Hence $\deg(\det \mathcal{E}) = \deg(\bigotimes_{i=0}^{r-1} \mathcal{L}_i) = \sum_{i=0}^{r-1} \deg \mathcal{L}_i$.

On the other hand, we also have that $\deg \mathcal{E}_{i+1} = \deg \mathcal{E}_i + \deg \mathcal{L}_i$ for all $i$. Summing together, we get $\deg \mathcal{E} = \sum_{i=0}^{r-1} \deg \mathcal{L}_i = \deg(\det \mathcal{E})$, done! □

**Lemma 3.7.** *Let $X$ be an arbitrary scheme, $D, E$ effective Cartier divisors with disjoint supports. Then there is an exact sequence*

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_X(D) \oplus \mathcal{O}_X(E) \longrightarrow \mathcal{O}_X(D+E) \longrightarrow 0 \tag{60}$$

*Proof.* Construct the maps locally. Take affine cover $U_i$ of $X$ trivializing both $\mathcal{O}_X(D)$ and $\mathcal{O}_X(E)$. Assume $D\big|_{U_i}$ and $E\big|_{U_i}$ are defined by regular sections $f_i$ and $g_i$ respectively. Since $\mathrm{Supp}\, D$ and $\mathrm{Supp}\, E$ are disjoint, $f_i$ and $g_i$ are coprime.

Define $\mathcal{O}_{U_i} \to \mathcal{O}_X(D)\big|_{U_i} \oplus \mathcal{O}_X(E)\big|_{U_i}$ sending 1 to $(f_i, g_i)$, and $\mathcal{O}_X(D)\big|_{U_i} \oplus \mathcal{O}_X(E)\big|_{U_i} \to \mathcal{O}_X(D+E)\big|_{U_i}$ sending $(a, b)$ to $g_i a - f_i b$. Obviously these maps give a short exact sequence and glue up to our desired exact sequence. □

**Proposition 3.5.** *Let $X$ be a regular projective curve. Then $K(X) \cong \mathrm{Pic}(X) \oplus \mathbb{Z}$.*

*Proof.* Suffices to show $K_0(X) \cong \mathrm{Pic}(X) \oplus \mathbb{Z}$. We may consider the rank map $\mathrm{rank} : K_0(X) \to \mathbb{Z}$. Denote the kernel by $A(X)$. Then it is clear that $K_0(X) \cong A(X) \cong \mathbb{Z}$.

Left to show that $A(X) \cong \mathrm{Pic}(X)$. Construct map $\alpha : \mathrm{Pic}(X) \to A(X)$ sending $\mathcal{L}$ sending $\mathcal{L}$ to $\overline{\mathcal{L} - \mathcal{O}_X}$. Check that $\alpha$ is a group morphism. For divisors $D$ and $E$, want to show $\alpha(\mathcal{O}_X(D+E)) = \alpha(\mathcal{O}_X(D)) + \alpha(\mathcal{O}_X(E))$.

In fact, as $X$ is projective, for all different points $x$ and $y$ in $X$, $x - y$ is a principle divisor. Hence we can pick effective divisors $D_1$, $D_2$, $E_1$ and $E_2$ pairwise disjoint such that $D = D_1 - D_2$ and $E = E_1 - E_2$. Then by Lemma 3.7, there is an exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D_2 - E_2) \longrightarrow \mathcal{O}_X(D_1 + E_1 - D_2 - E_2) \oplus \mathcal{O}_X \longrightarrow \mathcal{O}_X(D_1 + E_1) \longrightarrow 0 \tag{61}$$

so that $\overline{\mathcal{O}_X(D+E)} = \overline{\mathcal{O}_X(-D_2 - E_2) + \mathcal{O}_X(D_1 + E_1) - \mathcal{O}_X}$. Similarly, we have

$$\begin{aligned} \overline{\mathcal{O}_X(D)} &= \overline{\mathcal{O}_X(-D_2) + \mathcal{O}_X(D_1) - \mathcal{O}_X} \\ \overline{\mathcal{O}_X(E)} &= \overline{\mathcal{O}_X(-E_2) + \mathcal{O}_X(E_1) - \mathcal{O}_X} \\ \overline{\mathcal{O}_X(D_1 + E_1)} &= \overline{\mathcal{O}_X(D_1) + \mathcal{O}_X(E_1) - \mathcal{O}_X} \\ \overline{\mathcal{O}_X(-D_2 - E_2)} &= \overline{\mathcal{O}_X(-D_2) + \mathcal{O}_X(-E_2) - \mathcal{O}_X} \end{aligned} \tag{62}$$

Combining these equation, we get

$$\begin{aligned}\overline{\mathcal{O}_X(D+E)} &= \overline{\mathcal{O}_X(-D_2-E_2)+\mathcal{O}_X(D_1+E_1)-\mathcal{O}_X} \\ &= \overline{\mathcal{O}_X(-D_2)+\mathcal{O}_X(-E_2)-\mathcal{O}_X+\mathcal{O}_X(D_1)+\mathcal{O}_X(E_1)-\mathcal{O}_X-\mathcal{O}_X} \\ &= \overline{\mathcal{O}_X(D)+\mathcal{O}_X(E)-\mathcal{O}_X}\end{aligned} \tag{63}$$

Thus $\alpha$ is a group homomorphism. And $\det\circ\alpha(\mathcal{L}) = \mathcal{L}\otimes\mathcal{O}_X = \mathcal{L}$ so that $\alpha$ is injective. On the other hand, $A(X)$ is spanned by elements of the form $\overline{\mathcal{E}-r\mathcal{O}_X}$, where $\mathcal{E}$ is of rank $r$. In fact, by same argument as proof 3.2, we know $\overline{\mathcal{E}} = \sum_{i=0}^{r-1}\overline{\mathcal{L}_i}$ in $K_0(X)$. Hence $\overline{\mathcal{E}-r\mathcal{O}_X} = \sum_{i=0}^{r-1}\overline{\mathcal{L}_i-\mathcal{O}_X} = \alpha(\bigotimes_{i=0}^{r-1}\mathcal{L}_i)$. Conclude that $\alpha$ is isomorphic. □

**Lemma 3.8.** *Let $X$ be a regular projective curve, $\mathcal{E}$ and $\mathcal{E}'$ vector bundles on $X$. Then $\deg(\mathcal{E}\otimes\mathcal{E}') = \operatorname{rank}\mathcal{E}\cdot\deg\mathcal{E}' + \operatorname{rank}\mathcal{E}'\cdot\deg\mathcal{E}$.*

*Proof.* Take filtration $(\mathcal{L}'_1,\cdots,\mathcal{L}'_{r'})$ of $\mathcal{E}'$. Consider exact sequence $0\to\mathcal{E}'_i\otimes\mathcal{E}\to\mathcal{E}'_{i+1}\otimes\mathcal{E}\to\mathcal{L}'_{i+1}\otimes\mathcal{E}\to 0$. Hence $\deg(\mathcal{E}\otimes\mathcal{E}') = \sum\deg(\mathcal{L}'_i\otimes\mathcal{E})$. Suffice to show that $\deg(\mathcal{E}\otimes\mathcal{L}) = \deg\mathcal{E} + r\cdot\deg\mathcal{L}$ for all line bundle $\mathcal{L}$. Take filtration $(\mathcal{L}_1,\cdots,\mathcal{L}_r)$ of $\mathcal{E}$. Similarly, we get $\deg(\mathcal{E}\otimes\mathcal{L}) = \sum\deg(\mathcal{L}_i\otimes\mathcal{L}) = \sum(\deg\mathcal{L}_i+\deg\mathcal{L}) = \deg\mathcal{E}+r\cdot\deg\mathcal{L}$. □

**Corollary 3.7.** *Let $X$ be a regular projective curve, $\mathcal{E}$ vector bundle on $X$. Then $\deg\mathcal{E}^\vee = -\deg\mathcal{E}$.*

*Proof.* Assume $\operatorname{rank}\mathcal{E} = r$ and $g(X) = g$. Then we have that

$$\begin{aligned}\deg\mathcal{E}^\vee &= \chi(\mathcal{E}^\vee) - r(1-g) \\ &= -\chi(\mathcal{E}\otimes\omega_X) - r(1-g) \\ &= -\deg(\mathcal{E}\otimes\omega_X) - 2r(1-g) \\ &= -\deg\mathcal{E} - r\cdot\deg\omega_X - 2r(1-g) \\ &= -\deg\mathcal{E}\end{aligned} \tag{64}$$

□

## 3.4 Riemann-Roch for curves

Let $X$ be a smooth projective curve, $D = \sum_i n_iP_i$ a Weil divisor on $X$. Set $\ell(D) := \dim_k(\Gamma(X,\mathcal{O}_X(D))) = \dim_k(H^0(X,\mathcal{O}_X(D)))$. Since $X$ is projective and $\mathcal{O}_X(D)$ is coherent, get $\ell(D) < \infty$.

**Lemma 3.9.** *Let $X$ be a smooth projective curve, $D$ a Weil divisor on $X$. If $\ell(D)\neq 0$, then $\deg(D)\geq 0$. If $\ell(D)\neq 0$ and $\deg(D) = 0$, then $D\sim 0$ i.e. $\mathcal{O}_X(D)\cong\mathcal{O}_X$.*

*Proof.* If $\ell(D)\neq 0$, then $\mathcal{O}_X(D)\neq 0$ and nonzero section $s$ gives an effective divisor $\operatorname{div}(s)\sim D$ so that $\deg(D) = \deg(\operatorname{div}(s))\geq 0$. Since $\operatorname{div}(s)$ is effective, if moreover $\deg(D) = 0$, then $s$ is invertible so that $D\sim 0$. □

**Definition 3.5** (***Canonical Divisors***)**.** *A canonical divisor $K_X$ is any Cartier divisor corresponding to the canonical sheaf $\omega_X$. Set $g(X)(= p_g(X)) := \dim_k(\Gamma(X,\omega_X))$ to be the geometric genus of $X$.*

**Theorem 3.3** (***Riemann-Roch Theorem for Curve Case***)**.** *Let $X$ be a smooth projective curve, $D$ a Weil divisor on $X$. Then $\ell(D) - \ell(K_X - D) = \deg(D) + 1 - g(X)$.*

**Remark 3.5.** *By Serre's duality, $\ell(K_X - D) = \dim_k(H^1(X, \mathcal{O}_X(D)))$. In particular, assuming the theorem, for $D = 0$, we have that $g(X) = \ell(K_X) + 1 - \ell(0)$. Since $\ell(0) = \dim_k(\Gamma(X, \mathcal{O}_x)) = \dim_k(k) = 1$, get $g(x) = \ell(K_X)$, called the arithmetirc genus of $X$.*

*For $\mathcal{F}$ coherent sheaf on $X$, set $\chi(\mathcal{F}) := \sum_i (-1)^i \dim_k(H^i(X, \mathcal{F}))$, called the Euler characteristic of $\mathcal{F}$. By Serre duality, can rewrite Riemann-Roch Theorem as $\chi(\mathcal{O}_X(D)) = \deg(D) + 1 - g(X)$. Note that $1 - g(X) = \chi(\mathcal{O}_X)$, get $\chi(\mathcal{O}_X(D)) - \chi(\mathcal{O}_X) = \deg(D)$.*

*Proof.* By Serre duality, $\ell(K_X - D) = h^1(X, \mathcal{O}_X(D))$ so that $\ell(D) - \ell(K_X - D) = \chi(D)$. Then by definition of degree and Lemma 3.4, $\chi(D) = \deg(D) + 1 - g(X)$, done! □

**Example 3.1.** *Let $X$ be a curve which smooth and proper over field $k$, $D$ a Weil divisor on $X$.*
*(1)If $\deg(D) > 0$, then dimension of the complete linear system of $nD$ is $\dim_k(|nD|) = n\deg(D) - g(X)$ for $n$ large enough, since $\ell(K_X - nD) = 0$ for $n$ large enough.*
*(2)Let $D = K_X$, then $\ell(K_X) - 1 = \deg(K_X) + 1 - g(X)$. Get $\deg(K_X) = 2g(X) - 2$.*
*(3)If $g(X) = 1$, then $\deg(K_X) = 0$ and $\ell(K_X) = g(X) = 1$. Get $K_X \sim 0$.*
*(4)Let $X$ be an elliptic curve i.e. $g(X) = 1$ and the $k$-points of $X$ isn't empty, $P_0 \in X(k)$. Take $Pic^0(X) \subseteq Pic(X)$ to be the subgroup corresponding to divisor classes of degree zero. Then $X(k) \longrightarrow Pic^0(X) \quad P \longmapsto \mathcal{O}_X(P - P_0)$ is bijective inducing a group structure over $X(k)$. Injection is obvious. For surjection, let $D$ be a divisor of degree $0$, then $\deg(D + P_0) = \deg(P_0) = 1$. By Riemann-Roch, $\ell(D+P_0) - \ell(K_X - D - P_0) = 1$. While $\deg(K_X - D - P_0) < 0$, get $\ell(K_X - D - P_0) = 0$ and $\ell(D + P_0) = 1$. Thus there exists effective divisor $P \sim D + P_0$ so that $P \longmapsto \mathcal{O}_X(D)$.*
*(5)If $\deg(D) \geq 2g(X) + 1$, then $\mathcal{O}_X(D)$ is very ample. Thus $\mathcal{O}_X(D)$ is ample if and only if $\deg(D) > 0$.*

# 4 Ample Vector Bundles

## 4.1 Ample line bundles

**Proposition 4.1.** *Let $A$ be a ring, $X$ an $A$-scheme, $Y = \mathrm{Proj}(A[x_0, \cdots, x_d]) = \mathbb{P}^d_A$. Then*
*(1)Let $f : X \longrightarrow Y$ be an $A$-morphism. Then $f^*\mathcal{O}_Y(1)$ is an invertible sheaf on $X$, generated by $d + 1$ global sections.*
*(2)Conversely, if $\mathcal{L}$ is an invertible sheaf on $X$, generated by global sections $s_0, \cdots, s_d$, then there exists unique $A$-morphism $f : X \longrightarrow Y$ such that $\mathcal{L} \cong f^*\mathcal{O}_Y(1)$ and $f^*x_i$ is identified with $s_i$.*

*Proof.* (1): $\mathcal{O}_Y(1)$ is generated by $d+1$ global sections, inducing global sections $s_1, \cdots, s_d$ of $f^*\mathcal{O}_Y(1)$. While for all $x \in X$ and $y = f(x)$, then $(f^*\mathcal{O}_Y(1))_x = \mathcal{O}_Y(1)_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}$. Thus $s_{0x}, \cdots, s_{dx}$ generate $(f^*\mathcal{O}_Y)_x$.

(2): As assumption, X is covered by $\{X_{s_i}\}$. For all i, want to define $f_i : X_{s_i} \longrightarrow D_+(x_i)$. By Hartshorne Chapter II exercise 2.4, it is equivalent to give a ring homomorphism between global sections $A[\frac{x_0}{x_i}, \cdots, \widehat{\frac{x_i}{x_i}}, \cdots, \frac{x_d}{x_i}] \longrightarrow \mathcal{O}_{X_{s_i}}$, which can be defined by mapping $\frac{x_j}{x_i}$ to $\frac{s_j}{s_i}$, where $\frac{s_j}{s_i}$ is the unique element $a \in \mathcal{O}_X(X_{s_i})$ such that $s_j\big|_{X_{s_i}} = as_i\big|_{X_{s_i}}$. It is easy to check that $f_i$ satisfy conditions for gluing lemma. Get $f : X \longrightarrow Y$. $\square$

**Remark 4.1.** *Intuitively, $f$ should be $x \longmapsto (s_0(x) : \cdots : s_d(x))$. In addition, $f$ is a closed immersion if and only if for all i, $X_{s_i}$ is affine and the A-algebra is generated by $\{s_j/s_i\}$.*

**Definition 4.1** (***Immersions***)**.** *Let $f : X \longrightarrow Y$ be a morphism of schemes. We say that f is an immersion if f factors as $X \overset{open\ immersion}{\longrightarrow} Z \overset{closed\ immersion}{\longrightarrow} Y$.*

**Proposition 4.2.** *There are lots of properties of immersions.*
*(1)Closed immersions are immersions.*
*(2)Composition of immersions are immersions.*
*(3)Immersions are stable under base change.*

**Remark 4.2.** *Since (1),(2),(3) are in fact the properties (a),(b),(c) in Hartshorne Chapter II exercise 4.8, by the exercise, we immediately get that (d),(e),(f) also holds for immersion.*

**Definition 4.2** (***Quasi-projective***)**.** *Let A be a ring, X an A-scheme. We say that X is quasi-projective over A if $X \longrightarrow \operatorname{Spec} A$ factors as*

$$\begin{tikzcd} X \arrow[r, "immersion"] \arrow[dr] & \mathbb{P}^d_A \arrow[d] \\ & \operatorname{Spec} A \end{tikzcd}$$

In Midterm24.pdf, there is also a definition of immersion which is a little different from here and we would rename it.

**Definition 4.3** (***Locally Closed Immersions***)**.** *Let $f : X \longrightarrow Y$ be a morphism of schemes. We say that f is a locally closed immersion if f factors as $X \overset{closed\ immersion}{\longrightarrow} Z \overset{open\ immersion}{\longrightarrow} Y$.*

**Proposition 4.3.** *Let $f : X \longrightarrow Y$ be a morphism of schemes. If f is a immersion, then f is a locally closed immersion. On the other hand, if f is a locally closed immersion and suppose that either X is noetherian or f is quasi-compact, then f is an immersion.*

**Definition 4.4** (***Very Ample Invertible Sheaves***)**.** *Let A be a ring, X an A-scheme, $\mathcal{L}$ an invertible sheaf on X. We say that $\mathcal{L}$ is very ample relative to A if there exists A-immersion $f : X \longrightarrow \mathbb{P}^d_A$ such that $\mathcal{L} \cong \mathcal{O}_X(1) = f^*\mathcal{O}_{\mathbb{P}^d_A}(1)$.*

**Remark 4.3.** *$\mathcal{L}$ is very ample if and only if $\mathcal{L}$ can be generated by finitely many global sections such that the associated A-morphism $f : X \longrightarrow \mathbb{P}^d_A$ is an immersion. If moreover $X \longrightarrow \operatorname{Spec} A$ is proper, then f is a closed immersion. Thus X is projective over A if and only if X is proper over A and admits a very ample invertible sheaf.*

**Definition 4.5** (***Ample Invertible Sheaves***)**.** *Let X be a noetherian scheme, $\mathcal{L}$ an invertible sheaf on X. We say that $\mathcal{L}$ is ample if for all coherent sheaf $\mathcal{F}$ on X, there exists $n_0 > 0$ such that for all $n \geq n_0$, $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n$ is generated by global sections.*

**Remark 4.4.** *Serre's Theorem are saying that for $X \longrightarrow \operatorname{Spec} A$ proper with $A$ noetherian, a very ample invertible sheaf $\mathcal{L}$ on $X$ is ample. In fact, can remove proper by reducing to proper case with the following nontrivial fact which is Hartshorne Chapter II exercise 5.15.*

*Fact: Let $X$ be a noetherian scheme, $U \subseteq X$ open subset, $\mathcal{F}$ coherent sheaf on $U$. Then there exists coherent sheaf on $X$ whose restriction to $U$ is just $\mathcal{F}$.*

**Example 4.1.** *(1)Any invertible sheaf on an affine noetherian scheme is ample.*
*(2)If $\mathcal{L}$ is ample, then $\mathcal{L}^r$ is ample for all $r > 0$. Conversely, if $\mathcal{L}^r$ is ample for some $r > 0$, then $\mathcal{L}$ is ample.*
*(3)Let $X = \mathbb{P}^d_A$. Then $\mathcal{O}_X(n)$ is very ample/ample if and only if $n > 0$.*

**Lemma 4.1.** *Let $X$ be a noetherian scheme, $\mathcal{L}$ ample invertible sheaf on $X$. For all $x \in X$, there exists $n > 0$ and $s \in \mathcal{L}^n(X)$ such that $x \in X_s \subseteq X$ and $X_s$ is an affine open subset.*

*Proof.* Let $U \subseteq X$ be an affine open neighbourhood of $x$ such that $\mathcal{L}\big|_U$ is free. Our goal is to find an affine open subset inside $U$ of the form $X_s$ containing $x$. Consider $Z = X \setminus U$ with the reduced scheme structure and the ideal sheaf $\mathcal{I}$ on $Z$. For all $n > 0$, since $\otimes_{\mathcal{O}_X} \mathcal{L}^n$ is exact functor, $I\mathcal{L}^n \subseteq \mathcal{L}^n$ is a subsheaf. As $\mathcal{L}$ is ample, for $n$ large enough, $\mathcal{I}\mathcal{L}^n$ is generated by global sections. In particular, since $x \notin Z$, there exists global section $s \in \mathcal{I}\mathcal{L}^n(X) \subseteq \mathcal{L}^n(X)$ such that $s_x$ generates $(\mathcal{I}\mathcal{L}^n)_x \cong \mathcal{O}_{X,x} \otimes_{\mathcal{O}_{X,x}} \mathcal{L}^n_x \cong \mathcal{L}^n_x$. Thus $x \in X_s$. On the other hand, $X_s \subseteq U$ since if $y \in X_s$, then $(\mathcal{I}\mathcal{L}^n)_u$ contains $s_y$ which generates $\mathcal{L}^n_y$. Thus $X_s = X_s \cap U$ is affine. ☐

**Theorem 4.1.** *Let $A$ be a noetherian ring and $X$ be a scheme of finite type over $A$, $\mathcal{L}$ an invertible sheaf on $X$. If $\mathcal{L}$ is ample, then there exists $r > 0$ such that $\mathcal{L}^r$ is very ample. Conversely, if $\mathcal{L}^r$ very ample for some $r > 0$, then $\mathcal{L}$ is ample.*

*Proof.* Since $X$ is noetherian, we can cover $X$ by finitely many $\{X_{s_i}\}$ as in Lemma 4.1. Can also assume $s_i \in \mathcal{L}^n(X)$ for same $n$ and can even assume that $n = 1$ by replacing $\mathcal{L}$ by $\mathcal{L}^n$. Note that $X$ is of finite type over $A$, $\mathcal{O}_X(X_{s_i})$ is generated as $A$-algebra by finitely many $\{g_{ij}\}$. By Lemma 1.2, there exists $r > 0$ such that $g_{ij} \otimes s_i\big|^r_{X_{s_i}}$ extends to a global section $s_{ij} \in (\mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{L}^r)(X) = \mathcal{L}^r(X)$ for all $i, j$. Now $\{s_i^r\}$ generate $\mathcal{L}^r$ since $\{X_{s_i} = X_{s_i^r}\}$ cover $X$. Thus $\{s_i^r, s_{ij}\}$ also generate $\mathcal{L}^r$.

By Proposition 4.1, there is a morphism $f : X \longrightarrow \operatorname{Proj}(A[x_i, x_{ij}]) = \mathbb{P}$ such that $f^*\mathcal{O}_{\mathbb{P}}(1) \cong \mathcal{L}^n$. Now for all $i$, $\mathcal{O}_{\mathbb{P}}(D_+(x_i)) \longrightarrow \mathcal{O}_X(X_{s_i}) \quad x_{ij}/x_i \longmapsto g_{ij}$ is surjective, inducing by $g_{ij} \otimes s\big|^r_{X_{s_i}}$ extends to $s_{ij}$. Then $f$ is a closed immersion from $X$ to $\cup_i D_+(x_i) \subseteq \mathbb{P}$, since $X_{s_i}$ and $D_+(x_i)$ are both affine. Get $f$ is locally closed immersion. As $X$ is Noetherian, by Proposition 4.3, get $f$ is immersion. Thus $\mathcal{L}^r$ is very ample by remark 4.2.

Conversely, first to show very ampleness implies ampleness. Assume $\mathcal{L}^r$ corresponds to immersion $f : X \to \mathbb{P}$ and $\mathcal{L}^r = f^*\mathcal{O}_{\mathbb{P}}(1)$. Then for all $\mathcal{F} \in \mathbf{Coh}(X)$, $f_*\mathcal{F}$ is a coherent sheaf on $\mathbb{P}$. As $\mathcal{O}_{\mathbb{P}}(1)$ is ample, $f_*\mathcal{F}(n)$ is globally generated for $n$ large enough.

By Projection Formula, $f_*(\mathcal{F} \otimes f^*\mathcal{O}_{\mathbb{P}}(n)) \cong f_*\mathcal{F}(n)$. And there is a commutative diagram for each $x \in X$,

$$\begin{tikzcd} H^0(\mathbb{P}, f_*(\mathcal{F} \otimes f^*\mathcal{O}_{\mathbb{P}}(n))) \otimes \mathcal{O}_{\mathbb{P},f(x)} \arrow[r, two heads] \arrow[d] & f_*(\mathcal{F} \otimes f^*\mathcal{O}_{\mathbb{P}}(n))_{f(x)} \arrow[d, equal] \\ H^0(X, \mathcal{F} \otimes \mathcal{L}^{rn}) \otimes \mathcal{O}_{X,x} \arrow[r, "\alpha"] & (\mathcal{F} \otimes \mathcal{L}^{rn})_x \end{tikzcd} \tag{65}$$

where the right arrow is an equality because $f$ is immersion. Then $\alpha$ is surjective and conclude that $\mathcal{L}^r$ is ample and so is $\mathcal{L}$. $\square$

**Corollary 4.1.** *Let $A$ be a noetherian ring and $X$ be a scheme of finite type over $A$, $\mathcal{L}$ an ample invertible sheaf on $X$. Then $X$ is quasi-projective (resp. projective) over $A$ if and only if $X$ admits an ample invertible sheaf (resp. $X$ admits an ample invertible sheaf and $X$ is proper).*

**Remark 4.5.** *Let $A$ be a noetherian ring and $X$ be a scheme of finite type over $A$, $\mathcal{L}$ an ample invertible sheaf on $X$. If $X$ admits an ample invertible sheaf, then $X$ is separated.*

**Corollary 4.2.** *Let $A$ be a noetherian ring, $X$ proper over* $\operatorname{Spec} A$*, $\mathcal{L}$ ample invertible sheaf on $X$. Then for all $\mathcal{F}$ coherent sheaf on $X$, there exists $n_0$ relative to $\mathcal{F}$ such that $H^i(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n) = 0$ for all $i > 0$ and $n \geq n_0$.*

**Reason 4.1.** *Note that there exists $m > 0$ such that $\mathcal{L}^m$ is very ample. Since $X$ is proper, get closed immersion $i : X \hookrightarrow \mathbb{P}^r_A$ such that $\mathcal{L}^m = \mathcal{O}_X(1)$. Apply Theorem 2.7 to $\mathcal{F}, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}, \cdots, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{m-1}$.*

**Theorem 4.2** (***Serre Cohomological Criteria for Ampleness***)**.** *Let $A$ be a noetherian ring, $X$ proper over* $\operatorname{Spec} A$*, $\mathcal{L}$ invertible sheaf on $X$. Then the following conditions are equivalent*
*(1)$\mathcal{L}$ is ample*
*(2)for all $\mathcal{F}$ coherent sheaf on $X$, there exists $n_0$ relative to $\mathcal{F}$ such that $H^i(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n) = 0$ for all $i > 0$ and $n \geq n_0$.*
*(3)for all $\mathcal{F}$ coherent sheaf on $X$, there exists $n_0$ relative to $\mathcal{F}$ such that $H^1(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n) = 0$.*

*Proof.* Step 1. Since by Corollary 4.2, we have (1) $\Rightarrow$ (2), and (2) $\Rightarrow$ (3) is obvious. Only need to prove (3) $\Rightarrow$ (1). Moreover, suffices to prove that for all $\mathcal{F}$ coherent sheaf on $X$ and closed point $x \in X$, there exists open neighbourhood $U_x \ni x$ and $n_0(x) > 0$ such that for all $n \geq n_0(x)$ and $y \in U_x$, the stalk $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)_y$ is generated by global sections of $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n$, since $X$ noetherian, it can be covered by only finitely many $U_x$ so that we can take a uniform upper bound of $n_0(x)$.

Step 2. Claim that it suffices to prove that for all $\mathcal{F}$ coherent sheaf on $X$, closed point $x \in X$ and large enough $n$, there exists open neighbourhood $U_x \ni x$ depending on $n$ such that for all $y \in U_x$, the stalk $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n)_y$ is generated by global sections of $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n$. Assume we have proved this. Apply it to $\mathcal{F} = \mathcal{O}_X$, there exists large enough $n_1$ and $V \ni x$ such that for all $y \in V$, $(\mathcal{L}^{n_1})_y$ is generated by global sections. Then apply this to $\mathcal{F}$, there exists large enough $n_0$ and open neighbourhoods $U_0, U_1, \cdots, U_{n_1-1}$ of $x$ such that for all $i$ and $y \in U_i$,

$(\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^{n+i})_y$ is generated by global sections. Take $U_x = V\cap U_0\cap\cdots\cap U_{n_1-1}$. For all $n\geq n_0$ and $y\in U_x$, assume $n=n_0+mn_1+i$, then $(\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n)_y = (\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^{n_0+i})_y\otimes_{\mathcal{O}_{X,y}}(\mathcal{L}^{n_1})^m_y$ is generated by global sections.

Step 3. Let $x\in X$ be a closed point, $\mathcal{I}_x$ the ideal sheaf of $\{x\}\xhookrightarrow{i} X$, where $\{x\}$ has the same scheme structure as $\mathrm{Spec}(k(x))$ with structure sheaf denoted by $k(x)$ for convenience. We have short exact sequence

$$0\longrightarrow \mathcal{I}_X\longrightarrow \mathcal{O}_x\longrightarrow i_*k(x)\longrightarrow 0 \tag{66}$$

Tensor by $\mathcal{F}$, get

$$\mathcal{I}_x\mathcal{F}\longrightarrow \mathcal{F}\longrightarrow i_*k(x)\otimes_{\mathcal{O}_X}\mathcal{F}\longrightarrow 0 \tag{67}$$

Tensor by $\mathcal{L}^r$, get

$$\mathcal{I}_x\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n\longrightarrow \mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n\longrightarrow i_*k(x)\otimes_{\mathcal{O}_X}\mathcal{F}\otimes_{\mathcal{O}_x}\mathcal{L}^n\longrightarrow 0 \tag{68}$$

Note that, by hypothesis, $H^1(X,\mathcal{I}_X\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n)=0$ for all $n\geq n_0$. So $\Gamma(X,\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n)\longrightarrow \Gamma(X,i_*k(x)\otimes_{\mathcal{O}_X}\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n)$ is surjective for all $n\geq n_0$. Note that $i_*k(x)\otimes_{\mathcal{O}_X}\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n$ is only supported at $x$, we have that $\Gamma(X,i_*k(x)\otimes_{\mathcal{O}_X}\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n) = (i_*k(x)\otimes_{\mathcal{O}_X}\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n)_x = k(x)\otimes_{\mathcal{O}_{X,x}}(\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n)_x$. Denote $(\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n)_x$ by $M$. As $M/\mathfrak{m}_xM$ is generated by global sections, by Nakayama's Lemma, $M$ is generated by global sections.

As $\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n$ is coherent, take an affine open neighbourhood $U=\mathrm{Spec}\,A$ of $x$ so that $(\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n)\big|_U = \widetilde{N}$ for some finitely generated $A$-module. Assume $x$ corresponds to maximal ideal $\mathfrak{m}$ and $N$ is generated by $m_1,\cdots,m_l$. Since $N_\mathfrak{m}$ is generated by global sections, $\frac{m_k}{1} = \frac{\sum_j a_{kj}s_{j,x}}{b_k}$, where $a_{kj}\in A$, $b_k\notin\mathfrak{m}$ and $s_j$ are global sections. Then there exists $c_k\notin\mathfrak{m}$ such that $(b_km_k-\sum_j a_{kj}s_{j,x})c_k=0$. Consider $b=\prod_{k=1}^l b_kc_k$, then $\mathfrak{m}\in D(b)$. Obviously, for each $\mathfrak{p}\in D(b)$, $\frac{m_k}{1}$ can be generated by global sections in $N_\mathfrak{p}$. Thus for all $n\geq n_0$, there exists open neighbourhood $U_x\ni x$ depending on $n$ such that for all $y\in U_x$, $(\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{L}^n)_y$ is generated by global sections, done! □

**Remark 4.6.** *In fact, all finiteness results in Theorem 2.7 still hold for $X$ proper over $A$. Proofs can be seen in EGA or Stacks Project.*

**Proposition 4.4.** *Let $X$ be a variety, $\mathcal{L}$ and $\mathcal{M}$ invertible sheaves on $X$. Then*
*(1)If $\mathcal{L}$ and $\mathcal{M}$ are both ample, then $\mathcal{L}\otimes\mathcal{M}$ is ample.*
*(2)If $\mathcal{L}$ is ample, then $\mathcal{L}^m\otimes\mathcal{M}$ is ample for $m$ large enough.*

*Proof.* (1): For all coherent sheaf $\mathcal{F}$, as $\mathcal{L}$ is ample, there exists $n_1$ such that $\mathcal{F}\otimes\mathcal{L}^n$ is globally generated for all $n\geq n_1$. In addition, since $\mathcal{M}$ is ample, there exists $n_2$ such that $\mathcal{L}\otimes\mathcal{M}^n$ is globally generated for all $n\geq n_2$. Set $n_0=\max\{n_1+1,n_2\}$. Then for all $n\geq n_0$, $\mathcal{F}\otimes(\mathcal{L}\otimes\mathcal{M})^n = (\mathcal{F}\otimes\mathcal{L}^{n-1})\otimes(\mathcal{L}\otimes\mathcal{M}^n)$ is globally generated. Thus $\mathcal{L}\otimes\mathcal{M}$ is ample.

(2):Since $\mathcal{L}$ is ample, there exists $m_0$ such that $\mathcal{L}^m\otimes\mathcal{M}$ is globally generated for all $m\geq m_0$. We may take $m\geq m_0+1$. For all coherent sheaf $\mathcal{F}$, there exists $n_0$ such that $\mathcal{F}\otimes\mathcal{L}^n$ is globally generated for all $n\geq n_0$. Then $\mathcal{F}\otimes(\mathcal{L}^m\otimes\mathcal{M})^n = (\mathcal{F}\otimes\mathcal{L}^n)\otimes(\mathcal{L}^{m-1}\otimes\mathcal{M})^n$ is globally generated. Thus $\mathcal{L}^m\otimes\mathcal{M}$ is ample for $m$ large enough. □

**Proposition 4.5.** *Let $f : X \to Y$ be a proper and affine morphism between noetherian schemes, $\mathcal{L}$ ample invertible sheaf on $Y$. Then $f^*\mathcal{L}$ is ample invertible sheaf on $X$.*

*Proof.* Since $f$ is affine, the natural map $f^*f_*\mathcal{F} \twoheadrightarrow \mathcal{F}$ is surjective for all coherent sheaf $\mathcal{F}$, which can be affine locally defined as $M \otimes_A B \twoheadrightarrow M$. In addition, as $f$ is proper, $f_*\mathcal{F}$ is a coherent sheaf on $Y$.

Then by Projection Formula, we know $f_*(\mathcal{F} \otimes f^*\mathcal{L}^n) \cong f_*\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for $n$ large enough. Hence we have a surjection $\mathcal{O}_Y^{\oplus I} \twoheadrightarrow f_*\mathcal{F} \otimes \mathcal{L}^n$. Take pull back, we get $\mathcal{O}_X^{\oplus I} \twoheadrightarrow f^*f_*(\mathcal{F} \otimes f^*\mathcal{L}^n) \twoheadrightarrow \mathcal{F} \otimes f^*\mathcal{L}^n$ for $n$ large enough. Conclude that $f^*\mathcal{L}$ is ample. $\square$

**Remark 4.7.** *When $f$ is not affine, then we cannot take affine cover such that $f^*f_*\mathcal{F}$ is of the form $M \otimes_A B$ because $f_*\mathcal{F}$ locally is not given by $M$ as an $A$-module. So the natural map $f^*f_*\mathcal{F} \twoheadrightarrow \mathcal{F}$ is not surjective.*

**Corollary 4.3.** *Let $X$ be a noetherian scheme, $\mathcal{L}$ ample invertible sheaf on $X$. Assume $Y$ is a locally closed subscheme of $X$. Then $\mathcal{L}\big|_Y$ is ample.*

*Proof.* Since a locally closed immersion is composition of open immersion and closed immersion. It suffices to prove these two cases. When $f : Y \hookrightarrow X$ is open, by Hartshorne Chapter II exercise 5.15, for all $\mathcal{F} \in \mathbf{Coh}(Y)$, we can extend $\mathcal{F}$ to a coherent sheaf $\mathcal{F}'$ on $X$. Hence $\mathcal{F}' \otimes \mathcal{L}^n$ is globally generated for $n$ large enough. While $\mathcal{F}' \otimes \mathcal{L}^n$ and $\mathcal{F} \otimes \mathcal{L}^n\big|_Y$ have same stalks at $y \in Y$, it is clear that $\mathcal{F} \otimes \mathcal{L}^n\big|_Y$ is globally generated so that $\mathcal{L}\big|_Y$ is ample. On the other hand, when $f : Y \to X$ is closed, since $f$ is proper and affine, by Proposition 4.6, $\mathcal{L}\big|_Y$ is ample. $\square$

**Proposition 4.6.** *Let $X$ be complete scheme, $\mathcal{L}$ invertible sheaf on $X$. Then*
*(1)$\mathcal{L}$ is ample on $X$ if and only if $\mathcal{L}\big|_{X^{\mathrm{red}}}$ is ample on $X^{\mathrm{red}}$.*
*(2)$\mathcal{L}$ is ample on $X$ if and only if $\mathcal{L}\big|_{X_i}$ is ample on $X_i$ for all irreducible component $X_i$ of $X$.*

*Proof.* Since $f : X^{\mathrm{red}} \to X$ and $f_i : X_i \hookrightarrow X$ are closed immersions hence finite, by Proposition 4.5, if $\mathcal{L}$ is ample on $X$, then $\mathcal{L}\big|_{X^{\mathrm{red}}}$ is ample on $X^{\mathrm{red}}$ and $\mathcal{L}\big|_{X_i}$ is ample on $X_i$.

For the converse, if $\mathcal{L}\big|_{X^{\mathrm{red}}}$ is ample on $X^{\mathrm{red}}$, consider the following commutative diagram for all $\mathcal{F} \in \mathbf{Coh}(X)$ and $x \in X$

$$\begin{tikzcd} H^0(X, \mathcal{F} \otimes \mathcal{L}^n) \otimes \mathcal{O}_{X,x} \arrow[r, "\alpha"] \arrow[d, two heads] & (\mathcal{F} \otimes \mathcal{L}^n)_x \arrow[d, equal] \\ H^0(X^{\mathrm{red}}, f^*\mathcal{F} \otimes f^*\mathcal{L}^n) \otimes \mathcal{O}_{X^{\mathrm{red}},x} \arrow[r, two heads] & (f^*\mathcal{F} \otimes f^*\mathcal{L}^n)_x \end{tikzcd} \tag{69}$$

Then $\alpha$ is surjective so that $\mathcal{L}$ is ample on $X$. Similarly, if $\mathcal{L}\big|_{X_i}$ is ample on $X_i$ for all irreducible component $X_i$ of $X$, then for all $x \in X$, we can find irreducible component $X_i$ containing $x$. Consider the following commutative diagram for all $\mathcal{F} \in \mathbf{Coh}(X)$

$$\begin{tikzcd} H^0(X, \mathcal{F} \otimes \mathcal{L}^n) \otimes \mathcal{O}_{X,x} \arrow[r, "\alpha"] \arrow[d, two heads] & (\mathcal{F} \otimes \mathcal{L}^n)_x \arrow[d, equal] \\ H^0(X_i, f_i^*\mathcal{F} \otimes f_i^*\mathcal{L}^n) \otimes \mathcal{O}_{X_i,x} \arrow[r, two heads] & (f_i^*\mathcal{F} \otimes f_i^*\mathcal{L}^n)_x \end{tikzcd} \tag{70}$$

Thus $\mathcal{L}$ is ample on $X$. $\square$

## 4.2 Ample vector bundles

**Definition 4.6.** *Let $X$ be a noetherian scheme. A vector bundle $\mathcal{E}$ on $X$ is said to be ample if for all $\mathcal{F} \in \mathbf{Coh}(X)$, the sheaf $\mathcal{F} \otimes S^n\mathcal{E}$ is globally generated for $n$ large enough.*

**Proposition 4.7** (***Local Criteria***)**.** *Let $X$ be a noetherian scheme, $\mathcal{E}$ vector bundle on $X$. Then $\mathcal{E}$ is ample if and only if for all $\mathcal{F} \in \mathbf{Coh}(X)$ and closed point $x \in X$, $H^0(X, \mathcal{F} \otimes S^n\mathcal{E})$ generate $(\mathcal{F} \otimes S^n\mathcal{E})_x$ as $\mathcal{O}_{X,x}$-module.*

*Proof.* First take $n_{1,x}$ such that $H^0(X, S^n\mathcal{E})$ generate $(S^n\mathcal{E})_x$ for all $n \geq n_{1,x}$. Then there is an open neighbourhood $U_{1,x}$ of $x$ such that $S^{n_{1,x}}\mathcal{E}\big|_{U_{1,x}}$ is globally generated. And we can take $n_{2,x}$ large enough such that $H^0(X, \mathcal{F} \otimes S^i\mathcal{E} \otimes S^n\mathcal{E})$ generate $(\mathcal{F} \otimes S^i\mathcal{E} \otimes S^n\mathcal{E})$ for all $n \geq n_{2,x}$ and for all $0 \leq i \leq n_{1,x} - 1$. Then there is an open neighbourhood $U_{2,x}$ of $x$ such that $\mathcal{F} \otimes S^i\mathcal{E} \otimes S^{n_{2,x}}\mathcal{E}\big|_{U_{2,x}}$ are globally generated.

Set $U_x = U_{1,x} \cap U_{2,x}$ and $n_{0,x} = n_{2,x}$. Then for all $n \geq n_{0,x}$, we can write $n = n_{2,x} + r \cdot n_{1,x} + i$, where $r \geq 0$ and $0 \leq i \leq n_{1,x} - 1$. As $\mathcal{F} \otimes S^i\mathcal{E} \otimes S^{n_{2,x}}\mathcal{E} \otimes (S^{n_{1,x}}\mathcal{E})^r\big|_{U_{0,x}}$ is globally generated and there is a surjection

$$\mathcal{F} \otimes S^i\mathcal{E} \otimes S^{n_{2,x}}\mathcal{E} \otimes (S^{n_{1,x}}\mathcal{E})^r \twoheadrightarrow \mathcal{F} \otimes S^n\mathcal{E} \tag{71}$$

so that $\mathcal{F} \otimes S^n\mathcal{E}\big|_{U_x}$ is globally generated. Note that $X$ is quasi-compact so that $U_x$ cover $X$ and there is a finite subcovering $U_{x_1}, \cdots, U_{x_m}$. Take $n_0 = \max\{n_{0,x_1}, \cdots, n_{0,x_m}\}$. It is clear that $\mathcal{F} \otimes S^n\mathcal{E}$ is globally generated for all $n \geq n_0$. Conclude that $\mathcal{E}$ is ample. ☐

**Remark 4.8.** *Here the statement that $S^n\mathcal{E}\big|_{U_{1,x}}$ is globally generated might be a little misunderstood. What we mean is that $H^0(X, S^n\mathcal{E})$ generate $(S^n\mathcal{E})_y$ for all $y \in U_{1,x}$. But sometimes, it can be understood as the inverse image of $S^n\mathcal{E}$ along $U_{1,x} \hookrightarrow X$ is globally generated on $U_{1,x}$. And we cannot conclude $S^n\mathcal{E}$ is globally generated with the latter one.*

**Proposition 4.8.** *Let $X$ be a noetherian scheme, $\mathcal{E}$ ample vector bundle on $X$. If $\mathcal{E}'$ is a vector bundle which is a quotient of $\mathcal{E}$, then $\mathcal{E}'$ is ample.*

**Reason 4.2.** *Note that there is a surjection $S^n\mathcal{E} \twoheadrightarrow S^n\mathcal{E}'$.*

**Proposition 4.9.** *Let $X$ be a noetherian scheme, $\mathcal{E}_1$ and $\mathcal{E}_2$ ample vector bundles on $X$. Then for all $\mathcal{F} \in \mathbf{Coh}(X)$, $\mathcal{F} \otimes S^p\mathcal{E}_1 \otimes S^q\mathcal{E}_2$ is globally generated for $p + q$ large enough.*

*Proof.* Firstly, we can take $n_1$ such that $S^n\mathcal{E}_1$ is globally generated for all $n \geq n_1$ and $n_2$ such that $S^n\mathcal{E}_2$ is globally generated for all $n \geq n_2$. Then take $n_3$ such that $\mathcal{F} \otimes S^i\mathcal{E}_2 \otimes S^n\mathcal{E}_1$ is globally generated for all $n \geq n_3$ and $0 \leq i \leq n_2 - 1$, and $n_4$ such that $\mathcal{F} \otimes S^j\mathcal{E}_1 \otimes S^n\mathcal{E}_2$ is globally generated for all $n \geq n_4$ and $0 \leq j \leq n_1 - 1$.

Put $n_0 = n_3 + n_4$. Then for all $p + q \geq n_0$, we know $p \geq n_3$ or $q \geq n_4$. We may assume that $p \geq n_3$ and write $q = r \cdot n_2 + i$, where $0 \leq i \leq n_2 - 1$. Note that there is a surjection

$$\mathcal{F} \otimes S^p\mathcal{E}_1 \otimes S^{r \cdot n_2}\mathcal{E}_2 \otimes S^i\mathcal{E}_2 \twoheadrightarrow \mathcal{F} \otimes S^p\mathcal{E}_1 \otimes S^q\mathcal{E}_2 \tag{72}$$

And by our assumption, $\mathcal{F} \otimes S^i\mathcal{E}_2 \otimes S^p\mathcal{E}_1$ and $S^{r \cdot n_2}\mathcal{E}_2$ are both globally generated. Conclude that $\mathcal{F} \otimes S^p\mathcal{E}_1 \otimes S^q\mathcal{E}_2$ is globally generated for $p + q$ large enough. ☐

**Corollary 4.4.** *Let $X$ be a noetherian scheme, $\mathcal{E}_1$ and $\mathcal{E}_2$ vector bundles on $X$. Then $\mathcal{E}_1$ and $\mathcal{E}_2$ are ample if and only if $\mathcal{E}_1 \oplus \mathcal{E}_2$ is ample.*

*Proof.* Note that $S^n(\mathcal{E}_1 \oplus \mathcal{E}_2) = \bigoplus_{i=0}^n S^i\mathcal{E}_1 \otimes S^{n-i}\mathcal{E}_2$. If $\mathcal{E}_1$ and $\mathcal{E}_2$ are ample, then by Proposition 4.9, $\mathcal{E}_1 \oplus \mathcal{E}_2$ is ample. For the converse, since $\mathcal{E}_\infty$ and $\mathcal{E}_2$ are both quotient of $\mathcal{E}_1 \oplus \mathcal{E}_2$. By Proposition 4.8, done! □

**Corollary 4.5.** *Let $X$ be a noetherian scheme, $\mathcal{E}_1$ ample vector bundle on $X$. Then for all globally generated vector bundle $\mathcal{E}_2$ on $X$, $\mathcal{E}_1 \otimes \mathcal{E}_2$ is ample.*

*Proof.* Since $\mathcal{E}_2$ is globally generated, there is an surjection

$$\mathcal{O}_X^{\oplus r} \twoheadrightarrow \mathcal{E}_2 \to 0 \tag{73}$$

where $r$ is the dimension of global sections of $\mathcal{E}_2$. Tensoring by $\mathcal{E}_1$, we get

$$\mathcal{E}_1^{\oplus r} \twoheadrightarrow \mathcal{E}_1 \otimes \mathcal{E}_2 \to 0 \tag{74}$$

Then by Corollary 4.4, we know $\mathcal{E}_1^{\oplus r}$ is ample. Hence by Proposition 4.8, $\mathcal{E}_1 \otimes \mathcal{E}_2$ is ample. □

**Proposition 4.10.** *Let $X$ be a noetherian scheme, $\mathcal{E}$ vector bundle on $X$. If $\mathcal{E}$ is ample, then $S^n\mathcal{E}$ is ample for $n$ large enough. Conversely, if $S^n\mathcal{E}$ is ample for some $n > 0$, then $\mathcal{E}$ is ample.*

*Proof.* Firstly note that when $\mathcal{E}$ is ample, $S^n\mathcal{E}$ is globally generated for $n$ large enough. By Corollary 4.5, $\mathcal{E} \otimes S^n\mathcal{E}$ is ample for $n$ large enough. As $S^{n+1}\mathcal{E}$ is a quotient of $\mathcal{E} \otimes S^n\mathcal{E}$, by Proposition 4.8, $S^{n+1}\mathcal{E}$ is ample for $n$ large enough.

Conversely, if $S^n\mathcal{E}$ is ample for some $n > 0$, then for all coherent $\mathcal{F}$, consider $\mathcal{F} \otimes S^i\mathcal{E}$ for $0 \le i \le n-1$. We can take $m_0$ such that $\mathcal{F} \otimes S^i\mathcal{E} \otimes S^m(S^n\mathcal{E})$ is globally generated for all $m \ge m_0$. Note that there is a surjection $S^i\mathcal{E} \otimes S^m(S^n\mathcal{E}) \twoheadrightarrow S^{i+mn}\mathcal{E}$, we know $\mathcal{F} \otimes S^{i+mn}\mathcal{E}$ is globally generated. Thus $\mathcal{F} \otimes S^m\mathcal{E}$ is globally generated for all $m \ge m_0 n$. Conclude that $\mathcal{E}$ is ample. □

**Corollary 4.6.** *Let $X$ be a noetherian scheme, $\mathcal{E}$ vector bundle on $X$, $\mathcal{L}$ ample line bundle on $X$. Then $\mathcal{E}$ is ample if and only if $\mathcal{L}^\vee \otimes S^n\mathcal{E}$ is globally generated for some $n > 0$.*

**Reason 4.3.** *Immediately comes from Proposition 4.10 and Corollary 4.4.*

## 4.3 Serre criteria of ampleness

Given a vector bundle $\mathcal{E}$ on $X$. We claim that $\mathcal{E}$ is ample if and only if $\mathcal{L} := \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is ample line bundle on $\mathbb{P}(E)$, where $\mathbb{P}(\mathcal{E}) \xrightarrow{\pi} X$ is the projective bundle on $X$ associated to $\mathcal{E}$.

**Lemma 4.2.** *Let $X$ be a noetherian scheme. For any coherent sheaf $\mathcal{F}$ on $X$ and $n \ge 0$, we have an isomorphism*

$$\mathcal{F} \otimes S^n\mathcal{E} \xrightarrow{\sim} \pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n) \tag{75}$$

*In addition, $R^i\pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n) = 0$ for all $i > 0$.*

*Proof.* Note that there is a natural map $\mathcal{F} \to \pi_*\pi^*\mathcal{F}$ associated to identity $\pi^*\mathcal{F} \to \pi^*\mathcal{F}$. As $\pi_*\mathcal{L}^n \cong S^n(\mathcal{E})$, we get natural map $\mathcal{F} \otimes^{\text{pre}} S^n(\mathcal{E}) \to \pi_*\pi^*\mathcal{F} \otimes^{\text{pre}} \pi_*\mathcal{L}^n$ between presheaves. By universal property of sheafification, there is a natural map $\varphi : \mathcal{F} \otimes S^n\mathcal{E} \to \pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n)$ such that the following diagram commutes

$$\begin{tikzcd} \mathcal{F} \otimes S^n\mathcal{E} \arrow[r, "\varphi"] & \pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n) \\ \mathcal{F} \otimes^{\text{pre}} S^n(\mathcal{E}) \arrow[u] \arrow[r] & \pi_*\pi^*\mathcal{F} \otimes^{\text{pre}} \pi_*\mathcal{L}^n \arrow[u] \end{tikzcd} \tag{76}$$

Now we can reduce to affine case. Assume $X = \operatorname{Spec} A$ with $A$ noetherian and $\mathcal{E} = \mathcal{O}_X^{\oplus r}$. Then $\mathbb{P}(\mathcal{E}) \cong \mathbb{P}_A^{r-1}$.

Now first to prove the second statement by descending induction on $i$. Note that $R^i\pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n) = H^i(\mathbb{P}(\mathcal{E}), \widetilde{\pi^*\mathcal{F}} \otimes \mathcal{L}^n)$ and $\dim \mathbb{P}(\mathcal{E}) = r-1$. By Grothendieck Vanishing, $R^i\pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n) = 0$ for all $i > r-1$.

Suppose we have shown $R^{i+1}\pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n) = 0$ for all coherent $\mathcal{F}$ on $X$. For all short exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$, apply push forward functor and we get a long exact sequence

$$\cdots \longrightarrow R^i\pi_*(\pi^*\mathcal{F}' \otimes \mathcal{L}^n) \longrightarrow R^i\pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n) \longrightarrow R^i\pi_*(\pi^*\mathcal{F}'' \otimes \mathcal{L}^n) \longrightarrow 0 \tag{77}$$

Hence $R^i\pi_*(\pi^* \cdot \otimes \mathcal{L}^n)$ is a right exact functor on $\mathbf{Coh}(X)$. Since $X$ is noetherian and affine, we can find a short resolution of $\mathcal{F}$

$$\mathcal{O}_X^{\oplus n_1} \longrightarrow \mathcal{O}_X^{\oplus n_0} \longrightarrow \mathcal{F} \longrightarrow 0 \tag{78}$$

Thus it suffices to show $R^i\pi_*(\pi^*\mathcal{O}_X \otimes \mathcal{L}^n) = 0$. Note that $R^i\pi_*(\pi^*\mathcal{O}_X \otimes \mathcal{L}^n) = R^i\pi_*\mathcal{L}^n = H^i(\widetilde{\mathbb{P}(\mathcal{E})}, \mathcal{L}^n)$, which is 0 when $0 < i < r-1$ or $i = r-1$ and $n > -r$ by Theorem 2.6.

For the first statement, when $\mathcal{F} = \mathcal{O}_X$, the natural map exactly is the isomorphism $S^n\mathcal{E} \xrightarrow{\sim} \pi_*\mathcal{L}^n$. For general $\mathcal{F} \in \mathbf{Coh}(X)$, similarly consider the resolution, there is a commutative diagram with exact rows

$$\begin{tikzcd} (S^n\mathcal{E})^{\oplus n_1} \arrow[r] \arrow[d, "\sim"] & (S^n\mathcal{E})^{\oplus n_0} \arrow[r] \arrow[d, "\sim"] & \mathcal{F} \otimes S^n\mathcal{E} \arrow[r] \arrow[d, "\varphi"] & 0 \\ (\pi_*\mathcal{L}^n)^{\oplus n_1} \arrow[r] & (\pi_*\mathcal{L}^n)^{\oplus n_0} \arrow[r] & \pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n) \arrow[r] & 0 \end{tikzcd} \tag{79}$$

Then by 5 Lemma, $\varphi$ is isomorphic. $\square$

**Definition 4.7.** *Let $f : X \to Y$ be a morphism between noetherian schemes, $\mathcal{L}$ invertible sheaf on $X$. We say $\mathcal{L}$ is ample relative to $f$ (or say $f$-ample) if there exists affine open covering $U_i$ of $Y$ such that $\mathcal{L}\big|_{f^{-1}(U_i)}$ is ample on $f^{-1}(U_i)$.*

**Remark 4.9.** *In fact, when $Y$ is affine, by EGA II Corollary 4.6.6, $\mathcal{L}$ is $f$-ample if and only if $\mathcal{L}$ is ample on $X$.*

**Lemma 4.3.** *Let $f : X \to Y$ be a morphism between noetherian schemes, $\mathcal{L}$ $f$-ample invertible sheaf on $X$. Then for all coherent $\mathcal{F}$ on $Y$, the natural map $f^*f_*(\mathcal{F} \otimes \mathcal{L}^n) \to \mathcal{F} \otimes \mathcal{L}^n$ is surjective for $n$ large enough.*

*Proof.* Since the question is local, we may assume that $Y$ is affine. Then $\mathcal{L}$ is ample. Hence $\mathcal{F}\otimes\mathcal{L}^n$ is globally generated for $n$ large enough. Consider the following commutative diagram

$$\begin{tikzcd} H^0(X, f^*f_*(\mathcal{F}\otimes\mathcal{L}^n))\otimes\mathcal{O}_X \arrow[r, two heads] \arrow[d] & H^0(X,\mathcal{F}\otimes\mathcal{L}^n)\mathcal{O}_X \arrow[d, two heads] \\ f^*f_*(\mathcal{F}\otimes\mathcal{L}^n) \arrow[r, "\alpha"] & \mathcal{F}\otimes\mathcal{L}^n \end{tikzcd} \tag{80}$$

Thus the natural map $\alpha$ is surjective. □

**Proposition 4.11.** *Let $X$ be a noetherian scheme, $\mathcal{E}$ vector bundle on $X$ with tautological line bundle $\mathcal{L} := \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$. Then $\mathcal{E}$ is ample on $X$ if and only if $\mathcal{L}$ is ample on $\mathbb{P}(\mathcal{E})$.*

*Proof.* "$\Rightarrow$ ": For coherent $\mathcal{F}$ on $\mathbb{P}(\mathcal{E})$, want to show that $\mathcal{F}\otimes\mathcal{L}^n$ is globally generated. As $\pi_*\mathcal{L} = \mathcal{E}$, by definition, we know $\mathcal{L}$ is $\pi$-ample. Applying Lemma 4.3, $\pi^*\pi_*(\mathcal{F}\otimes\mathcal{L}^n)\to\mathcal{F}\otimes\mathcal{L}^n$ is surjective for $n$ large enough. Hence it suffices to show that $\pi^*\pi_*(\mathcal{F}\otimes\mathcal{L}^n)$ is globally generated.

Denote $\mathcal{G} := \pi_*(\mathcal{F}\otimes\mathcal{L}^{n_1})$, want to show $\pi^*\mathcal{G}$ is globally generated. Again by Lemma 4.3, $\pi^*\pi_*(\pi^*\mathcal{G}\otimes\mathcal{L}^n)\to\pi^*\mathcal{G}\otimes\mathcal{L}^n$ is surjective for $n$ large enough. It suffices to show that $\pi^*\pi_*(\pi^*\mathcal{G}\otimes\mathcal{L}^n)$ is globally generated. Then by Lemma 4.2, $\pi^*\pi_*(\pi^*\mathcal{G}\otimes\mathcal{L}^n)\cong\pi^*(\mathcal{G}\otimes S^n\mathcal{E})$, which is globally generated for $n$ large enough since $\mathcal{E}$ is ample.

"$\Leftarrow$ ": By local criteria, suffices to show $H^0(X,\mathcal{F}\otimes S^n\mathcal{E})$ generate $(\mathcal{F}\otimes S^n\mathcal{E})_x$ for all closed point $x$. By Nakayama's Lemma, it is equivalent to show $H^0(X,\mathcal{F}\otimes S^n\mathcal{E})$ generate $(\mathcal{F}\otimes S^n\mathcal{E})_x\otimes k(x) = (\mathcal{F}\otimes S^n\mathcal{E}\otimes k(x))_x$. Note that $\mathcal{F}\otimes S^n\mathcal{E}\otimes k(x)$ is a skyscraper sheaf, its global sections is same to the stalk at $x$. Hence we only need to show $H^0(X,\mathcal{F}\otimes S^n\mathcal{E})\to H^0(X,\mathcal{F}\otimes S^n\mathcal{E}\otimes k(x))$ is surjective for all closed point $x\in X$.

By Lemma 4.2, $H^0(X,\mathcal{F}\otimes S^n\mathcal{E}) = H^0(\mathbb{P}(\mathcal{E}), p^*\mathcal{F}\otimes\mathcal{L}^n)$ and $H^0(X,\mathcal{F}\otimes S^n\mathcal{E}\otimes k(x)) = H^0(\mathbb{P}(\mathcal{E}), p^*\mathcal{F}\otimes\mathcal{L}^n\otimes p^*k(x))$. Denote $Y := \pi^{-1}(x)$, which is a closed subscheme of $\mathbb{P}(\mathcal{E})$. Consider the following exact sequence

$$0\longrightarrow\mathcal{I}_Y\pi^*\mathcal{F}\longrightarrow\pi^*\mathcal{F}\longrightarrow\mathcal{O}_Y\otimes\pi^*\mathcal{F}\longrightarrow 0 \tag{81}$$

where $\mathcal{I}_Y\pi^*\mathcal{F}$ denotes the image of $\mathcal{I}_Y\otimes\pi^*\mathcal{F}\to\pi^*\mathcal{F}$. Tensoring by $\mathcal{L}^n$, get exact sequence

$$0\longrightarrow\mathcal{I}_Y\pi^*\mathcal{F}\otimes\mathcal{L}^n\longrightarrow\pi^*\mathcal{F}\otimes\mathcal{L}^n\longrightarrow\mathcal{O}_Y\otimes\pi^*\mathcal{F}\otimes\mathcal{L}^n\longrightarrow 0 \tag{82}$$

Note that $\operatorname{Spec} k(x)\to X$ is affine, by cohomology and base change, $\pi^*k(x)\cong\mathcal{O}_Y$. Hence it is equivalent to show that $H^1(\mathbb{P}(\mathcal{E}),\mathcal{I}_Y p^*\mathcal{F}\otimes\mathcal{L}^n)\to H^1(\mathbb{P}(\mathcal{E}),\mathcal{O}_Y\otimes p^*\mathcal{F}\otimes\mathcal{L}^n)$ is injective.

Now since $\mathcal{L}$ is ample, we find $r$ large enough such that $\mathcal{L}^r$ very ample with an immersion $\ell:\mathbb{P}(\mathcal{E})\hookrightarrow\mathbb{P}$ into some projective space. Take schematic image of $\mathbb{P}(\mathcal{E})$ in $\mathbb{P}$, denoted by $\overline{\mathbb{P}(\mathcal{E})}$. Denote $\mathcal{G}_i := \pi^*\mathcal{F}\otimes\mathcal{L}^i$ for all $0\le i\le r-1$. By Hartshorne Chapter II exercise 5.15, we can extend $\mathcal{G}_i$ to a coherent sheaf $\mathcal{G}_i'$ on $\overline{\mathbb{P}(\mathcal{E})}$. Denote $\overline{Y}$ to be the closure of $Y$ in $\overline{\mathbb{P}(\mathcal{E})}$. Then $\mathcal{I}_{\overline{Y}}\mathcal{G}_i'|_{\mathbb{P}(\mathcal{E})} = \mathcal{I}_Y\mathcal{G}_i = \mathcal{I}_Y\pi^*\mathcal{F}\otimes\mathcal{L}^i$.

Now for $Z := \overline{\mathbb{P}(\mathcal{E})}\setminus\mathbb{P}(\mathcal{E})$, by Hartshorne Chapter III exercise 2.3, there is a commutative diagram with exact rows

$$\begin{tikzcd}
H^1(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{I}_{\overline{Y}}\mathcal{G}_i'(m)) \arrow[r] \arrow[d, "\alpha"] & H^1(\mathbb{P}(\mathcal{E}), \mathcal{I}_{\overline{Y}}\mathcal{G}_i'(m)|_{\mathbb{P}(\mathcal{E})}) \arrow[r] \arrow[d, "\beta"] & H^2_Z(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{I}_{\overline{Y}}\mathcal{G}_i'(m)) \arrow[r] \arrow[d, "\gamma"] & \cdots \\
H^1(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{G}_i'(m)) \arrow[r] & H^1(\mathbb{P}(\mathcal{E}), \mathcal{G}_i'(m)|_{\mathbb{P}(\mathcal{E})}) \arrow[r] & H^2_Z(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{H}_i'(m)) \arrow[r] & \cdots
\end{tikzcd} \tag{83}$$

As $\overline{\mathbb{P}(\mathcal{E})}$ is proper, by Serre Theorem, $H^1(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{I}_{\overline{Y}}\mathcal{G}_i'(m))$ and $H^1(\overline{\mathbb{P}(\mathcal{E})}, \mathcal{G}_i'(m))$ are both zero for $m$ large enough. On the other side, since $Y$ is the fiber at $x$, $Y$ is proper so that $\overline{Y} = Y$ in $\mathbb{P}$. Thus $\overline{Y} \cap Z = \varnothing$ and $\mathcal{I}_{\overline{Y}} \cong \overline{\mathbb{P}(\mathcal{E})}$ in a neighbourhood of $Z$. Again by Hartshorne Chapter III exercise 2.3, we may replace $\overline{\mathbb{P}(\mathcal{E})}$ by this neighbourhood of $Z$, which tells us that $\gamma$ is isomorphic. Conclude that $\beta$ is injective for $m$ large enough and we are done! $\square$

**Proposition 4.12** (***Hartshorne's Cohomological Criteria for Ampleness***)**.** *Let $X$ be a noetherian scheme, $\mathcal{E}$ vector bundle on $X$. If for all coherent $\mathcal{F}$, we have $H^i(X, \mathcal{F} \otimes S^n\mathcal{E}) = 0$ for all $i > 0$ and $n$ large enough. Then $\mathcal{E}$ is ample.*

*Conversely, if $\mathcal{E}$ is ample and $X$ is proper, then for all coherent $\mathcal{F}$, we have $H^i(X, \mathcal{F} \otimes S^n\mathcal{E}) = 0$ for all $i > 0$ and $n$ large enough.*

*Proof.* "$\Rightarrow$": Similar to above argument, it suffices to show that $H^0(X, \mathcal{F} \otimes S^n\mathcal{E}) \to H^0(X, \mathcal{F} \otimes S^n\mathcal{E} \otimes k(x))$ is surjective for all closed point $x \in X$ and large enough $n$. Consider exact sequence

$$0 \to \mathcal{I}_{\{x\}}\mathcal{F} \otimes S^n\mathcal{E} \to \mathcal{F} \otimes S^n\mathcal{E} \to \mathcal{F} \otimes S^n\mathcal{E} \otimes k(x) \to 0 \tag{84}$$

Note that $\mathcal{I}_{\{x\}}\mathcal{F} \otimes S^n\mathcal{E}$ is coherent, we have $H^1(X, \mathcal{I}_{\{x\}}\mathcal{F} \otimes S^n\mathcal{E}) = 0$ for $n$ large enough, done!

"$\Leftarrow$": By Proposition 4.11, the tautological line bundle $\mathcal{L}$ is ample. As $X$ is proper, by Serre Theorem, for all coherent $\mathcal{F}$ on $X$, $H^i(\mathbb{P}(\mathcal{E}), \pi^*\mathcal{F} \otimes \mathcal{L}^n) = 0$ for all $i > 0$ and $n$ large enough. By Lemma 4.2, $R^i\pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n) = 0$. Note that $\Gamma(\mathbb{P}(\mathcal{E}), \cdot) = \Gamma(X, \cdot) \circ \pi_*$. By Grothendieck Spectral Sequence Theorem, we get

$$E_2^{pq} = H^p(X, R^q\pi_*(p^*\mathcal{F} \otimes \mathcal{L}^n)) \Rightarrow H^{p+q}(\mathbb{P}(\mathcal{E}), p^*\mathcal{F} \otimes \mathcal{L}^n) \tag{85}$$

Hence we get

$$\begin{aligned} H^i(X, \mathcal{F} \otimes S^n\mathcal{E}) &\cong H^i(X, \pi_*(\pi^*\mathcal{F} \otimes \mathcal{L}^n)) \\ &\cong H^i(\mathbb{P}(\mathcal{E}), \pi^*\mathcal{F} \otimes \mathcal{L}^n) \\ &= 0, \text{ for all } i > 0 \text{ and } n \text{ large enough} \end{aligned} \tag{86}$$

$\square$

**Corollary 4.7.** *Let $X$ be a noetherian proper scheme. For short exact sequence of vector bundles on $X$, $0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{E}'' \to 0$, if $\mathcal{E}'$ and $\mathcal{E}''$ are both ample, then $\mathcal{E}$ is ample.*

*Proof.* First by Corollary 4.5, we know $\mathcal{E}' \oplus \mathcal{E}'$ is ample. Hence by previous proposition, for all coherent $\mathcal{F}$, $H^{X, \mathcal{F} \otimes S^p\mathcal{E}' \otimes S^q\mathcal{E}''} = 0$ for all $i > 0$ and $p + q$ large enough. By Hartshorne Chapter II exercise 5.16, there is a locally free filtration

$$S^r\mathcal{E} = E^0 \supseteq E^1 \supseteq \cdots \supseteq E^r \supseteq E^{r+1} = 0 \tag{87}$$

such that $E^j/E^{j+1} \cong S^j\mathcal{E}' \otimes S^{r-j}\mathcal{E}''$. Note that a short exact sequence of locally free sheaves is still exact after tensored by any $\mathcal{O}_X$-module. One may check it locally. Then we get long exact sequence of cohomology

$$\cdots \to H^0(X, \mathcal{F} \otimes E^j) \to H^0(X, \mathcal{F} \otimes S^j\mathcal{E}' \otimes S^{r-j}\mathcal{E}'') \to H^1(X, \mathcal{F} \otimes E^{j+1}) \to \cdots \tag{88}$$

Since $H^i(X, \mathcal{F} \otimes S^j\mathcal{E}' \otimes S^{r-j}\mathcal{E}'') = 0$ for all $i > 0$ and $r$ large enough, we get that $H^i(X, \mathcal{F} \otimes E^j) = H^i(X, \mathcal{F} \otimes E^{j+1})$ for all $i > 1$. As $E^r = S^r\mathcal{E}'$, we get $H^i(X, \mathcal{F} \otimes S^r\mathcal{E}) = H^i(X, \mathcal{F} \otimes S^r\mathcal{E}') = 0$ for all $i > 2$ and $r$ large enough. For $i = 1$ case, note that there is an exact sequence

$$0 = H^1(X, \mathcal{F} \otimes E^r) \to H^1(X, \mathcal{F} \otimes E^{r-1}) \to H^1(X, \mathcal{F} \otimes S^{r-1}\mathcal{E}' \otimes \mathcal{E}'') = 0 \tag{89}$$

Hence $H^1(X, \mathcal{F} \otimes E^{r-1}) = 0$. Preceding step by step, we get $H^1(X, \mathcal{F} \otimes S^r\mathcal{E}) = 0$. Conclude that $\mathcal{E}$ is ample. □

**Proposition 4.13.** *Let $f : X \to Y$ be a proper and affine morphism between noetherian schemes, $\mathcal{E}$ ample vector bundle on $Y$. Then $\mathcal{E}\big|_X$ is ample on $X$.*

*Proof.* Since $f$ is affine, the natural map $f^*f_*\mathcal{F} \twoheadrightarrow \mathcal{F}$ is surjective for all coherent sheaf $\mathcal{F}$, which can be affine locally defined as $M \otimes_A B \twoheadrightarrow M$. In addition, as $f$ is proper, $f_*\mathcal{F}$ is a coherent sheaf on $Y$.

Then by Projection Formula, we know $f_*(\mathcal{F} \otimes S^n(f^*\mathcal{E})) \cong f_*\mathcal{F} \otimes S^n\mathcal{E}$ is globally generated for $n$ large enough. Hence we have a surjection $\mathcal{O}_Y^{\oplus I} \twoheadrightarrow f_*\mathcal{F} \otimes S^n\mathcal{E}$. Take pull back, we get $\mathcal{O}_X^{\oplus I} \twoheadrightarrow f^*f_*(\mathcal{F} \otimes S^n(f^*\mathcal{E})) \twoheadrightarrow \mathcal{F} \otimes S^n(f^*\mathcal{E})$ for $n$ large enough. Conclude that $f^*\mathcal{E}$ is ample. □

**Corollary 4.8.** *Let $X$ be a noetherian scheme, $\mathcal{E}$ ample vector bundle on $X$. Assume $Y$ is a locally closed subscheme of $X$. Then $\mathcal{E}\big|_Y$ is ample on $Y$.*

*Proof.* Since a locally closed immersion is composition of open immersion and closed immersion. It suffices to prove these two cases. When $f : Y \hookrightarrow X$ is open, by Hartshorne Chapter II exercise 5.15, for all $\mathcal{F} \in \mathbf{Coh}(Y)$, we can extend $\mathcal{F}$ to a coherent sheaf $\mathcal{F}'$ on $X$. Hence $\mathcal{F}' \otimes S^n\mathcal{E}$ is globally generated for $n$ large enough. While $\mathcal{F}' \otimes S^n\mathcal{E}$ and $\mathcal{F} \otimes S^n\mathcal{E}\big|_Y$ have same stalks at $y \in Y$, it is clear that $\mathcal{F} \otimes S^n\mathcal{E}\big|_Y$ is globally generated so that $\mathcal{E}\big|_Y$ is ample. On the other hand, when $f : Y \to X$ is closed, since $f$ is proper and affine, by Proposition 4.13, $\mathcal{E}\big|_Y$ is ample. □

**Proposition 4.14.** *Let $X$ be a noetherian scheme, $\mathcal{E}$ vector bundle on $X$. Then $\mathcal{E}$ is ample on $X$ if $\mathcal{E}\big|_{X^{\mathrm{red}}}$ is ample on $X^{\mathrm{red}}$*

**Reason 4.4.** *Note that $\mathbb{P}(\mathcal{E}\big|_{X^{\mathrm{red}}}) \cong \mathbb{P}(\mathcal{E})^{\mathrm{red}}$. Then this result immediately comes from Proposition 4.11.*

# 5 Vector Bundles on Curves

Let $C$ be a smooth projective curve of genus $g$.

## 5.1 Ample vector bundles on curves

**Lemma 5.1.** *For all invertible sheaf $\mathcal{L}$ on $C$ with $\deg \mathcal{L} \geq 2g$, $\mathcal{L}$ is globally generated.*

*Proof.* Consider the following short exact sequence for all closed point $x \in C$

$$0 \longrightarrow \mathcal{L} \otimes \mathcal{O}_C(-x) \longrightarrow \mathcal{L} \longrightarrow \mathcal{L} \otimes \mathcal{O}_x \longrightarrow 0 \tag{90}$$

Note that by Serre duality, $\dim H^1(C, \mathcal{L} \otimes \mathcal{O}_C(-x)) = \dim H^0(C, \mathcal{L}^\vee \otimes \mathcal{O}_C(x) \otimes \omega_C)$. While $\deg(\mathcal{L}^\vee \otimes \mathcal{O}_C(x) \otimes \omega_C) \leq 2g - 2 + 1 - 2g < 0$, by Corollary 3.3, it has no nonzero global section. Hence $H^1(C, \mathcal{L} \otimes \mathcal{O}_C(-x)) = 0$. And we get an exact sequence

$$0 \longrightarrow \Gamma(C, \mathcal{L} \otimes \mathcal{O}_C(-x)) \longrightarrow \Gamma(C, \mathcal{L}) \longrightarrow \Gamma(C, \mathcal{L} \otimes \mathcal{O}_x) \longrightarrow 0 \tag{91}$$

As $\Gamma(C, \mathcal{L} \otimes \mathcal{O}_x) = \mathcal{L}_x$. We get $\mathcal{L}$ is globally generated. □

**Proposition 5.1.** *For all coherent sheaf $\mathcal{F}$ on $X$, there exists $n_0$ such that for all $\mathcal{L}$ with $\deg \mathcal{L} > n_0$, we have that $\mathcal{F} \otimes \mathcal{L}$ is globally generated and $H^1(X, \mathcal{F} \otimes \mathcal{L}) = 0$.*

*Proof.* We first prove for invertible sheaf $\mathcal{M}$. Then by Lemma 5.1, we can take $n_1 = 2g - \deg M$. For a general coherent sheaf $\mathcal{F}$, since $C$ is projective, $\mathcal{F}$ is quotient of some direct sum of finitely many line bundles, say $\mathcal{G}$. Then there exists $n_0$ such that $\mathcal{G} \otimes \mathcal{L}$ is globally generated and $H^1(C, \mathcal{G} \otimes \mathcal{L}) = 0$ for all $\mathcal{L}$ with $\deg \mathcal{L} > n_0$. Hence $\mathcal{F} \otimes \mathcal{L}$ is globally generated. Consider the following exact sequence

$$0 \longrightarrow \ker \otimes \mathcal{L} \longrightarrow \mathcal{G} \otimes \mathcal{L} \longrightarrow \mathcal{F} \otimes \mathcal{L} \longrightarrow 0 \tag{92}$$

By Grothendieck Vanishing, we get exact sequence

$$\cdots \longrightarrow H^1(C, \ker \otimes \mathcal{L}) \longrightarrow H^1(C, \mathcal{G} \otimes \mathcal{L}) \longrightarrow H^1(C, \mathcal{F} \otimes \mathcal{L}) \longrightarrow 0 \tag{93}$$

Hence we also get $H^1(C, \mathcal{F} \otimes \mathcal{L}) = 0$, done! □

**Corollary 5.1.** *Let $\mathcal{L}$ be a line bundle on $C$. Then $\mathcal{L}$ is ample if and only if $\deg \mathcal{L} > 0$.*

*Proof.* Suppose $\mathcal{L}$ is ample and $\deg \mathcal{L} \leq 0$. Since $\mathcal{L}$ is ample, for a line bundle $\mathcal{M}$ with $\deg \mathcal{M} < 0$, we get $\mathcal{M} \otimes \mathcal{L}^n$ is globally generated for $n$ large enough. While $\deg(\mathcal{M} \otimes \mathcal{L}^n) < 0$, hence by Corollary 3.3, it has no nonzero global section, contradiction! Hence $\deg \mathcal{L} > 0$.

For the converse, note that since $C$ is projective, $\mathcal{L}$ is ample if and only if for all $\mathcal{F} \in$ **Coh**$(C)$, $H^1(X, \mathcal{F} \otimes \mathcal{L}^n) = 0$ for $n$ large enough. Then by Proposition 5.1, we are done. □

## 5.2 Filtration of vector bundles

Recall that for all vector bundle $\mathcal{E}$ on $C$, there is a filtration of subbundles by line bundles

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E} \tag{94}$$

such that each quotient $\mathcal{L}_i = \mathcal{E}_i / \mathcal{E}_{i-1}$ is a line bundle for all $1 \leq i \leq r$. We may use $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ to denote this filtration.

**Lemma 5.2.** *Let $\mathcal{E}$ be a vector bundle on $C$. Then for any nonzero section $s \in H^0(C, \mathcal{E})$, it naturally defines a line subbundle of $\mathcal{E}$, denoted by $[s]$.*

*Proof.* As $s$ induces a rational section $\tau : C \dashrightarrow \mathbb{P}(\mathcal{E})$ and $C$ is regular curve, the section is in fact regular. Note that there is a canonical inclusion $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(-1) \hookrightarrow \pi^*\mathcal{E}$, where $\pi : \mathbb{P}(\mathcal{E}) \to C$. Then $[s] := \tau^*\mathcal{O}_{\mathbb{P}(\mathcal{E})}(-1)$ is a line subbundle of $\mathcal{E}$. □

**Corollary 5.2.** *Let $\mathcal{E}$ be a vector bundle on $C$, $s$ nonzero section in $H^0(C, \mathcal{E})$. Then $[s] = \mathcal{O}_C(\mathrm{div}(s))$.*

*Proof.* Suffice to show $s$ is also a section of $[s]$ i.e. $s \in \Gamma(C, [s])$. By gluing property, we can reduce to affine case that $C = \operatorname{Spec} A$ and $\mathcal{E} = \mathcal{O}_C^{\oplus r}$. Assume $s = (s_1, \cdots, s_r)$. Then the canonical inclusion is given by

$$1 \longmapsto (\frac{x_1}{x_i}, \cdots, \frac{x_r}{x_i}) \tag{95}$$

And $(s_i, D_+(x_i)) \otimes 1 = (s_i \cdot \frac{s_1}{s_i}, \cdots, s_i \cdot \frac{s_r}{s_i}) = (\frac{s_1}{1}, \cdots, \frac{s_r}{1})$ is a section in $\Gamma(D(s_i), [s])$. Obviously, we can glue them up to section $s$ in $\Gamma(C, [s])$, done! □

**Remark 5.1.** *Then consider $\mathcal{E}' = \mathcal{E}/[s]$, which is also locally free. We have a filtration of $\mathcal{E}'$. Hence by lifting this filtration, we get a filtration of $\mathcal{E}$ starting with $[s]$.*

**Definition 5.1.** *Let $\mathcal{E}$ be a vector bundle on $C$, $s$ nonzero section in $H^0(C, \mathcal{E})$. Define $\deg s$ to be $\deg([s])$.*

**Lemma 5.3.** *Let $\mathcal{E}$ be a vector bundle on $C$. Then for all nonzero section $s \in H^0(C, \mathcal{E})$, we have $\deg s$ bounded above.*

*Proof.* First fix a filtration of $\mathcal{E}$ by line bundles $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$. Then for any nonzero section $s \in H^0(C, \mathcal{E})$, there exists $1 \leq i \leq r$ such that $[s] \subset \mathcal{E}_i$ but $[s] \not\subset \mathcal{E}_{i-1}$. Hence we get a nonzero morphism $[s] \to \mathcal{L}_i$, which is injective as $[s]$ is locally free of rank one. Hence $\deg s = \deg([s]) \leq \deg \mathcal{L}_i$. Conclude that $\deg s \leq \max_{1 \leq i \leq r}\{\deg \mathcal{L}_i\}$. □

**Remark 5.2.** *For $\mathcal{E}$ line bundle, we have $\deg s = \deg \mathcal{E}$ for all nonzero section $s$ by Lemma 3.5.*

**Definition 5.2.** *Let $\mathcal{E}$ be a vector bundle on $C$. Define $\mu_{\max}(\mathcal{E}) := \sup\{\deg s : s \in H^0(C, \mathcal{E}) \setminus \{0\}\}$. For nonzero section $s \in H^0(C, \mathcal{E})$, we say $s$ is maximal if $\deg s = \mu_{\max}(\mathcal{E})$ and the line subbundle $[s]$ is called maximal line subbundle of $\mathcal{E}$.*

**Definition 5.3.** *Let $\mathcal{E}$ be a vector bundle on $C$, $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ a filtration of $\mathcal{E}$. We say this filtration is maximal if $\deg \mathcal{L}_i = \mu_{\max}(\mathcal{E}/\mathcal{E}_{i-1})$ for all $1 \leq i \leq r$.*

**Lemma 5.4.** *Let $\mathcal{E}$ be a vector bundle on $C$ of rank 2, $(\mathcal{L}_1, \mathcal{L}_2)$ a maximal filtration of $\mathcal{E}$. Then $\deg \mathcal{L}_2 - \deg \mathcal{L}_1 \leq 2g$.*

*Proof.* Consider the following exact sequence

$$0 \longrightarrow \mathcal{L}_1 \longrightarrow \mathcal{E} \longrightarrow \mathcal{L}_2 \longrightarrow 0 \tag{96}$$

Tensoring by $\mathcal{L}_1^\vee$, we get

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \otimes \mathcal{L}_1^\vee \longrightarrow \mathcal{L}_2 \otimes \mathcal{L}_1^\vee \longrightarrow 0 \tag{97}$$

Suppose that $\deg \mathcal{L}_2 - \deg \mathcal{L}_1 \geq 2g+1$. Then $\deg(\mathcal{L}_2 \otimes \mathcal{L}_1^\vee) \geq 2g+1$. By definition of degree, we have $\deg(\mathcal{L}_2 \otimes \mathcal{L}_1^\vee) = \chi(\mathcal{L}_2 \otimes \mathcal{L}_1^\vee) - \chi(\mathcal{O}_C)$. Hence $\chi(\mathcal{L}_2 \otimes \mathcal{L}_1^\vee) \geq g+2$ and $h^0(\mathcal{L}_2 \otimes \mathcal{L}_1^\vee) \geq g+2$. Consider the following long exact sequence of cohomology

$$0 \to H^0(C, \mathcal{O}_C) \to H^0(C, \mathcal{E} \otimes \mathcal{L}_1^\vee) \to H^0(C, \mathcal{L}_2 \otimes \mathcal{L}_1^\vee) \to H^1(C, \mathcal{O}_C) \to \cdots \tag{98}$$

Note that $h^0(C, \mathcal{O}_C) + h^0(C, \mathcal{L}_2 \otimes \mathcal{L}_1^\vee) \leq h^0(C, \mathcal{E} \otimes \mathcal{L}_1^\vee) + h^1(C, \mathcal{O}_C)$. Hence $h^0(C, \mathcal{E} \otimes \mathcal{L}_1^\vee) \geq 3$. Consider $H^0(C, \mathcal{E} \otimes \mathcal{L}_1^\vee) \otimes \mathcal{O}_C \stackrel{w_x}{\to} (\mathcal{E} \otimes \mathcal{L}_1^\vee)_x$. Since $\mathcal{E} \otimes \mathcal{L}_1^\vee$ is locally free of rank 2, we can find a nonzero section $s \in \ker w_x$. Then $s$ defines a line subbundle $[s]$ of $\mathcal{E} \otimes \mathcal{L}_1^\vee$. As $\operatorname{coeff}_x \operatorname{div}(s) \geq 1$, we have $\deg[s] \geq 1$. Hence $\deg([s] \otimes \mathcal{L}_1) \geq \deg \mathcal{L}_1 + 1 > \deg \mathcal{L}_1$, contradicting the maximality of $\mathcal{L}_1$. □

**Corollary 5.3.** *Let $\mathcal{E}$ be a vector bundle on $C$ of rank $r$, $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ a maximal filtration of $\mathcal{E}$. Then $\deg \mathcal{L}_{i+1} - \deg \mathcal{L}_i \leq 2g$ for all $1 \leq i \leq r-1$.*

*Proof.* By induction on $r$. When $r = 2$, this is just previous lemma. For $r > 2$, consider the following exact sequence

$$0 \longrightarrow \mathcal{L}_1 \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}/\mathcal{L}_1 \longrightarrow 0 \tag{99}$$

The bundle $\mathcal{E}/\mathcal{L}_1$ has rank $r-1$, and the filtration $(\mathcal{L}_2/\mathcal{L}_1, \cdots, \mathcal{L}_r/\mathcal{L}_1)$ is maximal. By induction hypothesis, we have $\deg(\mathcal{L}_{i+1}/\mathcal{L}_1) - \deg(\mathcal{L}_i/\mathcal{L}_1) \leq 2g$ for all $2 \leq i \leq r-1$. This implies $\deg \mathcal{L}_{i+1} - \deg \mathcal{L}_i \leq 2g$ for all $1 \leq i \leq r-1$. □

## 5.3 Indecomposable vector bundles

**Definition 5.4.** *Let $\mathcal{E}$ be a vector bundle on $C$. We say $\mathcal{E}$ is decomposable if there exists a nontrivial decomposition $\mathcal{E} \cong \mathcal{E}' \oplus \mathcal{E}''$ for some vector bundles $\mathcal{E}'$ and $\mathcal{E}''$ on $C$. Otherwise, we say $\mathcal{E}$ is indecomposable.*

**Lemma 5.5.** *Let $\mathcal{E}$ be an indecomposable vector bundle on $C$ of rank $r$. Assume there is a short exact sequence*

$$0 \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}'' \longrightarrow 0 \tag{100}$$

*Then* $\operatorname{Hom}(\mathcal{E}', \mathcal{E}'' \otimes \omega_C) \neq 0$.

*Proof.* Since $\mathcal{E}$ is indecomposable, the above short exact sequence does not split. Hence $\operatorname{Ext}^1(\mathcal{E}', \mathcal{E}'') \neq 0$. While

$$\begin{aligned} \operatorname{Ext}^1(\mathcal{E}'', \mathcal{E}') &= H^0(C, \mathcal{E}xt^1(\mathcal{E}'', \mathcal{E}')) \\ &\cong H^1(C, \mathcal{H}om(\mathcal{E}'', \mathcal{E}')) \\ &\cong H^0(C, \mathcal{E}'' \otimes \mathcal{E}'^\vee \otimes \omega_C) \\ &\cong \operatorname{Hom}(\mathcal{E}', \mathcal{E}'' \otimes \omega_C) \end{aligned} \tag{101}$$

Conclude that $\operatorname{Hom}(\mathcal{E}', \mathcal{E}'' \otimes \omega_C) \neq 0$. □

**Lemma 5.6.** *Let $\mathcal{E}$ be a globally generated indecomposable vector bundle on $C$ of rank $r$. Then $\mathcal{E}$ has a maximal filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 - (i-1)(2g-2)$ for all $1 \leq i \leq r$.*

*Proof.* By definition, we can pick a maximal section $s \in H^0(C, \mathcal{E})$. Then $[s]$ is a maximal line subbundle of $\mathcal{E}$. Suppose now we have constructed a filtration of $\mathcal{E}$ of length $i$ starting with $[s]$.

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_i \subseteq \mathcal{E} \tag{102}$$

such that $\mathcal{L}_j$ is a maximal line subbundle of $\mathcal{E}/\mathcal{E}_{j-1}$ for all $1 \leq j \leq i$.

Set $\mathcal{E}' = \mathcal{E}/\mathcal{E}_i$. Since $\mathcal{E}$ is indecomposable, by previous lemma, we have $\mathrm{Hom}(\mathcal{E}_i, \mathcal{E}' \otimes \omega_C) \neq 0$. Hence there is a nonzero morphism $\mathcal{E}_i \to \mathcal{E}' \otimes \omega_C$. Pick $j_0$ such that $f\big|_{\mathcal{E}_{j_0}}$ is nonzero and $f\big|_{\mathcal{E}_j} = 0$ for all $j < j_0$. Thus $\mathcal{L}_{j_0} \to \mathcal{E}' \otimes \omega_C$ is a nonzero morphism.

Note that by definition of maximal line bundle, $\mathcal{L}_{j_0}$ has nonzero global section. Hence we can pick nonzero $s' \in H^0(C, \mathcal{L}_{j_0})$. Note that $s'$ is nowhere vanishing on $U := C \setminus \mathrm{div}(s')$, for all $x \in U$, $s'_x$ spans $\mathcal{L}_{j_0,x}$. As $f$ is nonzero, $f_x : \mathcal{L}_{j_0,x} \to \mathcal{E}'_x \otimes \omega_{C,x}$ is a nonzero except for finitely many closed points. Hence $f_x(s'_x) \neq 0$ except for finitely many closed points so that $f_* s' \neq 0$ and $[f_* s'] \geq [s']$. On the other hand, since $\mathcal{E}$ globally generated, $\mathcal{E}'$ is also globally generated so that there exists a nonzero section in $s'' \in H^0(C, \mathcal{E}')$ such that $[s'']$ is maximal. Thus

$$\begin{aligned} \deg[s''] &\geq \deg([f_* s'] \otimes \omega_C^\vee) \\ &= \deg[f_* s'] - (2g-2) \\ &\geq \deg[s'] - (2g-2) \\ &= \deg \mathcal{L}_{j_0} - (2g-2) \\ &\geq \deg \mathcal{L}_1 - j_0(2g-2) \\ &\geq \deg \mathcal{L}_1 - i(2g-2) \end{aligned} \tag{103}$$

Now take pull back $[s'']$ along the quotient map $\mathcal{E} \twoheadrightarrow \mathcal{E}'$, we get a vector subbundle $\mathcal{E}_{i+1}$ of $\mathcal{E}$ containing $\mathcal{E}_i$ and with $\deg(\mathcal{E}_{i+1}/\mathcal{E}_i) = \deg[s''] \geq \deg \mathcal{L}_1 - i(2g-2)$. By induction, we get a filtration of $\mathcal{E}$ of length $r$ starting with $[s]$ such that $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 - (i-1)(2g-2)$ for all $1 \leq i \leq r$. □

**Corollary 5.4.** *Let $\mathcal{E}$ be an indecomposable vector bundle on $C$ of rank $r$. Then $\mathcal{E}$ has a filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 - (i-1)(2g-2)$ for all $1 \leq i \leq r$.*

*Proof.* Since $\mathcal{O}_C(1)$ is ample, there exists $n_0$ such that $\mathcal{E}(n)$ is globally generated for all $n > n_0$. And it is clear that indecomposability is preserved by twisting by line bundles. Hence $\mathcal{E}(n)$ is a globally generated indecomposable vector bundle on $C$ for all $n > n_0$. By previous lemma, $\mathcal{E}(n)$ has a maximal filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 - (i-1)(2g-2)$ for all $1 \leq i \leq r$. Hence $\mathcal{E}$ has a filtration $(\mathcal{L}_1(-n), \cdots, \mathcal{L}_r(-n))$ with $\deg \mathcal{L}_i(-n) \geq \deg \mathcal{L}_1(-n) - (i-1)(2g-2)$ for all $1 \leq i \leq r$. □

**Remark 5.3.** *This is not a maximal filtration since $\mathcal{L}_i$ after twisting might be not given by a nonzero section.*

**Lemma 5.7.** *Let $\mathcal{E}$ be an indecomposable vector bundle on $C$ of rank $r$ and degree $d$. Then $\mathcal{E}$ has a filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \frac{d}{r} - (r-1)(3g-2)$ for all $1 \leq i \leq r$.*

*Proof.* Similar to Corollary 5.4, we may assume $\mathcal{E}$ is globally generated. Then by Corollary 4.5, $\mathcal{E}(1)$ is ample. By Lemma 5.6, $\mathcal{E}(1)$ has a maximal filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 - (i-1)(2g-2)$ for all $1 \leq i \leq r$. And by Corollary 5.3, $\deg \mathcal{L}_{i+1} - \deg \mathcal{L}_i \leq 2g$. Hence

$$-2g(i-1) \leq \deg \mathcal{L}_1 - \deg \mathcal{L}_i \leq (i-1)(2g-2) \tag{104}$$

Summing up, we get

$$-r(r-1)g \leq r \deg \mathcal{L}_1 - \sum_{i=1}^{r} \deg \mathcal{L}_i \leq r(r-1)(g-1) \tag{105}$$

Note that $\sum_{i=1}^{r} \deg \mathcal{L}_i = \deg \mathcal{E} = d$. Hence

$$-r(r-1)g \leq r \deg \mathcal{L}_1 - d \leq r(r-1)(g-1) \tag{106}$$

Moving terms and dividing by $r$, we get

$$\frac{r}{d} - (r-1)g \leq \deg \mathcal{L}_1 \leq \frac{d}{r} + (r-1)(g-1) \tag{107}$$

Therefore,

$$\begin{aligned} \deg \mathcal{L}_i &\geq \deg \mathcal{L}_1 - (r-1)(2g-2) \\ &\geq \frac{d}{r} - (r-1)g - (r-1)(2g-2) \\ &\geq \frac{d}{r} - (r-1)(3g-2) \end{aligned} \tag{108}$$

□

**Remark 5.4.** *In particular, when $g = 0$, we get $0 \leq \deg \mathcal{L}_1 - \frac{d}{r} \leq 1 - r$. Hence $r$ must equal to 1 and $\mathcal{E}$ is a line bundle. This is consistent with the Grothendieck's theorem that all vector bundles on a rational curve are direct sum of line bundles.*

**Definition 5.5.** *Define $\mathcal{E}(r, d)$ to be the set of isomorphism classes of indecomposable vector bundles on $C$ of rank $r$ and degree $d$.*

**Theorem 5.1** (***Atiyah***)**.** *There is an integer $N$ with respect to $g$, $r$ and $d$ such that $\mathcal{E}(n)$ is ample for all $\mathcal{E} \in \mathcal{E}(r, d)$.*

*Proof.* By Lemma 5.7, for all $\mathcal{E} \in \mathcal{E}(r, d)$, there is a filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ of $\mathcal{E}$ with $\deg \mathcal{L}_i \geq \frac{d}{r} - (r-1)(3g-2)$ for all $1 \leq i \leq r$. Hence $\deg \mathcal{L}_i(n) > 0$ for all $1 \leq i \leq r$ when $n > (r-1)(3g-2) - \frac{d}{r}$. By Corollary 5.1, $\mathcal{L}_i(n)$ is ample for all $1 \leq i \leq r$ when $n > (r-1)(3g-2) - \frac{d}{r}$. Hence $\mathcal{E}(n)$ is ample for all $\mathcal{E} \in \mathcal{E}(r, d)$ when $n > (r-1)(3g-2) - \frac{d}{r}$ by Corollary 4.6. □

### 5.4 Vector bundles on elliptic curves

This section is based on the famous paper, *Vector Bundles over An Elliptic Curve* by Atiyah.

Let $X$ be an elliptic curve over $\mathbb{C}$, i.e. a smooth projective curve of genus 1 with a chosen rational point $x_0$. And $\mathcal{E}$ be a vector bundle on $X$ of rank $r$ and degree $d$. Set $A = \mathcal{O}_X(x_0)$.

**Lemma 5.8.** $\deg A = 1$.

*Proof.* Considetr the following exact sequence

$$0 \longrightarrow A^\vee \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{x_0} \longrightarrow 0 \tag{109}$$

Then $\deg A = -\deg A^\vee = 0 - \deg \mathcal{O}_{x_0}$. While $\mathcal{O}_{x_0}$ is a skyscraper sheaf supported at $x_0$, its degree is 1. Hence $\deg A = 1$. □

**Remark 5.5.** *By Lemma 3.8,* $\deg(\mathcal{E} \otimes A) = \deg \mathcal{E} + r$. *Hence there is a canonical map*

$$\begin{aligned} \mathcal{E}(r,d) &\longrightarrow \mathcal{E}(r, d+nr) \\ \mathcal{E} &\longmapsto \mathcal{E} \otimes A^{\otimes n} \end{aligned} \tag{110}$$

*Clearly this map is bijective as $A$ is invertible. Hence we can identify $\mathcal{E}(r,d)$ with $\mathcal{E}(r,d+nr)$ for all $n \in \mathbb{Z}$. So we can just consider $\mathcal{E}(r,d)$ for $0 \le d < r$.*

**Proposition 5.2.** $\omega_X \cong \mathcal{O}_X$.

*Proof.* Note that $h^0(X, \omega_X) = g = 1$ and $\deg \omega_X = g - 1 - (1-g) = 0$. Hence there is a nonzero section $s \in H^0(X, \omega_X)$. Then similar to proof of Corollary 3.3, $\omega_X \cong \mathcal{O}_X(\mathrm{div}(s))$ of degree 0. Hence $\omega_X \cong \mathcal{O}_X$. □

**Lemma 5.9.** *Let $\mathcal{E}$ be an indecomposable vector bundle of rank $r$ with $h^0(X, \mathcal{E}) > 0$. Then $\mathcal{E}$ admits a maximal filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ such that $\mathcal{L}_i \ge \mathcal{L}_1$.*

*Proof.* The proof is similar to Lemma 5.6. By definition, we can pick a maximal section $s \in H^0(C, \mathcal{E})$. Then $[s]$ is a maximal line subbundle of $\mathcal{E}$. Suppose now we have constructed a filtration of $\mathcal{E}$ of length $i$ starting with $[s]$.

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_i \subseteq \mathcal{E} \tag{111}$$

such that $\mathcal{L}_j$ is a maximal line subbundle of $\mathcal{E}/\mathcal{E}_{j-1}$ for all $1 \le j \le i$.

Set $\mathcal{E}' = \mathcal{E}/\mathcal{E}_i$. Since $\mathcal{E}$ is indecomposable, by Lemma 5.5, we have $\mathrm{Hom}(\mathcal{E}_i, \mathcal{E}') \ne 0$. Hence there is a nonzero morphism $\mathcal{E}_i \to \mathcal{E}'$. Pick $j_0$ such that $f\big|_{\mathcal{E}_{j_0}}$ is nonzero and $f\big|_{\mathcal{E}_j} = 0$ for all $j < j_0$. Thus $\mathcal{L}_{j_0} \to \mathcal{E}'$ is a nonzero morphism.

Note that by definition of maximal line bundle, $\mathcal{L}_{j_0}$ has nonzero global section. Hence we can pick nonzero $s' \in H^0(X, \mathcal{L}_{j_0})$. Note that $s'$ is nowhere vanishing on $U := X \setminus \mathrm{div}(s')$, for all $x \in U$, $s'_x$ spans $\mathcal{L}_{j_0,x}$. As $f$ is nonzero, $f_x : \mathcal{L}_{j_0,x} \to \mathcal{E}'_x$ is a nonzero except for finitely many closed points. Hence $f_x(s'_x) \ne 0$ except for finitely many closed points so that $f_* s' \ne 0$ and $[f_* s'] \ge [s']$.

On the other hand, as $f_*s'$ is a nonzero section in $H^0(X,\mathcal{E})$, $H^0(X,\mathcal{E}) \neq 0$ and we can pick an $s''$ maximal. If $[f_x(s'_x)]$ itself is maximal, then we may take $s'' = f_x(s'_x)$ and hence $[s''] = [f_x(s'_x)]$. Otherwise, $\deg[s''] \geq \deg[f_x(s'_x)] + 1$ and $g(X) = 1$ so that $[s''] \otimes [f_x(s'_x)]^\vee$ has positive degree and hence nonzero global section. Thus

$$[s''] \geq [f_*s'] \geq [s'] = \mathcal{L}_{j_0} \geq \mathcal{L}_1 \tag{112}$$

Now take pull back $[s'']$ along the quotient map $\mathcal{E} \twoheadrightarrow \mathcal{E}'$, we get a vector subbundle $\mathcal{E}_{i+1}$ of $\mathcal{E}$ containing $\mathcal{E}_i$ and with $\deg(\mathcal{E}_{i+1}/\mathcal{E}_i) = \deg[s''] \geq \deg \mathcal{L}_1$. By induction, we get a filtration of $\mathcal{E}$ of length $r$ starting with $[s]$ such that $\mathcal{L}_i \geq \mathcal{L}_1$ for all $1 \leq i \leq r$. □

**Lemma 5.10.** *Let $\mathcal{E} \in \mathcal{E}(r,d)$, $0 \leq d < r$ and $h^0(X,\mathcal{E}) > 0$. Then $\mu_{\max}(\mathcal{E}) = 0$ and $\mathcal{E}$ has a trivial line subbundle of rank $h = h^0(X,\mathcal{E})$, denoted by $\mathcal{I}_{\mathcal{E}}$.*

*Proof.* By Lemma 5.9, $\mathcal{E}$ has a maximal filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ with $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1$ for all $1 \leq i \leq r$. If $\deg \mathcal{L}_1 > 0$, then $d = \deg \mathcal{E} = \sum_{i=1}^r \deg \mathcal{L}_i \geq r \cdot \deg \mathcal{L}_1 \geq r$, which contradicts the assumption that $0 \leq d < r$. Hence $\deg \mathcal{L}_1 = 0$.

Then for any $s \in H^0(X,\mathcal{L}_1)$, we have $0 \leq \deg s \leq \deg \mathcal{L}_1 = 0$. Consider $w_x : \Gamma(X,\mathcal{E}) \otimes \mathcal{O}_{X,x} \to \mathcal{E}_x$ for any $x \in X$. Then $w_x$ is injective otherwise there is a nonzero section $s \in H^0(X,\mathcal{E})$ such that $\deg s \geq \operatorname{coeff}_x \operatorname{div}(s) \geq 1$. Hence $\Gamma(X,\mathcal{E})$ generates a trivial line subbundle of rank $h$. □

**Lemma 5.11.** *Let $\mathcal{E} \in \mathcal{E}(r,r)$. Then $\mathcal{E}$ admits a maximal filtration $(\mathcal{L}, \cdots, \mathcal{L})$ with $\deg \mathcal{L} = 1$.*

*Proof.* As $r > 0$, we know $h^0(X,\mathcal{E}) \neq 0$. By Lemma 5.9, $\mathcal{E}$ has a maximal filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ with $\mathcal{L}_i \geq \mathcal{L}_1$ for all $1 \leq i \leq r$. Suppose $\deg \mathcal{L}_1 = 0$. Similar to proof of previous lemma, we can show that $\Gamma(X,\mathcal{E})$ generates a trivial subbundle $\mathcal{I}_{\mathcal{E}}$ of rank $h = h^0(X,\mathcal{E})$. Hence $r \geq h^0(X,\mathcal{E}) \geq \deg \mathcal{E} = r$ so that $h^0(X,\mathcal{E}) = r$ and $\mathcal{E} = \mathcal{I}_{\mathcal{E}}$, which contradicts the indecomposability of $\mathcal{E}$.

Now $\deg \mathcal{L}_1 \geq 1$. Note that $r = \deg \mathcal{E} \geq r \cdot \deg \mathcal{L}_1 \geq r$, which implies $\deg \mathcal{L}_i = 1$ for all $i$. Pick some nonzero global section $s$ of $\mathcal{L}_1$. Then $\deg(\operatorname{div}(s)) = \deg \mathcal{L}_1 = 1$. While there is a nonzero map $f_i : \mathcal{L}_1 \to \mathcal{L}_i$ for all $i \geq 1$, consider $f_i \circ s$. By argument similar to proof of Lemma 5.6, we know $f_i \circ s$ is a nonzero global section of $\mathcal{L}_i$. Note that $\deg(\operatorname{div}(f \circ s)) = \deg \mathcal{L}_i = 1 = \deg(\operatorname{div}(s))$ so that $f$ is nowhere vanishing and hence $f$ is invertible. Conclude that $\mathcal{L}_i \cong \mathcal{L}_1$ for all $i$. □

**Remark 5.6.** *In fact, for $\mathcal{E} \in \mathcal{E}(r,d)$ with $d \geq r$, by similar argument, we can show that $\mathcal{E}$ admits a maximal filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ with $\deg \mathcal{L}_1 \geq 1$.*

**Definition 5.6.** *An extension of vector bundles $0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{E}'' \to 0$ is called $\mathcal{E}''$-partial if there is a commutative diagram*

$$\begin{tikzcd} \mathcal{E} \arrow[r, "p"] & \mathcal{E}'' \arrow[d, two heads, "q"] \\ & \mathcal{E}''_1 \arrow[ul, "j"] \end{tikzcd} \tag{113}$$

*such that $q \circ p \circ j = \mathrm{id}_{\mathcal{E}''_1}$. Otherwise, we say this extension is $\mathcal{E}''$-complete.*

**Remark 5.7.** *Dually, An extension of vector bundles* $0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{E}'' \to 0$ *is called* $\mathcal{E}'$*-partial if the dual extension* $0 \to \mathcal{E}''^{\vee} \to \mathcal{E}^{\vee} \to \mathcal{E}'^{\vee} \to 0$ *is* $\mathcal{E}'^{\vee}$*-partial.*

**Lemma 5.12.** *Given an extension* $0 \to \mathcal{E}' \to \mathcal{E} \xrightarrow{p} \mathcal{E}'' \to 0$ *with* $\mathcal{E}''$ *trivial, it is* $\mathcal{E}''$*-complete if and only if* $p_* : H^0(X, \mathcal{E}) \to H^0(X, \mathcal{E}'')$ *is a zero map.*

*Proof.* If the extension is $\mathcal{E}''$-partial, then there is a commutative diagram

$$\begin{tikzcd} \mathcal{E} \arrow[r, "p"] & \mathcal{E}'' \arrow[d, two heads, "q"] \\ & \mathcal{E}''_1 \arrow[ul, "j"] \end{tikzcd} \tag{114}$$

such that $q \circ p \circ j = \mathrm{id}_{\mathcal{E}''_1}$. Hence $p_* \neq 0$ as $(q \circ p \circ j)_*$ is isomorphic.

Conversely, if $p_* \neq 0$, then we can pick a nonzero section $s \in H^0(X, \mathcal{E}'')$ such that $p_* s \neq 0$. Then $p_* s$ defines a line subbundle $[p_* s]$ of $\mathcal{E}''$. As $H^0(X, \mathcal{E}'') \cong \oplus H^0(X, \mathcal{O}_X)$, we have $p_* s$ is injective and $[p_* s]$ is a trivial line subbundle of $\mathcal{E}''$. And we can take orthogonal complement of $H^0(X, [p_* s])$ in $H^0(X, \mathcal{E}'')$, which implies that $[p_* s]$ is a direct summand of $\mathcal{E}''$. Take the projection $q : \mathcal{E}'' \twoheadrightarrow \mathcal{O}_X \cong [p_* s]$. Consider the following diagram

$$\begin{tikzcd} \mathcal{E} \arrow[r, "p"] & \mathcal{E}'' \arrow[d, two heads, "q"] \\ & {[p_* s]} \arrow[ul, "j"] \end{tikzcd} \tag{115}$$

where $j$ is the compostion of $[p_* s] \cong \mathcal{O}_X \xrightarrow{s} \mathcal{E}$. Clearly this diagram is commutative and $q \circ p \circ j = \mathrm{id}_{[p_* s]}$. □

**Corollary 5.5.** *Given an extension* $0 \to \mathcal{E}' \to \mathcal{E} \xrightarrow{p} \mathcal{E}'' \to 0$ *with* $\mathcal{E}''$ *trivial, it is* $\mathcal{E}''$*-complete if and only if the coboundary map* $\delta : H^0(X, \mathcal{E}'') \to H^1(X, \mathcal{E}')$ *is injective.*

*Proof.* Consider the following long exact sequence of cohomology

$$0 \to H^0(X, \mathcal{E}') \to H^0(X, \mathcal{E}) \xrightarrow{p_*} H^0(X, \mathcal{E}'') \xrightarrow{\delta} H^1(X, \mathcal{E}') \to \cdots \tag{116}$$

If the extension is $\mathcal{E}''$-complete, then by previous lemma, $p_*$ is a zero map. Hence $\delta$ is injective. Conversely, if $\delta$ is injective, then $p_*$ is a zero map. Hence the extension is $\mathcal{E}''$-complete by previous lemma. □

**Proposition 5.3.** *Given a vector bundle* $\mathcal{E}'$ *and trivial vector bundle* $\mathcal{E}''$ *of rank* $r$ *on* $X$*. The equivalence class of extensions of* $\mathcal{E}''$ *by* $\mathcal{E}'$ *is one-to-one corresponding with the set of coboundary maps* $\delta : H^0(X, \mathcal{E}'') \to H^1(X, \mathcal{E}')$*. In particular, if* $h^1(X, \mathcal{E}') = r$*, then there is a unique* $\mathcal{E}''$*-complete extension of* $\mathcal{E}''$ *by* $\mathcal{E}'$ *up to equivalence relation that two extension* $\mathcal{E}_1$ *and* $\mathcal{E}_2$ *are equivalent if there is a commutative diagram with exact rows*

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{E}' \arrow[r] \arrow[d, equal] & \mathcal{E}_2 \arrow[r] \arrow[d, "\sim"] & \mathcal{E}'' \arrow[r] \arrow[d, "\sim"] & 0 \\ 0 \arrow[r] & \mathcal{E}' \arrow[r] & \mathcal{E}_1 \arrow[r] & \mathcal{E}'' \arrow[r] & 0 \end{tikzcd} \tag{117}$$

*which is more general than the isomorphism class of extensions.*

*Proof.* It is well-known that the equivalence class of extensions of $\mathcal{E}''$ by $\mathcal{E}'$ is one-to-one corresponding with the set of extension classes in $\mathrm{Ext}^1(\mathcal{E}'', \mathcal{E}')$. Note that $\omega_X \cong \mathcal{O}_X$. As $\mathcal{E}''$ is trivial, by Serre duality, we have that

$$\begin{aligned}
\mathrm{Ext}^1(\mathcal{E}'', \mathcal{E}') &= H^0(X, \mathcal{E}xt^1(\mathcal{E}'', \mathcal{E}')) \\
&\cong H^1(X, \mathcal{H}om(\mathcal{E}'', \mathcal{E}')) \\
&\cong H^0(X, \mathcal{E}'' \otimes \mathcal{E}'^\vee \otimes \omega_X)^\vee \\
&\cong H^0(X, \mathcal{E}'' \otimes \mathcal{E}'^\vee) \\
&= H^0(X, \mathcal{E}''^\vee \otimes \mathcal{E}'^\vee) \\
&\cong H^0(X, \mathcal{H}om(\mathcal{E}'', \mathcal{H}om(\mathcal{E}', \mathcal{O}_X))) \\
&\cong \mathrm{Hom}(H^0(X, \mathcal{E}''), H^0(X, \mathcal{E}'^\vee)) \\
&\cong \mathrm{Hom}(H^0(X, \mathcal{E}''), H^1(X, \mathcal{E}'))
\end{aligned} \tag{118}$$

For the second part, as $h^1(X, \mathcal{E}') = r = h^0(X, \mathcal{E}'')$, the identity map gives $\mathcal{E}''$-complete extension. On the other hand, given any $\mathcal{E}''$-complete extension, consider the following long exact sequence of cohomology

$$0 \to H^0(X, \mathcal{E}') \to H^0(X, \mathcal{E}) \xrightarrow{p_*} H^0(X, \mathcal{E}'') \xrightarrow{\delta} H^1(X, \mathcal{E}') \to \cdots \tag{119}$$

By Corollary 5.5, $\delta$ is injective. Since $h^1(X, \mathcal{E}') = r = h^0(X, \mathcal{E}'')$, $\delta$ is an isomorphism. Assume there are two different $\delta_1$ and $\delta_2$. Then $\varphi := \delta_1^{-1} \circ \delta_2$ gives an automorphism of $H^0(X, \mathcal{E}'')$. As $\mathcal{E}''$ is trivial, $\varphi$ in fact an automorphism of $\mathcal{E}''$. Assume $\mathcal{E}_1$ is the extension corresponding to $\delta_1$. Take pull back

$$\begin{tikzcd}
\mathcal{E}_2 \arrow[r, two heads, "q"] \arrow[d, "\psi"] & \mathcal{E}'' \arrow[d, "\varphi"] \\
\mathcal{E}_1 \arrow[r, two heads, "p"] & \mathcal{E}''
\end{tikzcd} \tag{120}$$

It is clear to check that $\mathcal{E}_2$ is locally free and $\psi$ is isomorphic. As there is a commutative diagram

$$\begin{tikzcd}
\mathcal{E}' \arrow[r, "0"] \arrow[d] & \mathcal{E}'' \arrow[d, "\varphi"] \\
\mathcal{E}_1 \arrow[r, two heads, "p"] & \mathcal{E}''
\end{tikzcd} \tag{121}$$

By universal property, there is a unique map $\mathcal{E}' \to \mathcal{E}_2$. Easy to check that $0 \to \mathcal{E}' \to \mathcal{E}_2 \to \mathcal{E}'' \to 0$ is exact and hence an extension. Consider the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & H^0(X, \mathcal{E}') \arrow[r] \arrow[d, equal] & H^0(X, \mathcal{E}_2) \arrow[r] \arrow[d, "\psi"] & H^0(X, \mathcal{E}'') \arrow[r, "\delta"] \arrow[d, "\varphi"] & H^1(X, \mathcal{E}') \arrow[r] \arrow[d, equal] & \cdots \\
0 \arrow[r] & H^0(X, \mathcal{E}') \arrow[r] & H^0(X, \mathcal{E}_1) \arrow[r] & H^0(X, \mathcal{E}'') \arrow[r, "\delta_1"] & H^1(X, \mathcal{E}') \arrow[r] & \cdots
\end{tikzcd} \tag{122}$$

Hence $\delta = \delta_1 \circ \varphi = \delta_2$. This implies that the extension is unique up to isomorphism. □

**Lemma 5.13.** *Given an $\mathcal{E}'$-complete extension $0 \to \mathcal{E}' \to \mathcal{E} \xrightarrow{p} \mathcal{E}'' \to 0$ with $\mathcal{E}'$ trivial, then $h^0(X, \mathcal{E}) = h^0(X, \mathcal{E}'')$.*

*Proof.* Take the dual extension $0 \to \mathcal{E}''^{\vee} \to \mathcal{E}^{\vee} \to \mathcal{E}'^{\vee} \to 0$. By Lemma 5.12, we know $H^0(X, \mathcal{E}^{\vee}) \cong H^0(X, \mathcal{E}'^{\vee})$. Hence by Serre duality, $H^1(X, \mathcal{E}^{\vee}) \cong H^1(X, \mathcal{E}'^{\vee})$. Take dimension, we get $h^1(X, \mathcal{E}) = h^1(X, \mathcal{E}'')$. Note that $\deg \mathcal{E}' = 0$, by additivity of degree, $\deg \mathcal{E} = \deg \mathcal{E}''$. Hence by $h^0(X, \mathcal{E}) = h^0(X, \mathcal{E}'')$. □

In the following, we would give "uniqueness theorem "and "existence theorem "of $\mathcal{E}(r, d)$.

**Proposition 5.4** (***uniqueness***)**.** *Let $\mathcal{E} \in \mathcal{E}(r, d)$, $d \geq 0$. Then*
*(1)*$h^0(X, \mathcal{E}) = \begin{cases} 0 \text{ or } 1 & \text{if } d = 0 \\ d & \text{else} \end{cases}$
*(2)If $d < r$, then $\mathcal{E}' = \mathcal{E}/\mathcal{I}_{\mathcal{E}}$ is indecomposable with $h^0(X, \mathcal{E}') = h^0(X, \mathcal{E})$.*

*Proof.* For (1), if $d \geq r$, then by remark of Lemma 5.11, $\mathcal{E}$ admits a maximal filtration $(\mathcal{L}_1, \cdots, \mathcal{L}_r)$ with $\deg \mathcal{L}_1 \geq 1$. Hence for all $i$, $\deg \mathcal{L}_i \geq \deg \mathcal{L}_1 > 0$. And by Serre duality, $h^1(X, \mathcal{L}_i) = h^0(X, \mathcal{L}_i^{\vee}) = 0$ for all $i$. Note that

$$h^0(X, \mathcal{E}) \leq \sum_{i=1}^{r} h^0(X, \mathcal{L}_i) = \sum_{i=1}^{r} \deg \mathcal{L}_i = \deg \mathcal{E} = d \leq h^0(X, \mathcal{E}) \tag{123}$$

We get $h^0(X, \mathcal{E}) = d$. Suppose now that $d < r$. If $d = 0$ and $\Gamma(X, \mathcal{E}) = 0$ there is nothing to prove. Hence we may suppose that $\Gamma(X, \mathcal{E}) \neq 0$ even if $d = 0$. If $d > 0$, we always have $\Gamma(X, \mathcal{E}) \neq 0$. Consider the following exact sequence

$$(\mathcal{E}): \quad 0 \longrightarrow I_{\mathcal{E}} \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}' \longrightarrow 0. \tag{124}$$

Since $\mathcal{E}$ is indecomposable, $(\mathcal{E})$ is certainly $\mathcal{I}_{\mathcal{E}}$-complete. Hence by Lemma 5.13, $h^0(X, \mathcal{E}') = h^0(X, \mathcal{E}) =: h$, and so by duality $h^1(X, \mathcal{E}'^{\vee}) = h$. By Proposition 5.3 the extension $(\mathcal{E})$ corresponds to a monomorphism

$$\delta : \Gamma(\mathcal{E}'^{\vee}) \longrightarrow H^1(X, \mathcal{E}^{\vee}), \tag{125}$$

since both terms here have same dimension, $\delta$ must be an isomorphism. Suppose that $\mathcal{E}'$ is decomposable and $\mathcal{E}' = \mathcal{F} \oplus \mathcal{G}$ with $\mathcal{F} \neq 0$ and $\mathcal{G} \neq 0$. Then we may write $\delta = \delta_1 \oplus \delta_2$ where

$$\delta_1 : \Gamma(\mathcal{F}^{\vee}) \longrightarrow H^1(X, \mathcal{F}^{\vee}) \text{ and } \delta_2 : \Gamma(\mathcal{G}^{\vee}) \longrightarrow H^1(X, \mathcal{G}^{\vee}) \tag{126}$$

are isomorphisms, $f = \dim H^1(X, \mathcal{F}^{\vee})$, $g = \dim H^1(X, \mathcal{G}^{\vee})$. By Lemma 13, $\delta_1$ and $\delta_2$ correspond to extensions

$$\begin{aligned} (\mathcal{E}_1): \ 0 \longrightarrow \mathcal{I}_r \longrightarrow \mathcal{E}_1 \longrightarrow \mathcal{F} \longrightarrow 0 \\ (\mathcal{E}_2): \ 0 \longrightarrow \mathcal{I}_r \longrightarrow \mathcal{E}_2 \longrightarrow \mathcal{G} \longrightarrow 0 \end{aligned} \tag{127}$$

Thus $\delta_1 \oplus \delta_2$ corresponds to $(\mathcal{E}_1) \oplus (\mathcal{E}_2)$, and also to $(\mathcal{E})$. Hence, again by Proposition 5.3, $\mathcal{E} \cong \mathcal{E}_1 \oplus \mathcal{E}_2$, contradiction!

We have now proved (2) and (1) for $d \geq r$. It remains to prove (1) for $0 \leq d < r$. We proceed by induction on $r$. For $r = 1$, (1) is certainly true. Suppose it is true for all $r' < r$, in particular then for $r - h$. Since $\mathcal{E}' \in \mathcal{E}(r - h, d)$, $\dim \Gamma(\mathcal{E}') = d$ if $d > 0$, and $= 0$ or $1$ if $d = 0$. As we have already shown that $\dim \Gamma(\mathcal{E}') = \dim \Gamma(\mathcal{E})$, done! □

**Proposition 5.5** (***existence***)**.** *Let $\mathcal{E}' \in \mathcal{E}(r', d)$ with $d \geq 0$, and if $d = 0$ we assume $h^0(X, \mathcal{E}') \neq 0$. Then there exists $\mathcal{E} \in \mathcal{E}(r, d)$ such that $\mathcal{E}' \cong \mathcal{E}/\mathcal{I}_{\mathcal{E}}$ and $h^0(X, \mathcal{E}) = h^0(X, \mathcal{E}')$, which is unique up to isomorphism.*

*Proof.* Set $h = h^0(X, \mathcal{E}')$. Then for tivial vector bundle $\mathcal{I}_h$ of rank $h$, by Proposition 5.3, there is an $\mathcal{E}'$-complete extension

$$0 \longrightarrow \mathcal{I}_h \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}' \longrightarrow 0 \tag{128}$$

unique up to equivalence relation. By Lemma 5.13, $h^0(X, \mathcal{E}) = h^0(X, \mathcal{E}')$. Remains to show that $\mathcal{E}$ is indecomposable.

Suppose $\mathcal{E}$ is decomposable and $\mathcal{E} = \mathcal{F} \oplus \mathcal{G}$ with $\mathcal{F} \neq 0$ and $\mathcal{G} \neq 0$. Note that $\Gamma(X, \mathcal{I}_h) = \Gamma(X, \mathcal{E}) = \Gamma(X, \mathcal{F}) \oplus \Gamma(X, \mathcal{G})$, we get $\mathcal{I}_h$ is in fact $\mathcal{I}_{\mathcal{E}}$. Then $\Gamma(X, \mathcal{F})$ and $\Gamma(X, \mathcal{G})$ span trivial line subbundle $\mathcal{I}_{\mathcal{F}}$ and $\mathcal{I}_{\mathcal{G}}$ respectively. And $\mathcal{I}_{\mathcal{F}} \oplus \mathcal{I}_{\mathcal{G}} = \mathcal{I}_h$. While $\mathcal{E}' \cong \mathcal{E}/\mathcal{I}_h = \mathcal{E}/(\mathcal{I}_{\mathcal{F}} \oplus \mathcal{I}_{\mathcal{G}}) \cong \mathcal{F}/\mathcal{I}_{\mathcal{F}} \oplus \mathcal{G}/\mathcal{I}_{\mathcal{G}}$, as $\mathcal{E}'$ indecomposable, $\mathcal{F} = \mathcal{I}_{\mathcal{F}}$ or $\mathcal{G} = \mathcal{I}_{\mathcal{G}}$.

We may assume $\mathcal{F} = \mathcal{I}_{\mathcal{F}}$. There is a commutative diagram

$$\begin{tikzcd} \mathcal{F} \arrow[d, hook, "i"] & \\ \mathcal{I}_h \arrow[r, hook, "j"] & \mathcal{E} \arrow[ul, "\pi"'] \end{tikzcd} \tag{129}$$

such that $\pi \circ j \circ i = \mathrm{id}_{\mathcal{F}}$, contradicting that the extension is $\mathcal{I}_h$-complete. Conclude that $\mathcal{E}$ is indecomposable. □

**Theorem 5.2.** *There exists a vector bundle $\mathcal{F}_r \in \mathcal{E}(r, 0)$ with nonzero global sections, called the Atiyah bundle, which is unique up to isomorphism. Moreover, there is an exact sequence for all $r \geq 1$*

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{F}_r \longrightarrow \mathcal{F}_{r-1} \longrightarrow 0 \tag{130}$$

*Let $\mathcal{E} \in \mathcal{E}(r, 0)$. Then $\mathcal{E} \cong \mathcal{L} \otimes \mathcal{F}_r$ for some line bundle $\mathcal{L}$ of degree 0, which is also unique up to isomorphism. More precisely, $\mathcal{L}^r \cong \det \mathcal{E}$.*

*Proof.* We first prove the existence of $\mathcal{F}_r$ by induction on $r$. For $r = 1$, assume $\mathcal{E} \in \mathcal{E}(1, 0)$ with nonzero global section $s$. Then $\mathcal{E} \cong \mathrm{div}(s)$ is an effective divisor of degree 0, which must be isomorphic to $\mathcal{O}_X$. Suppose now we have proven the result for all $r' < r$. If $\mathcal{E} \in \mathcal{E}(r, 0)$ with nonzero global section, then $h^0(X, \mathcal{E}) = 1$ and by Proposition 5.4 there is an exact sequence

$$0 \longrightarrow \mathcal{I}_{\mathcal{E}} \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}' \longrightarrow 0 \tag{131}$$

where $\mathcal{E}' := \mathcal{E}/\mathcal{I}_{\mathcal{E}} \in \mathcal{E}(r-1, 0)$. By inductive hypothesis we must have $\mathcal{E}' \cong \mathcal{F}_{r-1}$. And by Proposition 5.5, we know extension of $\mathcal{F}_{r-1}$ by $\mathcal{O}_X$ is unique up to isomorphism. Hence the result also holds for $r$.

Now for any $\mathcal{E} \in \mathcal{E}(r, 0)$, then $\deg(\mathcal{E} \otimes A) = \deg \mathcal{E} + r \cdot \deg A = r$. Then by Lemma 5.11, $\mathcal{E} \otimes A$ admits a maximal filtration $(\mathcal{L}, \cdots, \mathcal{L})$ with $\deg \mathcal{L} = 1$. Note that $\mathcal{O}_X$ is a line subbundle of $\mathcal{E} \otimes A \otimes \mathcal{L}^\vee$, $\mathcal{E} \otimes A \otimes \mathcal{L}^\vee \in \mathcal{E}(r, 0)$ with nonzero global section so that isomorphic to $\mathcal{F}_r$. Take the line bundle to be $A^\vee \otimes \mathcal{L}$ is enough. In particular, we get $(\mathcal{O}_X, \cdots, \mathcal{O}_X)$ is a filtration of $\mathcal{F}_r$ and $\det \mathcal{E} \cong (\mathcal{A}^\vee \otimes \mathcal{L})^r$.

To show the uniqueness, suffices to prove if $\mathcal{F}_r \cong \mathcal{F}_r \otimes \mathcal{L}$, then $\mathcal{L}$ is tivial line bundle. As $(\mathcal{O}_X, \cdots, \mathcal{O}_X)$ is a filtration of $\mathcal{F}_r$, if $\mathcal{L}$ is not a trivial line bundle, then $h^0(X, \mathcal{L}) = 0$ and hence $h^0(X, \mathcal{F}_r \otimes \mathcal{L}) \leq r \cdot h^0(X, A^\vee \otimes \mathcal{L}) = 0$, contradicting to $\Gamma(X, \mathcal{F}_r) \neq 0$ □

**Corollary 5.6.** $\mathcal{F}_r \cong \mathcal{F}_r^\vee$.

*Proof.* According to proof of Theorem 5.2, we know $\mathcal{F}_r$ has a filtration $(\mathcal{O}_X, \cdots, \mathcal{O}_X)$. Hence $\mathcal{O}_X$ is a quotient of $\mathcal{F}_r$. Dually, we get $\mathcal{O}_X$ is a subbundle of $\mathcal{F}_r^\vee$. Thus $\mathcal{F}_r^\vee$ also has nonzero global sections. Conclude $\mathcal{F}_r \cong \mathcal{F}_r^\vee$. □

**Corollary 5.7.** *For any $s < r$, there is an exact sequence $0 \to \mathcal{F}_s \to \mathcal{F}_r \to \mathcal{F}_{r-s} \to 0$.*

*Proof.* First to show $\mathcal{F}_s$ is the unique subbundle of $\mathcal{F}$ of rank $s$, which is a successive extension of trivial line bundles. By Corollary 5.6, we know $\mathcal{F}_s \cong \mathcal{F}_s^\vee$ is such a subbundle of $\mathcal{F}_r \cong \mathcal{F}_r^\vee$. Suppose $\mathcal{E}_s$ is also such a subbundle. For $s = 1$, clearly $\mathcal{E}_1 \cong \mathcal{F}_1$. Induct on $s$ and $r$, suppose we have proven for all $s' < s$ and $r' > s'$. Note that $\mathcal{E}_s/\mathcal{F}_1$ is a subbundle of $\mathcal{F}_r/\mathcal{F}_1 \cong \mathcal{F}_{r-1}$. By inductive hypothesis, $\mathcal{E}_s/\mathcal{F}_1 \cong \mathcal{F}_{s-1}$. Now we get an extension

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{E}_s \longrightarrow \mathcal{F}_{s-1} \longrightarrow 0 \tag{132}$$

As $\mathcal{F}_1$ is trivial line bundle, the extension corresponds a map $H^0(X, \mathcal{F}_1^\vee) \to H^1(X, \mathcal{F}_{s-1}^\vee)$. Or equivalently by Serre duality, a map $H^0(X, \mathcal{F}_1) \to H^0(X, \mathcal{F}_{s-1})^\vee$. Note that $h^0(X, \mathcal{F}_i) = 0$ for all $i$. The map is either a zero map or an isomorphism. If it is a zero map, then $\mathcal{E}_s \cong \mathcal{F}_1 \oplus \mathcal{F}_{s-1}$. But then $\mathcal{O}_X^{\oplus 2} \subseteq \mathcal{F}_1 \oplus \mathcal{F}_{s-1}$ is a trivial subbundle of $\mathcal{F}_r$, contradicting to $h^0(X, \mathcal{F}_r) = 1$. Hence the map is an isomorphism so that $\mathcal{E}_s \cong \mathcal{F}_s$.

Finally, consider $\mathcal{E}_{r-s} := \mathcal{F}_r/\mathcal{F}_s$. Dually, we get $\mathcal{E}_{r-s}^\vee$ is a subbundle of $\mathcal{F}_r$. Remains to check $\mathcal{E}_{r-s}^\vee$ is a successive extension of trivial line bundles. As there is a filtration $\mathcal{E}_1^\vee \subset \cdots \subset \mathcal{E}_{r-s}^\vee$ and $\mathcal{E}_i^\vee/\mathcal{E}_{i-1}^\vee \cong \mathcal{F}_{r-i+1}/\mathcal{F}_{r-i} \cong \mathcal{F}_1$ is trivial line bundle, done! □

**Theorem 5.3.** *The fixed line bundle $A$ determines a one-to-one correspondence*

$$\alpha_{r,d} : \mathcal{E}(h, 0) \to \mathcal{E}(r, d) \tag{133}$$

*where $h$ is the greatest common divisor of $r$ and $d$. And for all $\mathcal{E} \in \mathcal{E}(h, 0)$, $\alpha_{r,d}$ is defined uniquely by the following properties*

- $\alpha_{r,0} = \mathrm{id}$
- $\alpha_{r,r+d}(\mathcal{E}) \cong \alpha_{r,d}(\mathcal{E}) \otimes A$
- *if $0 < d < r$, we have an exact sequence $0 \to \mathcal{I}_d \longrightarrow \alpha_{r,d}(\mathcal{E}) \longrightarrow \alpha_{r-d,d}(\mathcal{E}) \longrightarrow 0$, where $\mathcal{I}_d$ is the trivial bundle of rank $d$.*

*In particular,* $\det \alpha_{r,d}(\mathcal{E}) \cong \det \mathcal{E} \otimes A^d$.

*Proof.* With correspondence given by $\otimes A$, we may assume $d \geq 0$ first. When $d = 0$, we have $h = r$ so that we can just take $\alpha_{r,0}$ to be the identity map. If $d > 0$, note that in fact

Proposition 5.4 and Proposition 5.5 gives a one-to-one correspondence between $\mathcal{E}(r,d)$ and $\mathcal{E}(r-d,d)$, we have the following correspondence process that

$$\beta_{r,d} : \mathcal{E}(r,d) \longleftrightarrow \begin{cases} \mathcal{E}(r-d,d) & \text{if } r > d \\ \mathcal{E}(r,d-r) & \text{else} \end{cases} \tag{134}$$

Starting with $\mathcal{E}(r,d)$ and $\beta_{r,d}$ and reapeating the process, we get $\mathcal{E}(r,d) \xleftrightarrow{\beta_{r,d}} \mathcal{E}(r',d') \xleftrightarrow{\beta_{r',d'}} \cdots$. For each step, it is easy to see that $r'+d' < r+d$, $\gcd(r',d') = \gcd(r,d)$ and $d' \geq 0$. Hence the process would stop at $\mathcal{E}(h,0)$. Combining these correspondence together, we get one-to-one correspondence $\beta_{r,d} : \mathcal{E}(r,d) \longleftrightarrow \mathcal{E}(r_0,0)$. Take $\alpha_{r,d} = \beta_{r,d}^{-1}$. Then by definition, we know this correspondence satisfies those properties.

For uniqueness, suppose now we have another correspondence $\gamma_{r,d}$ satisfying those properties. Consider the correspondence $\mathcal{E}(r,d) \xleftrightarrow{\gamma_{r,d}^{-1}} \mathcal{E}(h,0) \xleftrightarrow{\gamma_{r,d+r}} \mathcal{E}(r,d+r)$. Then $\gamma_{r,d+r} \circ \gamma_{r,d}^{-1}(\mathcal{E}) \cong \mathcal{E} \otimes A = \alpha_{r,d+r} \circ \alpha_{r,d}^{-1}(\mathcal{E})$. And the last property also tells us that $\gamma_{r+d,d} \circ \gamma_{r,d}^{-1} = \alpha_{r+d,d} \circ \alpha_{r,d}^{-1}$. Conclude that $\gamma \cong \alpha$. □

**Theorem 5.4.** *We may regard $X$ as an abelian variety with identity the fixed point $x_0$. Then each $\mathcal{E}(r,d)$ can be identified with $X$ in such a way that*

$$\det : \mathcal{E}(r,d) \to \mathcal{E}(1,d) \textit{ corresponds to } H : X \to X \tag{135}$$

*where $H(x) = h \cdot x$ and $h = \gcd(r,d)$.*

**Reason 5.1.** *By Theorem 5.3, suffices to consider $\mathcal{E}(h,0)$. Then for all $\mathcal{E} \in \mathcal{E}(h,0)$, $\mathcal{E} \cong \mathcal{F}_h \otimes \mathcal{L}$ for some $\mathcal{L} \in \mathrm{Pic}^0(X)$. Hence we can identify $\mathcal{E}(h,0)$ with $\mathrm{Pic}^0(X)$. While there is isomorphism $X \to \mathrm{Pic}^0(X)$ sending $x$ to $\mathcal{O}_X(x) \otimes A^\vee$, we can identify $\mathcal{E}(h,0)$ with $X$. In particular, $\det \mathcal{E} \cong \det \mathcal{F}_h \otimes \mathcal{L}^h$ so that $\det$ corresponds to $H$.*

**Corollary 5.8.** *Notation as above, then for all $\mathcal{E} \in \mathcal{E}(r,d)$, we have that*
*(1)$\mathcal{E} \mapsto \det \mathcal{E}$ gives a one-to-one correspondence $\mathcal{E}(r,d) \longleftrightarrow \mathcal{E}(1,d)$.*
*(2)$\mathcal{E} \cong \mathcal{E}_A(r,d) \otimes \mathcal{L}$ for some $\mathcal{L} \in \mathrm{Pic}^0(X)$, where $\mathcal{E}_A(r,d) = \alpha_{r,d}(\mathcal{F}_h)$.*
*(3)$\mathcal{E}_A(r,d) \otimes \mathcal{L} \cong \mathcal{E}_A(r,d)$ if and only if $\mathcal{L}^r \cong \mathcal{O}_X$.*
*(4)$\mathcal{E}_A(r,d)^* \cong \mathcal{E}_A(r,-d)$.*

# 6 Stable Vector Bundles on Curves

## 6.1 Stability

Let $X$ be a regular projective curve over $\mathbb{C}$ of genus $g$.

**Definition 6.1.** *Let $f : \mathcal{E}_1 \to \mathcal{E}_2$ be a map between vector bundles on $X$. We say $f$ is of maximal rank if the induced map $\det f : \det \mathcal{E}_1 \to \det \mathcal{E}_2$ is a nonzero map.*

**Remark 6.1.** *As $\deg \mathcal{E}_i = \deg(\det \mathcal{E}_i)$, if $f : \mathcal{E}_1 \to \mathcal{E}_2$ is of maximal rank, then $\deg \mathcal{E}_1 \leq \deg \mathcal{E}_2$.*

**Definition 6.2.** *Let $\mathcal{F}$ be a coherent sheaf on $X$, $\mathcal{G} \subseteq \mathcal{F}$ subsheaf. We say $\mathcal{G}$ is saturated in $\mathcal{F}$ if $\mathcal{F}/\mathcal{G}$ is torsion free. Define the saturation of $\mathcal{G}$ in $\mathcal{F}$ is the minimal subsheaf $\mathcal{G}^{\mathrm{sat}}$ of $\mathcal{F}$ containing $\mathcal{G}$ such that $\mathcal{G}^{\mathrm{sat}}$ is saturated in $\mathcal{F}$.*

**Remark 6.2.** *As $X$ is integral, we can directly construct $\mathcal{G}^{\mathrm{sat}}$ by*

$$\mathcal{G}^{\mathrm{sat}} = \{s \in \mathcal{F} \big| \text{the image of } s \text{ in } \mathcal{F}/\mathcal{G}(U) \text{ is torsion}\} \tag{136}$$

*which is clearly the minimal saturated subsheaf containing $\mathcal{G}$.*

**Lemma 6.1.** *Let $\mathcal{F}$ be a coherent sheaf on $X$, $\mathcal{G} \subseteq \mathcal{F}$ subsheaf with saturation $\mathcal{G}^{\mathrm{sat}}$. Then* $\operatorname{rank} \mathcal{G} = \operatorname{rank} \mathcal{G}^{\mathrm{sat}}$.

*Proof.* As $X$ is integral, by Proposition 3.1, there is an open subset $U \subset X$ such that $(\mathcal{F}/\mathcal{G})\big|_U$ is locally free and hence torsion free. By our construction of saturation, it is clear that $\mathcal{G}^{\mathrm{sat}}\big|_U = \mathcal{G}\big|_U^{\mathrm{sat}} = \mathcal{G}\big|_U$. Thus $\operatorname{rank} \mathcal{G} = \operatorname{rank} \mathcal{G}^{\mathrm{sat}}$. □

Given a nonzero map $f : \mathcal{E} \to \mathcal{F}$ between vector bundles. Consider $\ker f$ and $\operatorname{im} f$, since they are subsheaves of $\mathcal{E}$ and $\mathcal{F}$ respectively, they are both torsion free. Moreover, as $X$ is a regular curve, we get $\ker f$ and $\operatorname{im} f$ are locally free. Take $\mathcal{E}' = \ker f$, $\mathcal{E}'' = \operatorname{im} f$, $\mathcal{F}'$ the saturation of $\mathcal{E}''$ in $\mathcal{F}$ and $\mathcal{F}'' = \mathcal{F}/\mathcal{F}'$. Then we get a commutative diagram consisting of vector bundles, with exact rows

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{E}' \arrow[r, hook] & \mathcal{E} \arrow[r, two heads] \arrow[d, "f"] & \mathcal{E}'' \arrow[r] \arrow[d, hook, "g"] & 0 \\
0 & \mathcal{F}'' \arrow[l] & \mathcal{F} \arrow[l, two heads] & \mathcal{F}' \arrow[l, hook'] & 0 \arrow[l]
\end{tikzcd} \tag{137}$$

By Lemma 6.1, we also have $\operatorname{rank} \mathcal{F}' = \operatorname{rank} \mathcal{E}''$.

**Definition 6.3.** *Let $\mathcal{F}$ be a coherent sheaf on $X$. Define the slope of $\mathcal{F}$ to be $\mu(\mathcal{F}) := \frac{\deg \mathcal{F}}{\operatorname{rank} \mathcal{F}}$.*

**Definition 6.4.** *We say a vector bundle $\mathcal{E}$ on $X$ is stable (resp. semi-stable) if for all proper subbundle $\mathcal{E}' \subset \mathcal{E}$, we have $\mu(\mathcal{E}') < \mu(\mathcal{E})$ (resp. $\mu(\mathcal{E}') \leq \mu(\mathcal{E})$).*

**Lemma 6.2.** *Let $\mathcal{E}$ be a semi-stable vector bundle of deg $d$ and rank $r$. If $\gcd(r, d) = 1$, then $\mathcal{E}$ is stable.*

*Proof.* Suffice to show for all subbundle $\mathcal{E}' \subseteq \mathcal{E}$, $\mu(\mathcal{E}') = \mu(\mathcal{E})$ if and only if $\mathcal{E}' = \mathcal{E}$. As $\mu(\mathcal{E}') = \mu(\mathcal{E})$, we get $\deg \mathcal{E}' \cdot \operatorname{rank} \mathcal{E} = \deg \mathcal{E} \cdot \operatorname{rank} \mathcal{E}'$. Since $\gcd(r, d) = 1$, we must have $r \mid \operatorname{rank} \mathcal{E}'$ so that $\mathcal{E}' = r$ and $\mathcal{E}' = \mathcal{E}$. □

**Proposition 6.1.** *There is an equivalent definition of stability (resp. semi-stability) that $\mathcal{E}$ is stable (resp. semi-stable) if for any nontrivial surjection $\mathcal{E} \twoheadrightarrow \mathcal{F}$ to vector bundle $\mathcal{F}$, we have $\mu(\mathcal{E}) < \mu(\mathcal{F})$ (resp. $\mu(\mathcal{E}) \leq \mu(\mathcal{F})$).*

*Proof.* Note that for any nontrivial surjection $\mathcal{E} \twoheadrightarrow \mathcal{F}$ to vector bundle $\mathcal{F}$, there is a short exact sequence

$$0 \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0 \tag{138}$$

where the kernel $\mathcal{E}'$ is automatically a proper subbundle of $\mathcal{E}$. This gives a one-to-one correspondence between proper subbundles and nontrivial surjections. In addition, as $\deg \mathcal{E} = \deg \mathcal{E}' + \deg \mathcal{F}$ and $\operatorname{rank} \mathcal{E} = \operatorname{rank} \mathcal{E}' + \operatorname{rank} \mathcal{F}$, we have

$$\begin{aligned} \deg \mathcal{E} \cdot \operatorname{rank} \mathcal{E}' > \deg \mathcal{E}' \cdot \operatorname{rank} \mathcal{E} &\iff \deg \mathcal{F} \cdot \operatorname{rank} \mathcal{E}' > \deg \mathcal{E}' \cdot \operatorname{rank} \mathcal{F} \\ &\iff \deg \mathcal{F} \cdot \operatorname{rank} \mathcal{E} > \deg \mathcal{E} \cdot \operatorname{rank} \mathcal{F} \end{aligned} \tag{139}$$

And respectively,

$$\begin{aligned} \deg \mathcal{E} \cdot \operatorname{rank} \mathcal{E}' \geq \deg \mathcal{E}' \cdot \operatorname{rank} \mathcal{E} &\iff \deg \mathcal{F} \cdot \operatorname{rank} \mathcal{E}' \geq \deg \mathcal{E}' \cdot \operatorname{rank} \mathcal{F} \\ &\iff \deg \mathcal{F} \cdot \operatorname{rank} \mathcal{E} \geq \deg \mathcal{E} \cdot \operatorname{rank} \mathcal{F} \end{aligned} \tag{140}$$

Conclude that the two definitions are equivalent. □

**Proposition 6.2.** *$\mathcal{E}$ is stable (resp. semi-stable) if and only if for all proper subbundle $\mathcal{E}' \subset \mathcal{E}$, $\deg(\mathcal{E}^\vee \otimes \mathcal{E}') < 0$ (resp. $\mathcal{E}' \subset \mathcal{E}$, $\deg(\mathcal{E}^\vee \otimes \mathcal{E}') \leq 0$).*

**Reason 6.1.** *Note that $\deg(\mathcal{E}^\vee \otimes \mathcal{E}') = \deg \mathcal{E}^\vee \cdot \operatorname{rank} \mathcal{E}' + \deg \mathcal{E}' \cdot \operatorname{rank} \mathcal{E}^\vee$. As $\operatorname{rank} \mathcal{E}^\vee = \operatorname{rank} \mathcal{E}$ and by Corollary 3.7, we have $\deg \mathcal{E}^\vee = -\deg \mathcal{E}$.*

**Proposition 6.3.** *Let $\mathcal{L}$ be a line bundle on $X$. Then $\mathcal{E}$ is stable (resp. semi-stable) if and only if $\mathcal{E} \otimes \mathcal{L}$ is stable (resp. semi-stable).*

*Proof.* Since $\mathcal{L}^\vee$ is also a line bundle, suffices to show if $\mathcal{E}$ stable (resp. semi-stable) then $\mathcal{E} \otimes \mathcal{L}$ stable (resp. semi-stable). In fact, for all proper subbundle $\mathcal{F} \subset \mathcal{E} \otimes \mathcal{L}$, we get $\mathcal{F} \otimes \mathcal{L}^\vee$ a proper subbundle of $\mathcal{E}$. Hence $\mu(\mathcal{F} \otimes \mathcal{L}^\vee) < \mu(\mathcal{E})$ (resp. $\mu(\mathcal{F} \otimes \mathcal{L}^\vee) \leq \mu(\mathcal{E})$). Note that

$$\begin{aligned} \mu(\mathcal{F} \otimes \mathcal{L}^\vee) &= \frac{\deg(\mathcal{F} \otimes \mathcal{L}^\vee)}{\operatorname{rank}(\mathcal{F} \otimes \mathcal{L}^\vee)} \\ &= \frac{\deg \mathcal{F} + \operatorname{rank} \mathcal{F} \cdot \deg \mathcal{L}^\vee}{\operatorname{rank} \mathcal{F}} \\ &= \mu(\mathcal{F}) + \deg \mathcal{L}^\vee \end{aligned} \tag{141}$$

and

$$\begin{aligned} \mu(\mathcal{E} \otimes \mathcal{L}) &= \frac{\deg(\mathcal{E} \otimes \mathcal{L})}{\operatorname{rank}(\mathcal{E} \otimes \mathcal{L})} \\ &= \frac{\deg \mathcal{E} + \operatorname{rank} \mathcal{E} \cdot \deg \mathcal{L}}{\operatorname{rank} \mathcal{E}} \\ &= \mu(\mathcal{E}) + \deg \mathcal{L} \end{aligned} \tag{142}$$

Thus $\mu(\mathcal{F}) < \mu(\mathcal{E} \otimes \mathcal{L})$ (resp. $\mu(\mathcal{F}) \leq \mu(\mathcal{E} \otimes \mathcal{L})$). □

**Proposition 6.4.** *Let $\mathcal{E}$ and $\mathcal{F}$ be two semi-stable vector bundles of same rank and degree. Assume at least one of them is stable. If $f : \mathcal{E} \to \mathcal{F}$ is a nonzero map, then $f$ is an isomorphism.*

*Proof.* Suppose $f$ is not isomorphism. By construction of the diagram 137, we get

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{E}' \arrow[r, hook] & \mathcal{E} \arrow[r, two heads] \arrow[d, "f"] & \mathcal{E}'' \arrow[r] \arrow[d, hook, "g"] & 0 \\ 0 & \mathcal{F}'' \arrow[l] & \mathcal{F} \arrow[l, two heads] & \mathcal{F}' \arrow[l, hook] & 0 \arrow[l] \end{tikzcd} \tag{143}$$

Since $\mathcal{E}$ is semi-stable, we have $\deg(\mathcal{E}^\vee \otimes \mathcal{E}') \leq 0$. As $0 = \deg(\mathcal{E}^\vee \otimes \mathcal{E}) = \deg(\mathcal{E}^\vee \otimes \mathcal{E}') + \deg(\mathcal{E}^\vee \otimes \mathcal{E}'')$, we get $\deg(\mathcal{E}^\vee \otimes \mathcal{E}'') \geq 0$. On the other hand, since $\mathcal{F}$ is semi-stable, we also have $\deg(\mathcal{F}^\vee \otimes \mathcal{F}') \leq 0$. As $g$ is of maximal rank, $\deg \mathcal{E}'' \leq \deg \mathcal{F}'$. While $\operatorname{rank} \mathcal{E}'' = \operatorname{rank} \mathcal{F}'$, we get $\deg(\mathcal{E}^\vee \otimes \mathcal{F}') \geq \deg(\mathcal{E}^\vee \otimes \mathcal{E}'')$. Since $\deg \mathcal{E} = \deg \mathcal{F}$ and $\operatorname{rank} \mathcal{E} = \operatorname{rank} \mathcal{F}$, we obtain

$$\begin{aligned} 0 &\geq \deg(\mathcal{F}^\vee \otimes \mathcal{F}') \\ &= \deg(\mathcal{E}^\vee \otimes \mathcal{F}') \\ &\geq \deg(\mathcal{E}^\vee \otimes \mathcal{E}'') \\ &\geq 0 \end{aligned} \tag{144}$$

As at least one of $\mathcal{E}$ and $\mathcal{F}$ is stable, at least one of the first and the last inequalities is strict, contradiction! Conclude that $f$ must be isomorphic. $\square$

**Definition 6.5.** *We say a vector bundle $\mathcal{E}$ on $X$ is simple if $h^0(X, \mathcal{E}nd(\mathcal{E})) = 1$. Or equivalently, the only elements in $\operatorname{End}(\mathcal{E})$ are multiplications by scalar.*

**Lemma 6.3.** *A simple vector bundle $\mathcal{E}$ is indecomposable.*

*Proof.* Suppose $\mathcal{E} \cong \mathcal{E}_1 \oplus \mathcal{E}_2$ for vector bundles $\mathcal{E}_1$ and $\mathcal{E}_2$. Then $\operatorname{End}(\mathcal{E}_1) \oplus \operatorname{End}(\mathcal{E}_2) \subseteq \operatorname{End}(\mathcal{E})$ so that $h^0(X, \mathcal{E}nd(\mathcal{E})) \geq 2$, contradiction! $\square$

**Corollary 6.1.** *A stable vector bundle $\mathcal{E}$ is simple and hence indecomposable.*

**Reason 6.2.** *By Proposition 6.4, the only elements in $\operatorname{End}(\mathcal{E})$ are zero map and isomorphisms. Hence $\operatorname{End}(\mathcal{E})$ is a division algebra finite over $\mathbb{C}$, which must be $\mathbb{C}$ as $\mathbb{C}$ is algebraically closed.*

**Proposition 6.5.** *Let $\mathcal{E}$ be a semi-stable vector bundle and $\mathcal{F}$ be a vector bundle with $\mu(\mathcal{E}) \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{E}) > \mu(\mathcal{F})$). If $f : \mathcal{E} \to \mathcal{F}$ is a nonzero morphism and $\mathcal{F}' := (\operatorname{im} f)^{\mathrm{sat}}$, then $\mu(\mathcal{F}') \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{F}') > \mu(\mathcal{F})$).*

*Proof.* By construction of the diagram 137, we get

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{E}' \arrow[r, hook] & \mathcal{E} \arrow[r, two heads] \arrow[d, "f"] & \mathcal{E}'' \arrow[r] \arrow[d, hook, "g"] & 0 \\ 0 & \mathcal{F}'' \arrow[l] & \mathcal{F} \arrow[l, two heads] & \mathcal{F}' \arrow[l, hook] & 0 \arrow[l] \end{tikzcd} \tag{145}$$

Hence $\mu(\mathcal{F}') \geq \mu(\mathcal{E}'')$. Since $\mathcal{E}$ is semi-stable, we get $\mu(\mathcal{E}'') \geq \mu(\mathcal{E})$. As $\mu(\mathcal{E}) \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{E}) > \mu(\mathcal{F})$), we are done! $\square$

**Proposition 6.6.** *If $\mathcal{E}$ is not stable, then there exists stable subbundle $\mathcal{E}' \subset \mathcal{E}$ with $\mu(\mathcal{E}') \geq \mu(\mathcal{E})$. Moreover, if $\mathcal{E}$ is not semi-stable, then there exists stable subbundle $\mathcal{E}' \subset \mathcal{E}$ with $\mu(\mathcal{E}') > \mu(\mathcal{E})$.*

*Proof.* Take $\mathcal{E}'$ to be the subbundle of $\mathcal{E}$ satisfying that $\mu(\mathcal{E}') \geq \mu(\mathcal{E})$ (resp. $\mu(\mathcal{E}') > \mu(\mathcal{E})$)with minimal rank. It is clear that $\mathcal{E}'$ is stable. $\square$

**Proposition 6.7.** *Let $\mathcal{F}$ be a vector bundle of rank $r$. Then $\mathcal{F}$ is not stable (resp. semi-stable) if and only if there exists stable vector bundle $\mathcal{E}$ with $\mu(\mathcal{E}) \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{E}) > \mu(\mathcal{F})$) and a nonisomorphic nonzero map $f : \mathcal{E} \to \mathcal{F}$.*

*Proof.* By Proposition 6.6, suffices to show if there exists stable vector bundle $\mathcal{E}$ with $\mu(\mathcal{E}) \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{E}) > \mu(\mathcal{F})$) and a nonisomorphic nonzero map $f : \mathcal{E} \to \mathcal{F}$, then $\mathcal{F}$ is not stable (resp. semi-stable). Suppose $\mathcal{F}$ is stable (resp. semi-stable).

Denote $\mathcal{F}' = (\operatorname{im} f)^{\mathrm{sat}}$. By Proposition 6.5, we have $\mu(\mathcal{F}') \geq \mu(\mathcal{F})$ (resp. $\mu(\mathcal{F}') > \mu(\mathcal{F})$). For the second case, this already contradicts to semi-stability of $\mathcal{F}$. For the first case, since $\mathcal{F}$ is stable, we get $\mathcal{F}'$ must be $\mathcal{F}$ itself. Consider the diagram 137

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{E}' \arrow[r, hook] & \mathcal{E} \arrow[r, two heads] \arrow[d, "f"] & \mathcal{E}'' \arrow[r] \arrow[d, hook, "g"] & 0 \\
0 & \mathcal{F}'' \arrow[l] & \mathcal{F} \arrow[l, two heads] & \mathcal{F}' \arrow[l, hook'] & 0 \arrow[l]
\end{tikzcd} \tag{146}$$

Then $\mu(\mathcal{F}) = \mu(\mathcal{F}') \geq \mu(\mathcal{E}'') \geq \mu(\mathcal{E}) \geq \mu(\mathcal{F})$. Hence all equalities hold. As $\mu(\mathcal{E}) = \mu(\mathcal{E}'')$ and $\mathcal{E}$ is stable, we get $\mathcal{E}' = \ker f = 0$. Now $\mathcal{E}$ and $\mathcal{F}$ have same rank and degree. By Proposition 6.4, $f$ is an isomorphism, contradiction! □

## 6.2 Relation to ampleness

For this section, you may refer to the paper *Ample Vector Bundles on Curves*, Hartshorne.

**Proposition 6.8.** *The following conditions are equivalent:*
*(1)Every stable vector bundle of positive degree on $X$ is ample.*
*(2)Every vector bundle, all of whose quotients bundles have positive degree, is ample.*

*Proof.* "$\Rightarrow$ ": Let $\mathcal{E}$ be a vector bundle, all of whose quotient bundles have positive degree. Induction on $\operatorname{rank}\mathcal{E}$. When $\operatorname{rank}\mathcal{E} = 1$, since line bundle of positive degree on $X$ is ample, there is nothing to prove. Suppose $\operatorname{rank}\mathcal{E} > 1$. If $\mathcal{E}$ is stable, then by assumption, $\mathcal{E}$ is ample.

Now $\mathcal{E}$ is not stable. By Proposition 6.6, there is a stable subbundle $\mathcal{E}' \subset \mathcal{E}$ with $\mu(\mathcal{E}') \geq \mu(\mathcal{E})$. Since $\mu(\mathcal{E}) > 0$, we get $\mu(\mathcal{E}') > 0$ so that $\mathcal{E}'$ has positive degree. By assumption, $\mathcal{E}'$ is ample. Consider the following exact sequence

$$0 \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E} \longmapsfrom \mathcal{E}'' := \mathcal{E}/\mathcal{E}' \longrightarrow 0 \tag{147}$$

Note that quotient bundles of $\mathcal{E}''$ are also quotient bundles of $\mathcal{E}$ and hence have positive degree. By inductive hypothesis, we know $\mathcal{E}''$ is ample. Then by Corollary 4.7, we get $\mathcal{E}$ is ample.

"$\Leftarrow$ ": Let $\mathcal{E}$ be a stable vector bundle with positive degree. For all nontrivial quotient bundle $\mathcal{E} \twoheadrightarrow \mathcal{E}''$, as $\mathcal{E}$ is stable, we know $\mu(\mathcal{E}'') > \mu(\mathcal{E}) > 0$ so that $\mathcal{E}''$ has positive degree. Then by assumption, $\mathcal{E}$ is ample. □

To relate stability and ampleness, we need a numerative criterion for ampleness.

**Theorem 6.1** (***Seshadri***)**.** *Let $X$ be a complete scheme, $D$ Cartier divisor on $X$. Then $D$ is ample on $X$ if and only if there exists $\varepsilon > 0$ such that for all integral curve $C \subset X$, we have that $\frac{D \cdot C}{m(C)} > \varepsilon$, where $m(C) := \max_{p \in C}\{\mathrm{multi}_p(C)\}$.*

*Proof.* "$\Rightarrow$ ": Take $m$ large enough such that $mD$ is very ample. Then $mD$ associates with an immersion $X \hookrightarrow \mathbb{P}^N$. Since $X$ is proper, the immersion is closed. Take hyperplane $H \subseteq \mathbb{P}^N$

such that $H \cap X = mD$. Then $(mD \cdot C)_X = (H \cdot C)_{\mathbb{P}^N} = \deg C \geq m(C)$. Hence $\frac{(D \cdot C)_X}{m(C)} \geq \frac{1}{m}$ and we may take $\varepsilon = \frac{1}{2m}$.

"$\Leftarrow$ ": We may assume $X$ is integral. Induct on $\dim X = n$. For $n = 1$, then there is no integral closed subscheme of higher dimension. As $D \cdot C > \varepsilon \cdot m(C) \geq 0$, by Nakai-Moisheson criterion, $D$ is ample. When $n > 1$, by induction hypothesis, we know $D^s \cdot Y > 0$ for all integral closed subscheme $Y \subset X$ of dimension $s < n$. Again by Nakai-Moisheson criterion, suffices to show $D^n > 0$.

Take a smooth closed point of $X$ and consider the blow up $f : X' \to X$ at $p$. Assume $E$ is the exceptional divisor and $D' = f^*D$ the total transform of $D$. Then $f^{-1}(p) \cong \mathbb{P}^{n-1}$ and $\mathcal{O}_X(E)\big|_E = \mathcal{O}_E(-1)$. Set $G := aD' - E$ where $a = \frac{1}{\varepsilon}$. Want to show $G$ is nef.

For all integral curve $C' \subset X'$. If $C' \subseteq E$, then $G \cdot C' = aD' \cdot C' - E \cdot C' = -E \cdot C' = \deg C' > 0$. Or if $C' \cap E' \neq \varnothing$ but $C' \not\subset E$, then $G \cdot C' = aD' \cdot C' - E \cdot C' = aD \cdot C - \mathrm{multi}_p(C) > 0$, where $C$ is the image of $C$ under $f$. Otherwise, $C' \cap E = \varnothing$, then $G \cdot C' = aD' \cdot C' - E \cdot C' = aD \cdot C > 0$. Thus by numerative criterion for nefness, $G$ is nef.

Now $G$ is nef. Note that $(D')^n = D^n$, $E^n = (E^{n-1})_E = (-1)^{n-1}$ and $(D')^i \cdot E^{n-i} = 0$ for all $0 < i < n$.

$$\begin{aligned} 0 \leq G^n &= (aD' - E)^n \\ &= (aD')^n - E^n \\ &= a^n D^n - 1 \end{aligned} \tag{148}$$

which implies that $D^n > 0$. □

In Hartshorne's paper, he proved the following deep theorem with results of Narasimhan and Seshadri relating stable vector bundles to certain group representations. Here we quote it without proof.

**Theorem 6.2.** *Let $X$ be a complete smooth complex curve of genus $g \geq 2$, $\mathcal{E}$ stable vector bundle on $X$. Then $S^n\mathcal{E}$ is semi-stable for all $n \geq 1$.*

**Lemma 6.4.** *Let $f : X \to Y$ be a finite surjection between integral noetherian schemes. sometimes called finite cover. Then we have*
*(1)*$R^i f_*\mathcal{F} = 0$ *for all coherent* $\mathcal{F}$ *and* $i > 0$
*(2)*$\mathrm{rank}(f_*\mathcal{F}) = \deg f \cdot \mathrm{rank}\,\mathcal{F}$ *for all coherent* $\mathcal{F}$*, where* $\deg f = [K(X) : K(Y)]$.
*(3)*$\deg(f_*\mathcal{F}) = \deg \mathcal{F} + \mathrm{rank}\,\mathcal{F} \cdot \deg(f_*\mathcal{F})$ *for all coherent* $\mathcal{F}$.

*Proof.* For (1), we may assume $Y = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$ with $A$ finite over $B$. Then $R^i f_*\mathcal{F} = \widetilde{H^i(X, \mathcal{F})}$, which by Serre vanishing is zero.

For (2), we can also $Y = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$ with $A$ finite over $B$. Assume $\mathcal{F}$ is given by $M$ finite over $B$. Then $\deg f = [\mathrm{Frac}(B) : \mathrm{Frac}(A)]$, $\mathrm{rank}(f_*\mathcal{F}) = \dim_{\mathrm{Frac}(A)}(M \otimes_A \mathrm{Frac}(A))$ and $\mathrm{rank}\,\mathcal{F} = \dim_{\mathrm{Frac}(B)}(M \otimes_B \mathrm{Frac}(B))$. As $f$ is surjective, the corresponding $\varphi : A \to B$ is injective.

Note that there is a natural map $\psi : B \otimes_A \mathrm{Frac}(A) \to \mathrm{Frac}(B)$. It is clear that $\psi$ is injective. And since $\varphi$ is integral, for all $b \in B$, we have $b^n + a_1 b^{n-1} + \cdots + a_{n-1} b + a_n = 0$

with $a_i \in A$. Then $b(b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1}) = -a_n \in A$ so that $\psi((b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1}) \otimes \frac{1}{-a_n}) = \frac{1}{b}$. Thus $\psi$ is isomorphic and we get

$$\begin{aligned} \operatorname{rank}(f_*\mathcal{F}) &= \dim_{\operatorname{Frac}(A)}(M \otimes_A \operatorname{Frac}(A)) \\ &= \dim_{\operatorname{Frac}(A)}(M \otimes_B (B \otimes_A \operatorname{Frac}(A))) \\ &= \dim_{\operatorname{Frac}(A)}(M \otimes_B \operatorname{Frac}(B)) \\ &= [\operatorname{Frac}(B) : \operatorname{Frac}(A)] \cdot \dim_{\operatorname{Frac}(B)}(M \otimes_B \operatorname{Frac}(B)) \\ &= \deg f \cdot \operatorname{rank} \mathcal{F} \end{aligned} \tag{149}$$

For (3), this immediately comes from (1) and (2) as following

$$\begin{aligned} \deg(f_*\mathcal{F}) &= \chi(Y, f_*\mathcal{F}) - \operatorname{rank}(f_*\mathcal{F}) \cdot \chi(Y, \mathcal{O}_Y) \\ &= \chi(X, \mathcal{F}) - \deg f \cdot \operatorname{rank} \mathcal{F} \cdot \chi(Y, \mathcal{O}_Y) \\ &= \deg \mathcal{F} + \operatorname{rank} \mathcal{F} \cdot \chi(X, \mathcal{O}_X) - \deg f \cdot \operatorname{rank} \mathcal{F} \cdot \chi(Y, \mathcal{O}_Y) \\ &= \deg \mathcal{F} + \operatorname{rank} \mathcal{F} \cdot \chi(Y, f_*\mathcal{O}_Y) - \operatorname{rank}(f_*\mathcal{O}_X) \cdot \operatorname{rank} \mathcal{F} \cdot \chi(Y, \mathcal{O}_Y) \\ &= \deg \mathcal{F} + \operatorname{rank} \mathcal{F} \cdot \deg(f_*\mathcal{F}) \end{aligned} \tag{150}$$

done! □

From now on, we again assume $X$ to be a smooth projective curve over $\mathbb{C}$.

**Proposition 6.9.** *Let $\mathcal{E}$ be a vector bundle on $X$ of positive degree satisfying that $S^n\mathcal{E}$ is semi-stable for all $n$. Then $\mathcal{E}$ is ample.*

*Proof.* Assume $\operatorname{rank} \mathcal{E} = r$ and $\deg \mathcal{E} = d$. Consider the associated projective bundle $\pi : \mathbb{P}(\mathcal{E})$. Then by Proposition 4.11, suffices to show the tautological line bundle $\mathcal{L} := \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is ample. And by Theorem 6.1, want to find $\varepsilon > 0$ such that $\mathcal{L} \cdot C > \varepsilon \cdot m(C)$ for all integral curve $C \subseteq \mathbb{P}(\mathcal{E})$.

If $\pi(C)$ is a single point, then $C$ is contained in some fiber of $\pi$, which is isomorphic to $\mathbb{P}^{r-1}_{\mathbb{C}}$. Hence $\mathcal{L} \cdot C = \deg C \geq m(C)$. When $\pi(C)$ is not a single point, then $\dim(\pi(C)) = 1$ so that $\pi(C) = X$ and $\pi\big|_C : C \to X$ is a finite surjection. Then by Lemma 6.4, $\operatorname{rank}((\pi\big|_C)_*(\mathcal{L}^m\big|_C)) = \deg(\pi\big|_C)$ and $\deg((\pi\big|_C)_*(\mathcal{L}^m\big|_C)) = \deg(\mathcal{L}^m\big|_C) + \deg((\pi\big|_C)_*\mathcal{O}_C)$. Consider the following surjection

$$S^m\mathcal{E} \cong (\pi\big|_C)_*(\mathcal{L}^m) \twoheadrightarrow (\pi\big|_C)_*(\mathcal{L}^m\big|_C) \tag{151}$$

Note that pushforward preserves torsion-freeness and $X$ is a regular curve, $(\pi\big|_C)_*(\mathcal{L}^m\big|_C)$ is locally free. By assumption, $S^m\mathcal{E}$ is semi-ample so that $\mu(S^m\mathcal{E}) \leq \mu((\pi\big|_C)_*(\mathcal{L}^m\big|_C))$. Then we have

$$\begin{aligned} \frac{dm}{r} &\leq \frac{\deg(\mathcal{L}^m\big|_C) + \deg((\pi\big|_C)_*\mathcal{O}_C)}{\deg(\pi\big|_C)} \\ \frac{dm \cdot \deg(\pi\big|_C)}{r} &\leq \deg(\mathcal{L}^m\big|_C) + \deg((\pi\big|_C)_*\mathcal{O}_C) \\ \frac{d \cdot \deg(\pi\big|_C)}{r} &\leq \frac{1}{m} \cdot (\deg(\mathcal{L}^m\big|_C) + \deg((\pi\big|_C)_*\mathcal{O}_C)) \end{aligned} \tag{152}$$

Note that $m \cdot \deg(\mathcal{L}\big|_C) = \deg(\mathcal{L}^m\big|_C) = \chi(\mathcal{L}^m\big|_C) - \chi(\mathcal{O}_C)$. By definition of intersection number, $\mathcal{L} \cdot C = \deg(\mathcal{L}\big|_C)$. Hence

$$\frac{d \cdot \deg(\pi\big|_C)}{r} \leq \mathcal{L} \cdot C + \frac{1}{m} \cdot \deg((\pi\big|_C)_* \mathcal{O}_C) \tag{153}$$

Taking $m$ converging to $\infty$, we get

$$\frac{d \cdot \deg(\pi\big|_C)}{r} \leq \mathcal{L} \cdot C \tag{154}$$

Note that $\deg(\pi\big|_C) \geq m(C)$, we are done! □

**Remark 6.3.** *It is not clear for me to see why* $\deg(\pi\big|_C) \geq m(C)$.

**Theorem 6.3.** *Vector bundle $\mathcal{E}$ is ample if and only if every quotient bundle of $\mathcal{E}$ has positive degree.*

*Proof.* If $\mathcal{E}$ is ample, then clearly every quotient bundle of $\mathcal{E}$ is also ample. And ample vector bundle automatically has positive degree. Conversely, by Proposition 6.8, suffices to prove every stable vector bundle of positive degree is ample. Consider stable vector bundle $\mathcal{E}$ of positive degree. When genus $g \geq 2$, by Theorem 6.2, $S^n\mathcal{E}$ is semi-stable for all $n \geq 1$. Then by Proposition 6.9, $\mathcal{E}$ is ample. For $g = 1$, this is a result by Atiyah. And for $g = 0$, $X$ is a rational curve. Note that stable vector bundle is indecomposable, while the only indecomposable vector bundle is line bundle and line bundle of positive degree is ample, done! □

**Lemma 6.5.** *Let $\mathcal{E}$ be a vector bundle on $X$, $s \in \Gamma(X, \mathcal{E})$ not vanishing at any singular point of $X$. Then $\mathcal{E}$ has a line subbundle $\mathcal{L}$ such that $s \in \Gamma(X, \mathcal{L})$. Moreover, if $s$ is not invertible, $\mathcal{L}$ is ample.*

*Proof.* Note that $s$ would give a rational section $\tau : X \dashrightarrow \mathbb{P}(\mathcal{E})$ defined on $X$ minus finitely many regular points. Hence $\tau$ would extend to a regular section. Then just as what we did in Lemma 5.2, we get a line subbundle $\mathcal{L}$ of $\mathcal{E}$. In addition, $s$ is a global section of $\mathcal{L}$. If $s$ is not invertible, then $\deg \mathcal{L} = \deg(\mathrm{div}(s)) \geq 1 > 0$ so that $\mathcal{L}$ is ample. □

**Proposition 6.10.** *Let $Y$ be a proper variety over algebraically closed field $k$, $\mathcal{E}$ vector bundle over $Y$ globally generated. Then $\mathcal{E}$ is ample if and only if every quotient line bundle of $\mathcal{E}\big|_C$ is ample for all curve $C \subset Y$.*

*Proof.* If $\mathcal{E}$ is ample, since $C \hookrightarrow Y$ is closed immersion, $\mathcal{E}\big|_C$ is still ample by Corollary 4.3. Hence every quotient line bundle of $\mathcal{E}\big|_C$ is also ample. Conversely, suppose that every quotient line bundle of $\mathcal{E}\big|_C$ is ample for all curve $C \subset Y$.

First consider a line bundle $\mathcal{E}$. As $\mathcal{E}$ is globally generated, it induces a morphism $\varphi : Y \to \mathbb{P}^N$ such that $\mathcal{E} \cong \varphi^*(\mathcal{O}(1))$. Suppose $\varphi$ is not quasi-finite, then there exists some fiber $Y'$ of dimension $\geq 1$ which would contain some curve $C$. Hence $\mathcal{E}\big|_C \cong \varphi^*(\mathcal{O}(1)\big|_{\varphi(C)})$ is trivial and of degree 0. While by assumption $\mathcal{E}\big|_C$ is ample, contradiction. Thus $\varphi$ is quasi-finite. As $Y$ is proper, we know $\varphi$ is in fact finite so that by Proposition 4.5, we know $\mathcal{E}$ is ample.

When $Y$ is a curve, we prove the result by induction on $\operatorname{rank}\mathcal{E}$. By inductive hypothesis, every nontrivial quotient bundle of $\mathcal{E}$ is ample. Hence we suffices to find an ample line subbundle of $\mathcal{E}$. By Lemma 6.5, only need to find a section which does not vanish at any singular point of $Y$ but not invertible.

Set $n = h^0(Y, \mathcal{E})$. As $\mathcal{E}$ is globally generated, there is a surjection $\psi : \mathcal{O}_Y^{\oplus n} \twoheadrightarrow \mathcal{E}$. Consider the Grassmannian scheme $\mathrm{Gr}(\operatorname{rank}\mathcal{E}, n)$ over $k$ with universal quotient bundle $\mathcal{U}$. Then $\psi$ would corresponds to a morphism $f : Y \to \mathrm{Gr}(\operatorname{rank}\mathcal{E}, n)$ satisfying that $\mathcal{E} \cong f^*\mathcal{U}$.

If $f(Y)$ is a single point, then similarly we get $\mathcal{E}$ trivial so that trivial line bundle is quotient line bundle of $\mathcal{E}$, contradicting to ample assumption. Then $\dim f(Y) = 1$ so that we can find $y \in Y$ such that $f(y)$ is not image of any singular point of $Y$. Note that for any two points $z$ and $w$ in $\mathrm{Gr}(\operatorname{rank}\mathcal{E}, n)$, $z = w$ if and only if

$$\Gamma(\mathrm{Gr}(\operatorname{rank}\mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_z) = \Gamma(\mathrm{Gr}(\operatorname{rank}\mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_w) \tag{155}$$

Thus $\Gamma(\mathrm{Gr}(\operatorname{rank}\mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_{f(y)}) \cap \Gamma(\mathrm{Gr}(\operatorname{rank}\mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_{f(y')}) \subsetneq \Gamma(\mathrm{Gr}(\operatorname{rank}\mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_{f(y')})$ for singular $f(y')$. Since $k$ is infinite, we get

$$\Gamma(\mathrm{Gr}(\operatorname{rank}\mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_{f(x)}) \subsetneq \cup_{y' \text{ singular}} \Gamma(\mathrm{Gr}(\operatorname{rank}\mathcal{E}, n), \mathcal{U} \otimes \mathcal{I}_{f(y')}) \tag{156}$$

So we may pick global section $s$ of $\mathcal{U}$ not vanishing at any $f(y')$. Pull back to $Y$, we get global section $f^*s$ of $\mathcal{E}$ not vanishing at any singular $y'$.

Now if $Y$ is an arbitrary proper variety, consider the associated projective bundle $\pi : \mathbb{P}(\mathcal{E}) \to Y$. By Proposition 4.11, suffices to show the tautological line bundle $\mathcal{L} := \mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is ample. Let $C'$ be any curve in $\mathbb{P}(\mathcal{E})$. If $C'$ is contained in some fiber, then $\mathcal{L}\big|_C$ is ample on $C'$. Otherwise, $\pi\big|_{C'} : C' \to \pi(C')$ is a finite surjection and $\pi(C')$ is a curve in $Y$. By argument above, $\mathcal{E}\big|_{\pi(C')}$ is ample and hence $(\pi\big|_{C'})^*\mathcal{E}$ is ample on $C'$. As there is a surjection $\pi^*\mathcal{E} \twoheadrightarrow \mathcal{L}$, we also get $\mathcal{L}\big|_C$ ample. Since $\mathcal{L}$ is globally generated, by argument above, $\mathcal{L}$ is ample and we are done! $\square$

# 7 Moduli Space of Vector Bundles

## 7.1 Higher direct image

Recall two local criterion for flatness.

**Proposition 7.1.** *Let $(A, \mathfrak{m})$ be a local ring with residue field $k$, $M$ finitely generated $A$-module. Then the following conditions are equivalent:*
*(1)$M$ is free*
*(2)$M$ is projective*
*(3)$M$ is flat*
*(4)*$\mathrm{Tor}_1^A(M, k) = 0$.

*Proof.* (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) is clear. Suffices to prove (4) $\Rightarrow$ (1). As $M$ is finitely generated, by Nakayama's Lemma, we can pick representative elements of a basis of $M/\mathfrak{m}M$ over $k$, which also generate $M$. And these elements induces a surjection $A^{\oplus n} \twoheadrightarrow M$ with kernel $N$. Since $\mathrm{Tor}_1^A(M,k)=0$, the following sequence is exact

$$0 \longrightarrow N \otimes_A k \longrightarrow A^{\oplus n} \otimes_A k \longrightarrow M \otimes_A k \longrightarrow 0 \tag{157}$$

Note that by our choice, $A^{\oplus n} \otimes_A k \to M \otimes_A k$ is an isomorphism. Hence $N \otimes_A k = 0$ so that by Nakayama's Lemma again, $N = 0$. Conclude that $M \cong A^{\oplus n}$ is free. □

**Proposition 7.2.** *Let $f : (A, \mathfrak{m}) \to (B, \mathfrak{n})$ be a local homomorphism between local rings, $M$ finitely generated $B$-module. Then $M$ is flat over $A$ if and only if $\mathrm{Tor}_1^A(M, A/\mathfrak{m}) = 0$.*

**Remark 7.1.** *This result is much more tricky. Proof can be seen in Stacks Project 00MK.*

**Corollary 7.1.** *Let $f : (A, \mathfrak{m}) \to (B, \mathfrak{n})$ be a local homomorphism between local rings, $M$ finitely generated $B$-module. Assume $B$ is flat over $A$. Then $M$ is flat over $A$ if and only if $\mathrm{Tor}_1^A(M, B \otimes_A A/\mathfrak{m}) = 0$.*

*Proof.* By Proposition 7.2, suffices to show if $\mathrm{Tor}_1^A(M, B\otimes_A A/\mathfrak{m}) = 0$, then $\mathrm{Tor}_1^A(M, A/\mathfrak{m}) = 0$. Consider the following exact sequence

$$0 \longrightarrow \mathrm{Tor}_1^A(M, A/\mathfrak{m}) \longrightarrow \mathfrak{m} \otimes_A M \longrightarrow M \longrightarrow A/\mathfrak{m} \otimes_A M \longrightarrow 0 \tag{158}$$

Since $B$ is flat over $A$, tensoring by $B$, we get exact sequence

$$0 \longrightarrow \mathrm{Tor}_1^A(M, A/\mathfrak{m}) \otimes_A B \longrightarrow \mathfrak{m} \otimes_A M \otimes_A B \longrightarrow M \otimes_A B \longrightarrow B \otimes_A A/\mathfrak{m} \otimes_A M \longrightarrow 0 \tag{159}$$

On the other hand, we also have the following exact sequence

$$0 \to \mathrm{Tor}_1^A(M, B \otimes_A A/\mathfrak{m}) \otimes_A B \to \mathfrak{m} \otimes_A M \otimes_A B \to M \otimes_A B \to B \otimes_A A/\mathfrak{m} \otimes_A M \to 0 \tag{160}$$

Hence we get $\mathrm{Tor}_1^A(M, A/\mathfrak{m}) \otimes_A B = 0$. While flat local homomorphism is automoatically faithfully flat, by faithfully flat descent, we get $\mathrm{Tor}_1^A(M, A/\mathfrak{m}) = 0$. □

**Definition 7.1.** *Let $f : X \to Y$ be a morphism, $\mathcal{F}$ sheaf module on $X$. We say $\mathcal{F}$ is flat over $Y$ is for all $x \in Y$, $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. In particular, is $\mathcal{O}_X$ is flat over $Y$, we say $X$ is flat over $Y$.*

Given $X$ flat over $Y$, $\mathcal{F}$ sheaf module on $X$. If there is a locally free resolution $\mathcal{L}^\bullet$ of $\mathcal{F}$, then for all $y \in Y$, restricting to fibered product $X(y) := X \times_Y \operatorname{Spec} \mathcal{O}_{Y,y}$, we get locally free resolution $\mathcal{L}^\bullet\big|_{X(y)}$ of $\mathcal{F}\big|_{X(y)}$ by faltness.

Let we introduce some results to better describe derived pushforward.

**Lemma 7.1.** *The derived pushforward $R^i f_* \mathcal{F}$ can be identified with the sheafification of the presheaf*

$$U \longmapsto H^i(f^{-1}(U), \mathcal{F}) \tag{161}$$

*Proof.* Take injective resolution $0 \to \mathcal{F} \to \mathcal{I}^\bullet$. By definition, $R^i f_*\mathcal{F}$ is the $i$th cohomology of the following complex

$$0 \longrightarrow f_*\mathcal{I}^0 \xrightarrow{\psi_0} f_*\mathcal{I}^1 \xrightarrow{\psi_1} f_*\mathcal{I}^2 \xrightarrow{\psi_2} \cdots \tag{162}$$

Hence $R^i f_*\mathcal{F}$ is the sheafification of presheaf $U \mapsto (\ker\psi_i)(U)/(\operatorname{im}\psi_{i-1})(U)$. Applying $\Gamma(U,\cdot)$, then we get

$$0 \longrightarrow \Gamma(f^{-1}(U),\mathcal{I}^0) \xrightarrow{\psi_0(U)} \Gamma(f^{-1}(U),\mathcal{I}^1) \xrightarrow{\psi_1(U)} \Gamma(f^{-1}(U),\mathcal{I}^2) \xrightarrow{\psi_2(U)} \cdots \tag{163}$$

Note that $\operatorname{im}\psi_i$ is the sheafification of the presheaf $U \mapsto \operatorname{im}\psi_i(U)$. Checking stalkwise, it is clear that $R^i f_*\mathcal{F}$ is the sheafification of $U \mapsto (\ker\psi_i)(U)/\operatorname{im}\psi_{i-1}(U) = H^i(\Gamma(f^{-1}(U),\mathcal{I}^\bullet)) = H^i(f^{-1}(U),\mathcal{F})$. $\square$

**Lemma 7.2.** *Let $X$ and $Y$ be algebraic varieties over field $k$, $\mathcal{F}$ coherent sheaf on $X$. Consider the projections $p : X \times Y \to X$ and $q : X \times Y \to Y$. Set $\mathcal{G} := p^*\mathcal{F}$. Then for all $y \in Y$, there is a canonical isomorphism $\varinjlim_{U\ni y}\mathcal{G}(X \times U) \cong \mathcal{F}(X) \otimes_k \mathcal{O}_{Y,y}$.*

*Proof.* Since the problem is local on $Y$, we can assume $Y = \operatorname{Spec} B$ is affine. Cover $X$ with affine open subsets $V_i = \operatorname{Spec} A_i$. Assume $\mathcal{F}\big|_{V_i}$ is defined by $A_i$-module $M_i$. Then $\mathcal{G}\big|_{V_i\times Y} = \widetilde{M_i \otimes_k} B$ so that

$$\begin{aligned}
\varinjlim_{U\ni y}\mathcal{G}(V_i \times U) &= \varinjlim_{U=D(b)\ni y} \mathcal{G}(V_i \times U) \\
&= \varinjlim_{U=D(b)\ni y} M_i \otimes_k B_b \\
&\cong M_i \otimes_k \varinjlim_{U=D(b)\ni y} B_b \\
&= M_i \otimes_k \mathcal{O}_{Y,y}
\end{aligned} \tag{164}$$

Hence the result holds when $X$ is affine. Now by definition of sheaf, there is an exact sequence

$$0 \longrightarrow \mathcal{G}(X \times Y) \longrightarrow \prod_i \mathcal{G}(V_i \times Y) \longrightarrow \prod_{i,j} \mathcal{G}((V_i \cap V_j) \times Y) \tag{165}$$

As taking direct limit is exact, we get exact sequence

$$0 \longrightarrow \varinjlim_{U\ni y}\mathcal{G}(X \times U) \longrightarrow \prod_i \varinjlim_{U\ni y}\mathcal{G}(V_i \times U) \longrightarrow \prod_{i,j} \varinjlim_{U\ni y}\mathcal{G}((V_i \cap V_j) \times U) \tag{166}$$

On the other hand, since $\mathcal{O}_{Y,y}$ is flat over $k$, we have exact sequence

$$0 \longrightarrow \mathcal{F}(X) \otimes_k \mathcal{O}_{Y,y} \longrightarrow \prod_i \mathcal{F}(V_i) \otimes_k \mathcal{O}_{Y,y} \longrightarrow \prod_{i,j} \mathcal{F}(V_i \cap V_j) \otimes_k \mathcal{O}_{Y,y} \tag{167}$$

While by argument above, we already know $\varinjlim_{U\ni y}\mathcal{G}(V_i \times U) \cong \mathcal{F}(V_i) \otimes_k \mathcal{O}_{Y,y}$ and $\varinjlim_{U\ni y}\mathcal{G}((V_i \cap V_j) \times U) \cong \mathcal{F}(V_i \cap V_j) \otimes_k \mathcal{O}_{Y,y}$. Conclude that $\varinjlim_{U\ni y}\mathcal{G}(X \times U) \cong \mathcal{F}(X) \otimes_k \mathcal{O}_{Y,y}$. $\square$

**Proposition 7.3.** *Let $X$ and $Y$ be algebraic varieties over field $k$, $\mathcal{F}$ coherent sheaf on $X$. Consider the projections $p : X \times Y \to X$ and $q : X \times Y \to Y$. Set $\mathcal{G} := p^*\mathcal{F}$. Then $R^i q_* \mathcal{G} \cong \mathcal{O}_Y \otimes_k H^i(X, \mathcal{F})$.*

*Proof.* Take injective resolution $0 \to \mathcal{F} \to \mathcal{I}^\bullet$ and $0 \to \mathcal{G} \to \mathcal{J}^*$. Then there exists chain map $p^*\mathcal{I}^\bullet \to \mathcal{J}^\bullet$. And for all open subset $U \subseteq Y$, there is a map of cohomology groups $H^i((p^*\mathcal{I}^\bullet)(X \times U)) \to H^i(\mathcal{J}^\bullet(X \times U))$. Then there is a sequence of maps

$$
\begin{aligned}
\mathcal{O}_Y(U) \otimes_k H^i(X, \mathcal{F}) &\xrightarrow{\sim} \mathcal{O}_Y(U) \otimes_k H^i(\mathcal{I}^\bullet(X)) \\
&\xrightarrow{\sim} H^i(\mathcal{I}^\bullet(X) \otimes_k \mathcal{O}_Y(U)) \\
&\xrightarrow{\sim} H^i(\mathcal{I}^\bullet(X) \otimes_{\mathcal{O}_X(X)} (\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(U))) \\
&\longrightarrow H^i((p^{-1}\mathcal{I}^\bullet)(X \times U) \otimes_{(p^{-1}\mathcal{O}_X)(X \times U)} \mathcal{O}_{X \times Y}(X \times U)) \\
&\longrightarrow H^i((p^*\mathcal{I}^\bullet)(X \times U)) \\
&\longrightarrow H^i(\mathcal{J}^\bullet(X \times U)) \\
&\xrightarrow{\sim} H^i(q^{-1}(U), \mathcal{G})
\end{aligned}
\tag{168}
$$

By Lemma 7.1 and universal property of sheafification, we know this map induces a map $\mathcal{O}_Y \otimes_k H^i(X, \mathcal{F}) \to R^i q_* \mathcal{G}$. Remains to show this map is isomorphic. Suffices to check stalkwise. For all $y \in Y$, take colimit $\varinjlim_{U \ni y}$ for the the sequence 168, only need to check the following sequence of maps are isomorphic

$$
\begin{aligned}
H^i(\mathcal{I}^\bullet(X) \otimes_{\mathcal{O}_X(X)} \varinjlim_{U \ni y} (\mathcal{O}_X(X) \otimes_k \mathcal{O}_Y(U))) &\longrightarrow H^i(\varinjlim_{U \ni y} (p^*\mathcal{I}^\bullet)(X \times U)) \\
&\longrightarrow \varinjlim_{U \ni y} H^i(X \times U, \mathcal{G})
\end{aligned}
\tag{169}
$$

Since $X$ and $Y$ are varieties, we can compute cohomology with Čech complex. In addition, since the problem is local on $Y$, we can assume $Y = \operatorname{Spec} B$ is affine. Cover $X$ with affine open subsets $V_i = \operatorname{Spec} A_i$. Consider the Čech complex

$$
0 \longrightarrow \prod_i \mathcal{G}(V_i \times Y) \longrightarrow \prod_{i,j} \mathcal{G}((V_i \cap V_j) \times Y) \longrightarrow \cdots \tag{170}
$$

Applying $\varinjlim_{U \ni y}$, get

$$
0 \longrightarrow \prod_i \varinjlim_{U \ni y} \mathcal{G}(V_i \times Y) \longrightarrow \prod_{i,j} \varinjlim_{U \ni y} \mathcal{G}((V_i \cap V_j) \times Y) \longrightarrow \cdots \tag{171}
$$

By Lemma 7.2, this is isomorphic to

$$
0 \longrightarrow \prod_i \mathcal{F}(V_i) \otimes_k \mathcal{O}_{Y,y} \longrightarrow \prod_{i,j} \mathcal{F}(V_i \cap V_j) \otimes_k \mathcal{O}_{Y,y} \longrightarrow \cdots \tag{172}
$$

Since exact functor commutes with taking cohomology, we know $H^i(X \times U, \mathcal{G}) \cong H^i(X) \otimes_k \mathcal{O}_{Y,y}$. It is clear that this is same to the map given by the sequence. Conclude that $R^i q_* \mathcal{G} \cong \mathcal{O}_Y \otimes_k H^i(X, \mathcal{F})$. □

**Lemma 7.3.** *Let $f : X \hookrightarrow Y$ be a closed immersion between varieties, $\mathcal{F}$ coherent on $X$. Then $f_*\mathcal{F}$ is coherent on $Y$ and $R^if_*\mathcal{F} = 0$ for all $i > 0$.*

*Proof.* Since the problem is local on $Y$, we may assume that $Y = \operatorname{Spec} A$ is affine so that $X = \operatorname{Spec}(A/I)$ for some ideal $I$. Then $R^if_*\mathcal{F} \cong \widetilde{H^i(X, \mathcal{F})}$. By Serre Vanishing Theorem, $H^i(X, \mathcal{F}) = 0$ for all $i > 0$. When $i = 0$, by definition, we know $H^0(X, \mathcal{F})$ is finitely generated over $A/I$ so that is also finitely generated over $A$. $\square$

**Theorem 7.1.** *Let $f : X \to Y$ be a projective morphism between varieties, $\mathcal{F}$ coherent on $X$. Then $R^if_*\mathcal{F}$ is coherent on $Y$ for all $i$.*

*Proof.* Since $f$ factor through $X \overset{\ell}{\hookrightarrow} \mathbb{P}^N_Y \overset{\pi}{\to} Y$ and $R^if_* = R^i\pi_* \circ (\ell_*)$. By Lemma 7.3, we know $\ell_*\mathcal{F}$ is coherent. Hence we only need prove for $X = \mathbb{P}^N_S$ projective space over $Y$. We may also assume $Y = \operatorname{Spec} A$ is affine. In fact by Theorem 2.7, we get $H^i(X, \mathcal{F})$ is finite over $A$ for all $i \geq 0$, done! $\square$

**Remark 7.2.** *There is another proof for a generalized version of this result in Stacks Project 02O5, which only ask the morphism to be proper.*

**Lemma 7.4.** *Let $f : X \to Y$ be a projective morphism between varieties. Then for all coherent $\mathcal{F}$ on $Y$, the natural map $f^*f_*(\mathcal{F}(n)) \to \mathcal{F}(n)$ is surjective for $n$ large enough.*

**Reason 7.1.** *In fact, this a special case of Lemma 4.3.*

**Theorem 7.2.** *Let $f : X \to Y$ be a projective morphism between varieties, $\mathcal{F}$ coherent on $X$. Then $R^if_*(\mathcal{F}(m)) = 0$ for $n$ large enough.*

**Reason 7.2.** *By similar argument as proof of Theorem 7.1, this result also comes from Theorem 2.7.*

Now assume $Y = X \times S$ for some projective smooth curve $X$ with two projections $p$ and $q$. If $S$ is affine, by Corollary 1.1, there is a locally free resolution $\mathcal{L}_\bullet \to \mathcal{F} \to 0$. In particular, we can choose all $\mathcal{L}_i$ of the form $\mathcal{O}_Y(-m_i)^{\oplus r_i}$ with $m_i$ and $r_i$ positive. Note that $\mathcal{O}_Y(-m_i) = p^*\mathcal{O}_X(-m_i)$ and $H^0(X, \mathcal{O}_X(-m_i)) = 0$. By Proposition 7.3, we get $R^iq_*\mathcal{L}_i = 0$ for $i \neq 1$ and $R^1q_*\mathcal{L}_i$ locally free.

**Lemma 7.5.** *In this setting, $R^jq_*\mathcal{F} = 0$ for all $j \geq 2$.*

*Proof.* Split the resolution into short exact sequences $0 \to \mathcal{Z}_i \to \mathcal{L}_i \to \mathcal{Z}_{i-1} \to 0$, where $\mathcal{Z}_{-1} = \mathcal{F}$ and $\mathcal{Z}_i = \ker(\mathcal{L}_i \to \mathcal{Z}_{i-1})$ for all $i \geq 0$. Push forward along $q$, we get long exact sequences

$$\begin{aligned} 0 &\longrightarrow q_*\mathcal{Z}_i \longrightarrow 0 \longrightarrow q_*\mathcal{Z}_{i-1} \longrightarrow R^1q_*\mathcal{Z}_i \longrightarrow R^1q_*\mathcal{L}_i \longrightarrow R^1q_*\mathcal{Z}_{i-1} \\ &\longrightarrow R^2q_*\mathcal{Z}_i \longrightarrow 0 \longrightarrow R^2q_*\mathcal{Z}_i \longrightarrow R^3q_*\mathcal{Z}_{i-1} \longrightarrow 0 \longrightarrow \cdots \end{aligned} \tag{173}$$

Hence $R^jq_*\mathcal{Z}_i \cong R^{j+1}q_*\mathcal{Z}_{i+1}$ for all $j \geq 2$. While $\dim Y$ is finite and $R^jq_*\mathcal{Z}_i$ is given by $H^j(X, \mathcal{Z}_i)$, $R^jq_*\mathcal{Z}_i = 0$ for all $j > \dim Y$. Conclude $R^jq_*\mathcal{F} = 0$ for all $j \geq 2$. $\square$